

Welcome to *General Intelligence and Seed AI* version 2.3. The purpose of this document is to describe the principles, paradigms, cognitive architecture, and cognitive components needed to build a complete mind possessed of general intelligence, capable of self-understanding, self-modification, and recursive self-enhancement.

Multi-page version: <http://singinst.org/GISAI/>

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Preface

"General Intelligence and Seed AI" is a publication of the Singularity Institute for Artificial Intelligence, Inc., a nonprofit corporation. You can contact the Singularity Institute at institute@singinst.org. To support the Singularity institute, visit <http://singinst.org/donate.html>. The Singularity Institute is a 501(c)(3) public charity and your donations are tax-deductible to the full extent of the law. The seed AI project is presently in the design/conceptualization stage and no code has yet been written; additional funding is required before the project can be launched.

"General Intelligence and Seed AI" is written in informal style. Academic readers, readers seeking a more technical explanation, or readers who prefer a more formal style, may wish to read "Levels of Organization in General Intelligence" instead.

This is a near-book-length explanation. If you need well-grounded knowledge of the subject, then we highly recommend reading "GISAI" straight through. However, if you need answers immediately, see the Singularity Institute pages on AI for introductory articles.

GISAI is a work in progress. As of version 2.3, the sections "Paradigms" and "Mind" are complete and self-contained. The section "Cognition" is in progress and may contain references to unimplemented topics. As additional topics are published, the minor version number (second digit) increases.

Words defined in the Glossary look like this: "A °seed AI is an AI capable of self-understanding, self-modification, and recursive self-enhancement."

Executive Summary and Introduction

Please bear in mind that the following is an introduction *only*. It contains some ideas which must be introduced in advance to avoid circular dependencies in the actual explanations, and a general summary of the cognitive architecture so you know where the ideas fit in. In particular, I am not expecting you to read the introduction and immediately shout: "Aha! This is the Secret of AI!" Some important ideas are described, yes, but just because an idea is *necessary* doesn't make it *sufficient*. Too many of AI's past failures have come of the trophy-hunting mentality, asking which buzzwords the code can be described by, and not asking what the code actually *does*.

This document is about general intelligence - what it is, and how to build one. The desired end result is a self-enhancing mind or "seed AI". Seed AI means that - rather than trying to build a mind immediately capable of human-equivalent or transhuman reasoning - the goal is to build a mind capable of enhancing itself, and then re-enhancing itself with that higher intelligence, until the goal point is reached. "The task is not to build an AI with some astronomical level of intelligence; the task is building an AI which is capable of improving *itself*, of understanding and rewriting its own source code. The task is not to build a mighty oak tree, but a humble seed." (From 1.1: Seed AI.)

General intelligence itself is *huge*. The human brain, created by millions of years of evolution, is composed of a hundred billion neurons connected by a hundred trillion synapses, forming more than a hundred neurologically distinguishable areas. We should not expect the problem of AI to be easy. Subproblems of cognition include attention, memory, association, abstraction, symbols, causality, subjunctivity, expectation, goals, actions, introspection, caching, and learning, to cite a non-exhaustive list. These features are *not* "emergent". They are °*complex functional adaptations*, evolved systems with multiple components and sophisticated internal architectures, whose functionality must be deliberately duplicated within an artificial mind. If done right, cognition can support the thoughts implementing abilities such as analysis, design, understanding, invention, self-awareness, and the other facets which together sum to an intelligent mind. An intelligent mind with access to its own source code can do all kinds of neat stuff, but we'll get into that later.

Different schools of AI are distinguished by different kinds of underlying "mindstuff". Classical AI consists of "predicate calculus" or "propositional logic", which is to say suggestively named °LISP tokens, plus directly coded procedures intended to imitate human formal logic. Connectionist AI consists of neurons implemented on the token level, with each neuron in the input and output layers having a programmer-determined interpretation, plus intervening layers which are usually *not* supposed to have a direct interpretation, with the overall network being trained by an external algorithm to perform perceptual tasks. (Although more biologically realistic implementations are emerging.) Agent-based AI consists of hundreds of humanly-written pieces of code which do whatever the programmer wants, with interactions ranging from handing data structures around to tampering with each other's behaviors.

Seed AI inherits connectionism's belief that error tolerance is a good thing. Error tolerance leads to the ability to mutate. The ability to mutate leads to evolution. Evolution leads to rich complexity - "mindstuff" with lots of tentacles and interconnections. However, connectionist theory presents a dualistic opposition between °stochastic, error-tolerant neurons and the crystalline fragility of code or assembly language. This conflates two logically distinct ideas. It's possible to have crystalline neural networks in which a single error breaks the chain of causality, or stochastic code in which (for example) multiple, mutable implementations of a function point have tweakable weightings. Seed AI strongly emphasizes the necessity of rich complexity in cognitive processes, and mistrusts classical AI's direct programmatic implementations.

However, seed AI also mistrusts that connectionist position which holds higher-level cognitive processes to be sacrosanct and opaque, off-limits to the human programmer, who is only allowed to fool around with neuron behaviors and training algorithms, and not the actual network patterns. Seed AI does prefer learned concepts to preprogrammed ones, since learned concepts are richer. Nonetheless, I think it's permissible, if risky, to preprogram concepts in order to bootstrap the AI to the point where it can learn. More to the point, it's okay to have an architecture where, even though the higher levels are stochastic or self-organizing or emergent or learned or whatever, the programmer can still see and modify what's going on. And it is *necessary* that the designer know what's happening on the higher levels, at least in general terms, because cognitive abilities are *not* emergent and do *not* happen by accident. Both classical AI and connectionist AI propose a kind of magic that avoids the difficulty of actually implementing the higher layers of cognition. Classical AI states that a LISP token named "goal" is a goal. Connectionist AI declares that it can all be done with neurons and training algorithms. Seed AI admits the necessity of confronting the problem directly.

In the human brain, there's at least one multilevel system where the higher levels, though stochastic, still have known interpretations: the visual processing system. Feature extraction by the visual cortex and associated areas doesn't proceed in a strict hierarchy with numbered levels (seed AI mistrusts that sort of thing), but there are definitely lower-level features (such as retinal pixels), mid-level features (such as edges and surface textures), and high-level features (such as 3D shapes and moving objects). Together, the pixels and attached interpretations constitute the cognitive object that is a visual description. It's also possible to run the feature-extraction system in reverse, activate a high-level feature and have it draw in the mid-level features which draw in the low-level features. Such "reversible patterns" are necessary-but-not-sufficient to memory recall and directed imagination. Memory and imagination, when implemented via this method, can hold rich concepts that mutate interestingly and mix coherently. A mental image of a red sausage can mutate directly to a mental image of a blue sausage without *either* storing the perception of redness in a single crystalline token *or* mutating the image pixel by independent pixel. °David Marr's paradigm of the "two-and-a-half dimensional world", multilevel holistic descriptions, is writ large and held to apply not just to sensory feature extraction but to categories, symbols, and other concepts. If seed AI has a "mindstuff", this is it.

Seed AI also emphasizes the problem of sensory modalities (such as the visual cortex, auditory cortex, and sensorimotor cortex in humans), previously considered a matter for specialized robots. A sensory modality consists of data structures suited to representing the "pixels" and features of the target domain, and codelets or processing stages which extract mid-level and high-level features of that domain. Sensory modalities grant superior intuitions and visualizational power in the target domain, which itself is sufficient reason to give a self-modifying AI a sensory modality for source code. Sensory modalities can also provide useful metaphors and concrete substrate for abstract reasoning about other domains; you can play chess using your visual cortex, or imagine a "branching" if-then-else statement. Sensory modalities provide a source of computational "raw material" from which concepts can form. Finally, a sensory modality provides intuitions for understanding concrete problems in a training domain, such as

source code. This makes it possible for the AI to *learn* the art of abstraction - moving from concrete problems, to categorizing sensory data, to conceptualizing complex methods, and so on - instead of being expected to swallow high-level thought all at once.

Sensory modalities are the foundations of intelligence - a term carefully selected to reflect necessity but not sufficiency; after you build the foundations, there's still a lot of house left over. In particular, a codic modality does not write source code, just as the visual cortex does not design skyscrapers. When I speak of a "codic" sensory modality, I am not extending the term "sensory modality" to include an autonomous facility for writing source code. I am using "modality" in the original sense to describe a system almost exactly analogous to the visual cortex, just operating in the domain of *source code* instead of *pixels*.

Sensory modalities - visual, spatial, codic - are the bottom layer of the AI, the layer in which representations and behaviors are specified directly by the programmer. (Although avoiding the crystalline fragility of classical AI is still a design goal.) The next layer is *concepts*. Concepts are pieces of mindstuff, which can either describe the mental world, or can be *applied* to alter the mental world. (Note that successive concepts can be applied to a single target, building up a complex visualization.) Concepts are contained in long-term memory. Categories, symbols, and most varieties of declarative memory are concepts. Concepts are more powerful if they are learned, trained, or otherwise created by the AI, but can be created by the programmer for bootstrapping purposes. (If, of course, the programmer can hack the tools necessary to modify the concept level.) The underlying substrate of the concept can be code, assembly language, or neural nets, whichever is least fragile and is easiest to understand and mutate; this issue is discussed later, but I currently lean towards code. (Not *raw* code, of course, but code as it is understood by the AI.)

Concepts, when retrieved from long-term memory, built into a structure, and activated, create a *thought*. The archetypal example of a thought is building words - symbols - into a grammatical sentence and "speaking" them within the mind. Thoughts exist in the RAM of the mind, the "working memory" created by available workspace in the sensory modalities. During their existence, thoughts can modify that portion of the world-model currently being examined in working memory. (Not every sentence spoken within the mind is supposed to describe reality; thoughts can also create and modify "subjunctive" ("what-if") hypotheses.) Thoughts are identified with - supposed to implement the functionality of - the human "stream of consciousness".

The three-layer model of intelligence is necessary, but not sufficient. Building an AI "with sensory modalities, concepts, and thoughts" is no guarantee of intelligence. The AI must have the *right* sensory modalities, the *right* concepts, and the *right* thoughts.

Evolution is the cause of intelligence in humans. Intelligence is an evolutionary advantage because it enables us to model, predict, and manipulate reality, including that portion of reality consisting of other humans and ourselves. In our physical Universe, reality tends to organize itself along lines that might be called "holistic" or "reductionist", depending on whether you're looking up or looking down. "Which facts are likely to reappear? The simple facts. How to recognize them? Choose those that seem simple. Either this simplicity is real or the complex elements are indistinguishable. In the first case we're likely to meet this simple fact again either alone or as an element in a complex fact. The second case too has a good chance of recurring since nature doesn't randomly construct such cases." (Robert M. Pirsig, "Zen and the Art of Motorcycle Maintenance", p. 238.)

Thought takes place within a causal, goal-oriented, "reductholistic" world-model, and seeks to better understand the world or invent solutions to a problem. Some methods include: "Holistic analysis: Taking a known high-level characteristic of a known high-level object ("birds fly"), and using "heuristics (thought-level knowledge learned from experience) to try and construct an explanation for the characteristic; an explanation consists of a low-level structure which gives rise to that high-level characteristic in a manner consistent with all known facts about the high-level object ("a bird's flapping wings push it upwards"). Causal analysis: Taking a known fact ("my telephone is ringing") and using heuristics to construct a causal sequence which results in that fact ("someone wants to speak to me"). Holistic design: Taking a high-level characteristic as a design goal ("go fast"), using heuristics to reduce the search space by reasoning about constraints and opportunities in possible designs ("use wheels"), and then testing ideas for specific low-level structures that attempt to satisfy the goals ("bicycles").

Both understanding and invention are fundamentally and messily recursive; whether a bicycle works depends on the design of the wheels, and whether a wheel works depends on whether that wheel consists of steel, rubber or tapioca pudding. Hence the need for "heuristics that bind high-level characteristics to low-level properties. Hence the need to recurse on finding new heuristics or more evidence or better tools or greater intelligence or higher self-awareness before the ultimate task can be solved. Solving a problem gives rise to lasting self-development as well as immediate solutions.

When a sufficiently advanced AI can bind a high-level characteristic like "word-processing program" through the multiple layers of design to individual lines of code, "we can write a word-processing program given the verbal instruction of "Write a word-processing program." (Of course, following verbal instructions also assumes speech recognition and language processing - not to mention a very detailed knowledge of what a word-processing program is, what it does, what it's for, how humans will use it, and why the program shouldn't erase the hard drive.) When the AI, perhaps given a sensory modality for atoms and molecules, can understand all the extant research on molecular manipulation, "we can work out a sequence of steps which will result in the construction of a general nanotechnological assembler, or tools to build one. When the AI can bind a high-level characteristic like "useful intelligence" through the multiple layers of designed cognitive processes to individual lines of code, we can redesign "vis own source code and increase vis intelligence.

Developing such a seed AI may require a tremendous amount of programmer effort and programmer creativity; it is entirely possible that a seed AI is the most ambitious software project in history, not just in terms of the end result, but in terms of the sheer *depth* of internal design complexity. To bring the problem into the range of the humanly solvable, it is necessary that development be broken up into stages, so that the first stages of the AI can assist with later stages. The usual aphorism is that 10% of the code implements 90% of the functionality, which suggests one approach. Seed AI adds the distinction between learned concepts and programmer-designed concepts. If so, the first stage might be an AI with simplified modalities, preprogrammed simple concepts, low-level goal definitions, and perhaps even programmer-assisted development of the stream-of-consciousness reflexes needed for coherent thought. Such an AI would hopefully be capable of manipulating code in simple ways, thus rendering the source code for concepts (and in fact its own source code) subject to the type of flexible and useful mutations needed to learn rich concepts or evolve more optimized code. The skeleton AI helps us fill in the flesh on the skeleton.

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Have you got all that?

Good.

Take a deep breath.

We're ready to begin.

1: Paradigms

- 1.1: Seed AI
 - 1.1.1: The AI Advantage
- 1.2: Thinking About AI

1.1: Seed AI

It is probably impossible to write an AI in immediate possession of human-equivalent abilities in every field; transhuman abilities even more so, since there's no working model. The task is not to build an AI with some astronomical level of intelligence; the task is building an AI which is capable of improving *itself*, of understanding and rewriting its own source code. The task is not to build a mighty oak tree, but a humble seed.

As the AI rewrites itself, it moves along a trajectory of intelligence. The task is not to build an AI at some specific point on the trajectory, but to ensure that the trajectory is open-ended, reaching human equivalence and transcending it. Smarter and smarter AIs become better and better at rewriting their own code and making themselves even smarter. When writing a seed AI, it's not just what the AI can do now, but what it will be able to do later. And the problem isn't just writing good code, it's writing code that the seed AI can understand, since the eventual goal is for it to rewrite its own assembly language. (1).

If "recursive self-enhancement" is to avoid running out of steam, it's necessary for code optimization or architectural changes to result in an increment of actual intelligence, of *smartness*, not just speed. Running an optimizing compiler over its own source code (2) may result in a faster optimizing compiler. Repeating the procedure a second time accomplishes nothing, producing an identical set of binaries, since the same algorithm is being run - only faster. A human who fails to solve a problem in one year (or solves it suboptimally) may benefit from another ten years to think about the problem; even so, an individual human may eventually run out of ideas. An individual human who fails to solve a problem in a hundred years may, if somehow transformed into an Einstein, solve it within an hour. Faster unintelligent algorithms accomplish little or nothing; faster intelligent thought can make a small difference; *better* intelligent thought makes the problem new again.

If each rung on the ladder of recursive self-enhancement involves a leap of sufficient magnitude, then each rung should open up enough new vistas of self-improvement for the next rung to be reached. If not, of course, the seed AI will have optimized itself and used up all perceived opportunities for improvement without generating the insight needed to see new *kinds* of opportunities. In this case the seed AI will have stalled, and it will be time for the human programmers to go to work nudging it over the bottleneck. Ultimately, the AI must cross, not only the gap that separates the mythical average human from Einstein, but the gap that separates *homo sapiens neanderthalis* from *homo sapiens sapiens*. The leap to true *understanding*, when it happens, will open up at least as many possibilities as would be available to a human researcher with access to his own neural source code.

A surprisingly frequent objection to self-enhancement is that intelligence, when defined as "the ability to increase intelligence", is a circular definition - one which would, they say, result in a sterile and uninteresting AI. Even if this were the definition (it isn't), and the definition were circular (it wouldn't be), the cycle could be broken simply by grounding the definition in chess-playing ability or some similar test of ability. However, intelligence is *not* defined as the ability to increase intelligence; that is simply the form of intelligent behavior we are most interested in. Intelligence is not *defined* at all. What intelligence *is*, if you look at a human, is more than a hundred "cytoarchitecturally distinct areas of the brain, all of which work together to create intelligence. Intelligence is, in short, *modular*, and the tasks performed by individual modules are *different in kind* from the nature of the overall intelligence. If the overall intelligence can turn around and look at a module as an isolated process, it can make clearly defined performance improvements - improvements that eventually sum up to improved overall intelligence - without ever confronting the circular problem of "making itself more intelligent". Intelligence, from a design perspective, is a goal with many, many subgoals. An intelligence seeking the goal of improved intelligence does not confront "improved intelligence" as a naked fact, but as a very rich and complicated fact adorned with less complicated subgoals.

Presumably there is an ultimate limit to the intelligence that can be achieved on a given piece of hardware, but if the seed AI can design better hardware, the cycle continues. To be concrete, if a seed AI is smart enough to chart a path from modern technological capabilities to "nanotechnology" - to the hardware described in K. Eric Drexler's *Nanosystems* - this should be enough computing power to provide thousands or millions of times the raw capacity of a human brain. (3). Whether the cognitive and technological trajectory beyond this point continues forever or tops out at some ultimate physical limit is basically irrelevant from a human perspective; nanotechnology plus thousands of times human brainpower should be *far* more than enough to accomplish whatever you wanted a transhuman for in the first place.

This scenario often meets with the objection that a lone AI can accomplish nothing; that technological advancement requires an entire civilization, with exchanges between thousands of scientists or millions of humans. This actually *understates* the problem. To think a single thought, it is necessary to duplicate far more than the genetically programmed functionality of a single human brain. After all, even if the functionality of a human were perfectly duplicated, the AI might do nothing but burble for the first year - that's what human infants do.

Perceptions have to coalesce into concepts. The concepts have to be strung together into thoughts. Enough good thoughts have to be repeated often enough for the sequences to become "cached", for the often-repeated subpatterns to become reflex. Enough of these infrastructural reflexes must accumulate for one thought to give rise to another thought, in a connected chain, forming a stream of consciousness. Unless we want to sit around for years listening to the computer go ga-ga, the functionality of infancy must be either encapsulated in a virtual world that runs in computer time, or bypassed using a skeleton set of preprogrammed concepts and thoughts. (Hopefully, the "skeleton thoughts" will be replaced by real, *learned* thoughts as the seed AI practices thinking.)

Human *scientific* thought relies on millennia of accumulated knowledge, the how-to-think "heuristics" discovered by hundreds of geniuses. While a seed AI may be able to absorb some of this knowledge by surfing the 'Net, there will be other dilemmas, unique to seed AIs, that it must solve on its own.

Finally, the autonomic processes of the human mind reflect millions of years of evolutionary optimization. Unless we want to expend an equal amount of programming effort, the functionality of evolution itself must be replaced - either by the seed AI's self-tweaking of those algorithms, or by replacing processes that are autonomic in humans with the deliberate decisions of the seed AI.

That's a gargantuan job, but it's matched by equally powerful tools.

1.1.1: The AI Advantage

The traditional advantages of computer programs - not "AI", but "computer programs" - are threefold: The ability to perform repetitive tasks *without getting bored*; the ability to perform algorithmic tasks at *greater linear speeds* than our 200-"hertz" neurons permit; and the ability to perform complex algorithmic tasks *without making mistakes* (or rather, without making those classes of mistakes which are due to distraction or running out of short-term memory). All of which, of course, has nothing to do with intelligence.

The toolbox of seed AI is yet unknown; nobody has built one. This page is more about building the first stages, the task of getting the seed AI to say "Hello, world!" But, if this can be done, what advantages would we expect of a general intelligence with access to its own source code?

The ability to design *new sensory modalities*. In a sense, any human programmer is a blind painter - worse, a painter born without a visual cortex. Our programs are painted pixel by pixel, and are accordingly sensitive to single errors. We need to consciously keep track of each line of code as an abstract object. A seed AI could have a "codic cortex", a sensory modality devoted to code, with intuitions and instincts devoted to code, and the ability to abstract higher-level concepts from code and intuitively visualize complete models detailed in code. A human programmer is very far indeed from vis ancestral environment, but an AI can always be at home. (But remember: A codic modality doesn't write code, just as a human visual cortex doesn't design skyscrapers.)

The ability to *blend conscious and autonomic thought*. Combining "Deep Blue" with "Kasparov" doesn't yield a being who can consciously examine a billion moves per second; it yields a Kasparov who can wonder "How can I put a queen here?" and blink out for a fraction of a second while a million moves are automatically examined. At a higher level of integration, Kasparov's conscious perceptions of each consciously examined chess position may incorporate data culled from a million possibilities, and Kasparov's dozen examined positions may not be consciously simulated moves, but "skips" to the dozen most plausible futures five moves ahead. (5).

Freedom from human failings, and especially human politics. The tendency to rationalize untenable positions to oneself, in order to win arguments and gain social status, seems so *natural* to us; it's hard to remember that rationalization is a "complex functional adaptation, one that would have no reason to exist in "minds in general". A synthetic mind has no political instincts (6); a synthetic mind could run the course of human civilization without politically-imposed dead ends, without "observer bias", without the tendency to rationalize. The reason we humans instinctively think that progress requires multiple minds is that we're used to human geniuses, who make one or two breakthroughs, but then get stuck on their Great Idea and oppose all progress until the next generation of brash young scientists comes along. A genius-equivalent mind that doesn't age and doesn't rationalize could encapsulate that cycle within a single entity.

Overpower - the ability to devote more raw computing power, or more efficient computing power, than is devoted to some module in the original human mind; the ability to throw more brainpower at the problem to yield intelligence of higher quality, greater quantity, faster speed, even *difference in kind*. Deep Blue eventually beat Kasparov by pouring huge amounts of computing power into what was essentially a glorified search tree; imagine if the basic component processes of human intelligence could be similarly overclocked...

Self-observation - the ability to capture the execution of a module and play it back in slow motion; the ability to watch one's own thoughts and trace out chains of causality; the ability to form concepts about the self based on fine-grained introspection.

Conscious learning - the ability to deliberately construct or deliberately improve concepts and memories, rather than entrusting them to autonomic processes; the ability to tweak, optimize, or debug learned skills based on deliberate analysis.

Self-improvement - the ubiquitous glue that holds a seed AI's mind together; the means by which the AI moves from crystalline, programmer-implemented skeleton functionality to rich and flexible thoughts. In the human mind, "stochastic concepts" - combined answers made up of the average of many little answers - leads to error tolerance; error tolerance lets concepts mutate without breaking; mutation leads to evolutionary growth and rich complexity. An AI, by using probabilistic elements, can achieve the same effect; another route is deliberate observation and manipulation, leading to deliberate "mutations" with a vastly lower error rate. What are these mutations or manipulations? A blind search can become a heuristically guided search and vastly more useful; an autonomic process can become conscious and vastly richer; a conscious process can

become autonomic and vastly faster - there is no sharp border between conscious learning and tweaking your own code. And finally, there are high-level redesigns, not "mutations" at all, alterations which require too many simultaneous, non-backwards-compatible changes to ever be implemented by evolution.

If all of that works, it gives rise to *self-encapsulation* and *recursive self-enhancement*. When the newborn mind fully understands vis own source code, when ve fully °understands the intelligent reasoning that went into vis own creation - and when ve is capable of °inventing that reason independently, so that the mind contains its own design - the cycle is closed. The mind causes the design, and the design causes the mind. Any increase in intelligence, whether sparked by hardware or software, will result in a better mind; which, since the design was (or could have been) generated by the mind, will propagate to cause a better design; which, in turn, will propagate to cause a better mind. (7). And since the seed AI will encapsulate not only the functionality of human individual intelligence but the functionality of evolution and society, these causes of intelligence will be subject to improvement as well. We might call it a "civilization-in-a-box", an entity with more "hardware" intelligence than Einstein (8) and capable of codifying abstract thought to run at the linear speed of a modern computer.

A successful seed AI would have *power*. A genuine civilization-in-a-box, thinking at a millionfold human speed, might fold centuries of technological progress into mere hours. I won't beat the point to death. I've done so in my other writings - Staring into the Singularity, in particular. It's just that the fundamental *purpose* of transhuman AI differs from that of traditional AI.

The academic purpose of modern prehuman AI is to write programs that demonstrate some aspect of human thought - to hold a mirror up to the brain. The commercial purpose of prehuman AI is to automate tasks too boring, too fast, or too expensive for humans. It's possible to dispute whether an academic implementation actually captures an aspect of human intelligence, or whether a commercial application performs a task that deserves to be called "intelligent".

In transhuman AI, if success isn't *blatantly obvious* to everyone except trained philosophers, the effort has failed. The ultimate purpose of transhuman AI is to create a °Transition Guide; an entity that can safely develop °nanotechnology and any subsequent ultratechnologies that may be possible, use transhuman °Friendliness to see what comes next, and use those ultratechnologies to see humanity safely through to whatever life is like on the other side of the Singularity. This might consist of assisting all humanity in upgrading to the level of superintelligent Powers, or creating an operating system for all the quarks in the Solar System, or something completely unknowable. I believe that, as the result of creating a °Friendly superintelligence, involuntary death, pain, coercion, and stupidity will be erased from the human condition; and that humanity, or whatever we become, will go on to fulfill to the maximum possible extent whatever greater destiny or higher goals exist, if any do.

To return to Earth: There will undoubtedly be many milestones, many interim subgoals and interim successes, along the path to superintelligence. The key point is that while embodying some aspect of cognition may be useful or necessary, it is not an end in itself. Treating facets of cognition as ends in themselves has led traditional AI to develop a sort of "trophy mentality", a tendency to value programs according to whether they fit surface descriptions. (One gets the impression that if you asked certain AI researchers to write the next Great English Novel, they'd write a 20-page essay on toaster ovens and then tear off through the streets, shouting: "Eureka! It's in English! It's in English!") My hope is that the lofty but utilitarian goals of seed AI will lead to the habit of looking at every piece of the design and saying: "Sure, it sounds neat, but how does it contribute materially to general intelligence?" After all, if an aspect of cognition is duplicated faithfully but without understanding its overall purpose, it's a matter of pure faith to expect it to contribute anything.

But that brings us to the next section, "Thinking About AI".

1.2: Thinking About AI

AI has, in the past, failed repeatedly. The shadow cast by this failure falls over all proposals for new AI projects. The question is always asked: "Why won't your project fail, like all the other projects? Why did the previous projects fail? Does your theory of general intelligence explain the previous failures while predicting success for your own efforts?" Actually, anyone can explain away previous failures and predict success; all you have to do is assert that some particular new characteristic is the One Great Idea, necessary and sufficient to intelligence. The real question is whether a new approach to AI makes the failure of previous efforts seem massively inevitable, the predictable result of historical factors; whether the approach provides a theory of previous failures that is satisfyingly obvious in retrospect, makes earlier errors look like natural mistakes that any growing civilization might make, and thus "swallows" the historical failures in a new theory which leaves no dangling anxieties.

Okay. I won't go quite that far. Still, AI has an embarrassing tendency to predict success where none materializes, to make mountains out of molehills, and to assert that some simpleminded pattern of suggestively-named °LISP tokens completely explains some incredibly high-level thought process. Why?

Consider the symbol your mind contains for 'light bulb'. In your mind, the sounds of the spoken words "light bulb" are reconstructed in your auditory cortex. A picture of a light bulb is loaded into your visual cortex. Furthermore, the auditory and visual cortices are far more complex, and intelligent, than the algorithm your computer uses to play sounds and MPEG files. Your auditory cortex has evolved specifically to process incoming speech sounds, with better fineness and resolution than it displays on other auditory tasks. Your visual cortex does not simply contain a 2D pixel array. The visual cortex has specialized processes that extract David Marr's "two-and-a-half dimensional world" - edge detection, corner interpretation, surfaces, shading, movement - and processes that extract from this a model of 3D objects in a 3D world. "About 50 percent of the cerebral cortex of primates is devoted exclusively to visual processing, and the estimated territory for humans is nearly comparable." (°MITECS, "Mid-Level Vision".)

In the *semantic net* or *Physical Symbol System* of classical AI, a light bulb would be represented by an atomic LISP token named light-bulb.

NOTE: I say "LISP tokens", not "LISP symbols", despite convention and accepted usage. Calling the lowest level of the system "symbols" is a

Some of the problem may be explained by history; back when AI was being invented, in the 1950s and 1960s, researchers had tiny little machines that modern pocket calculators would sneer at. These early researchers chose to believe they could succeed with "symbols" composed of small LISP structures, cognitive "processes" with the complexity of one subroutine in a modern class library. They were wrong, but the need to *believe* produced approaches and paradigms that sank AI for decades.

Previous AI has been conducted under the Physicist's Paradigm. The development of physics over the past few centuries - at least, the dramatic, stereotypical part - has been characterized by the discovery of simple equations that neatly account for complex phenomena. In physics, the task is finding a single bright idea that explains everything. Newton took a single assumption (masses attract each other with a force equal to the product of the masses divided by the square of the distance) and churned through some calculus to show that, if an apple falls towards the ground at a constant acceleration, then this explains why planets move in elliptical orbits. The search for a similar fits-on-a-T-Shirt unifying principle to fully explain a brain with hundreds of "cytoarchitecturally distinct areas has wreaked havoc on AI.

"Heuristics are compiled hindsight; they are judgemental rules which, if only we'd had them earlier, would have enabled us to reach our present state of achievement more rapidly." (Douglas Lenat, 1981.) The heuristic learned from past failures of AI might be titled "**Necessary, But Not Sufficient**". Whenever neural networks are mentioned in press releases, the blurb always includes the phrase "neural networks, which use the same parallel architecture found in the human brain". Of course, the "neurons" in neural networks are usually nothing remotely like biological neurons. But the main thing that gets overlooked is that it would be equally true (not very) to say that neural networks use the same parallel architecture found in an *earthworm's* brain. Regardless of whether neural networks are Necessary, they are certainly Not Sufficient. The human brain requires millions of years of evolution, thousands of modules, hundreds of thousands of adaptations, *on top of* the simple bright idea of "Hey, let's build a neural network!"

The Physicist's Paradigm lends itself easily to our need for drama. One great principle, one bold new idea, comes along to overthrow the false gods of the old religion... and set up a new bunch of false gods. As always when trying to prove a desired result from a flawed premise, the simplest path involves the Laws of Similarity and Contagion. For example, the "neurons" in neural networks involve associative links of activation. Therefore, the extremely subtle and high-level associative links of human concepts must be explained by this low-level property. Similarly, any instance of human deduction which can be written down (after the fact) as a syllogism must be explained by the blind operation of a ten-line-of-code process - even if the human thoughts blatantly involve a rich visualization of the subject matter, with the results yielded by direct examination of the visualization rather than formal deductive reasoning.

In AI, the *one great simple idea* usually operates on a low level, in accordance with the Physicist's Paradigm. Reasoning from similarity of surface properties is used to assert that high-level cognitive phenomena are explained by the low-level phenomenon, which (it is claimed) is both Necessary and Sufficient. This cognitive structure is a full-blown fallacy; it contains the social drama (one brilliant idea, new against old) and the rationalization (reasoning by similarity of surface properties, sympathetic magic) necessary to bear any amount of emotional weight. And that's how AI research goes wrong.

There are several ways to avoid making this class of mistake. One is to have the words "Necessary, But Not Sufficient" tattooed on your forehead. One is an intuition of causal analysis that says "This cause does not have sufficient complexity to explain this effect." One is to be instinctively wary of attempts to implement cognition on the token level. (One is learning enough evolutionary psychology to recognize and counter ideology-based thoughts directly, but that's moving off-topic...)

One is introspection. Human introspection currently has a bad reputation in cognitive science, looked on as untrustworthy, unscientific, and easy to abuse. This is totally true. Still, you can't build a mind without a working model. It is necessary to *know*, intuitively, that classical-AI propositional logic - syllogisms, property inheritance, et cetera - is inadequate to explain your deduction that dropping an anvil on a car will break it. You should be able to *see*, introspectively, that there's more than that going on. You can visualize an anvil smashing into your car's hood, the metal crumpling, and the windshield shattering. (9). Clearly visible is vastly more mental *material*, more cognitive "stuff", than classical-AI propositional logic involves.

The revolt against the Physicist's Paradigm can be formalized as the Law of Pragmatism:

The Law of Pragmatism

Any form of cognition which can be mathematically formalized, or which has a provably correct implementation, is too simple to contribute materially to intelligence.

The key words are "contribute materially". An architecture can be *necessary* to thought without accounting for the *substance* of thought. The Law of Pragmatism says that if a neural network's rules are simple enough to be formalized mathematically, then the substance of any intelligent answers produced by that network will be attributable to the specific pattern of weightings. If the pattern of weightings is created by a mathematically formalizable learning method, then the substance of intelligence will lie, not in the learning method, but in the intricate pattern of regularities within the training instances.

We can't be *certain* that the Law of Pragmatism will hold in the future, but it's definitely a heuristic in the Lenatian sense; if only we'd known it in the 1950s, so much error could have been avoided. The Law of Pragmatism is one of the tools used to determine whether an idea is Necessary, But Not Sufficient. (11).

°GISAI proposes a mind which contains modules vaguely analogous to human sensory modalities (auditory cortex, visual cortex, etc.). This does not mean that you can design any old system which can be described as "containing modular sensory modalities" and then dash off a press release about how your company is building an AI containing modular sensory modalities. That's the trophy mentality I was talking about earlier. A modular, modality-based system is Necessary, But Not Sufficient; it is also necessary to have the *right* modules, in the *right* sensory modalities, using the *right*

representation and the *right* intuitions to process the *right* base of experience to produce the *right* concepts that support the *right* thoughts within the *right* larger architecture.

When you think of a light bulb, the syllables and phonemes of "light bulb" are loaded into your auditory cortex; if you're a visual person, a generic picture of a light bulb - the default exemplar - appears in your visual cortex. Let's suppose that some AI has reasonably sophisticated analogues of the auditory cortex and visual cortex, capable of perceiving higher-level features as well as the raw binary data. This is clearly *necessary*; is it *sufficient* to understand light bulbs in the same way as a human?

No. Not even close. When you hear the phrase "triangular light bulb", you visualize a triangular light bulb.

NOTE: Please halt, close your eyes, and visualize a triangular light bulb. Please? Pretty please with sugar on top?

How do these two symbols *combine*? You know that light bulbs are fragile; you have a built-in comprehension of real-world physics - sometimes called "naive" physics - that enables you to understand fragility. You understand that the bulb and the filament are made of different materials; you can somehow attribute non-visual properties to pieces of the three-dimensional shape hanging in your visual cortex. If you try to design a triangular light bulb, you'll design a fluorescent triangular loop, or a pyramid-shaped incandescent bulb; in either case, unlike the default visualization of "triangle", the result will not have sharp edges. You know that sharp edges, on glass, will cut the hand that holds it.

Look at all that! It requires a temporal, four-dimensional understanding of the light bulb. It requires an appreciation, a set of intuitions, for cause and effect. It requires that you be capable of spotting a problem - a conflict with a goal - which requires means for representing conflicts, and cognitive reflexes derived from a goal system.

Look at *yourself* "looking at all that". It requires introspection, reflection, self-perception. It requires an entire self-sensory modality - representations, intuitions, cached reflexes, expectations - focused on the mind doing the thinking.

For you to read this paragraph, and think about it, requires a stream of consciousness. For you to think about light bulbs implies that you codified your past experiences of actual light bulbs into the representation used by your long-term memory. The visual image of the light bulb, appearing in your visual cortex, implies that a default exemplar for "light bulb" was abstracted from experience, stored under the symbol for "light bulb", and triggered by that symbol's auditory tag of 'light bulb'. And this exemplar can even be combined with the learned symbol for "triangle". You have formed an adjective, "triangular", consisting of characteristics which can be applied to modify the visual and design substance of the light-bulb concept. For you to visualize a light-bulb smashing, with an accompanying tinkling noise, requires synchronization of recollection and reconstruction across multiple sensory modalities.

I've mentioned many features in the last paragraphs; *none* of them are emergent. *None* of them will magically pop into existence on the high level "if only the simple low-level equation can be found". In a human, these features are °complex functional adaptations, generated by millions of years of evolution. For an AI, that means you *sit down and write the code*; that you change the design, or add design elements (special-purpose low-level code that directly implements a high-level case is usually a Bad Thing), specifically to yield the needed result.

In short, the design in GISAI is simply far *larger*, as a system architecture, than any design which has been previously attempted. It's large enough to resemble systems of the complexity described in the 471 articles in °The MIT Encyclopedia of the Cognitive Sciences. (12). You'll appreciate this better after reading the rest of the document, of course, but when you have done so, I expect that seed AI will look too *different* from past failures for one to reflect on the other. Fish and fowl, apples and oranges, elephants and typewriters. There is still the possibility that any given seed AI project will fail, or even that seed AI itself will fail - but if so, it will fail for different reasons.

2: Mind

- 2.1: World-model
- Interlude: The Consensus and the Veil of Maya
- 2.2: Sensory modalities
- 2.3: Concepts
 - 2.3.1: Modality-level, concept-level, thought-level
 - 2.3.2: Abstraction is information-loss; abstraction is not information-loss
 - 2.3.3: The concept of "three"
 - 2.3.4: Concept combination and application
 - 2.3.5: Thoughts are created by concept structures
- Interlude: Represent, Notice, Understand, Invent
- 2.4: Thoughts
 - 2.4.1: Building the world-model
 - 2.4.2: Creativity and invention
 - 2.4.3: Thoughts about thoughts
 - 2.4.4: The legitimate use of the word "I"

2.1: World-model

Intelligence is an evolutionary advantage because it enables us to model, predict, and manipulate reality. This includes not only Joe Caveman (or rather, Pat Hunter-Gatherer) inventing the bow and arrow, but Chris Tribal-Chief outwitting his (13) political rivals and Sandy Spear-Maker realizing that the reason her spears keep breaking is that she's being too impatient while making them. That is, the "reality" we model includes not just things, but other humans, and the self. (14).

A chain of reasoning is important because it ends with a conclusion about how the world works, or about how the world can be altered. The "world", for these purposes, includes the internal world of the AI; when designing a bicycle, the hypothesis "a round object can traverse ground without bumping" is a statement about the external world. The hypotheses "it'd be a good idea to think about round objects", or "the key problem is to figure out how to interface with the ground", or even "I feel like designing a bicycle", are statements about the internal world.

From an external perspective, cognitive events matter only insofar as they affect external behavior. Just so, from an internal perspective, the effect on the world-model is the punchline, the substance. This is not to say that every line of code must make a change to the world-model, or that the world-model is composed exclusively of high-level beliefs about the real world. The thought sequences that construct a what-if scenario - a "subjunctive fantasy world" - are altering a world-model, even if it's not *the* model of *the* world. A "vague feeling that there's some kind of as-yet unnamed similarity between two pictures" is part of the content of the AI's beliefs about the world. The code that produces that intuition may undergo many internal iterations, acting on data structures with no obvious correspondence to the world-model, before producing an understandable output.

What makes a pattern of bytes - or neurons - a "model"? And what makes a particular statement in that model "true" or "false"? (15). The best definition I've found is derived from looking at the *cause* of our intelligence: "Intelligence is an evolutionary advantage because it enables us to model, predict, and manipulate reality." Models are useful because they correspond to external reality.

I distinguish four levels of *binding*:

- A *sensory* binding occurs when there is a mapping between the model's data structures and characteristics of external reality.
- A *predictive* binding occurs when the model can be used to correctly predict future sensory inputs. (This presumes some kind of sensory device targeted on external reality.)
- A *decisive* binding occurs when the model can predict the effects of several possible *actions* on external reality, and choose whichever action gives the best result (according to some goal system). By modeling the future *given* each of several possible actions, it becomes possible to choose between futures - that is, between future sensory inputs. (If the model is sufficiently accurate.)
- A *manipulative* binding occurs when a future can be hypothesized, and a sequence of actions invented which results in that future. Given a desirable future - that is, a high-level property of the model which is defined by the goal system as an end in itself ("supergoal"), or which is a means to an end ("subgoal") - it is possible to invent the actions required to bring the model into correspondence with that future. If the model is correct, taking the specified external actions will actually result in the desired external reality.
 - *Qualitative* actions are selected from a finite set. If this set is small enough that all possible actions can be modeled - and *are* modeled - then there is no fundamental distinction between a "decisive binding", and a manipulative binding that uses qualitative actions.
 - *Quantitative* actions have one or more real (i.e., floating-point) parameters. Since this will usually make an exhaustive, "blind" search either theoretically or practically impossible - particularly if the fit must be exact - some conscious "heuristic, or a reversible feature of a sensory modality, must be used to derive the numerical action required from the numerical outcome specified. (Note that adding a continuous time parameter to a simple on-or-off qualitative action makes it a quantitative action.)
 - *Structural* actions have multiple elements (quantitative or qualitative), possibly with links or interactions (quantitative or qualitative). Emitting a string of characters - "foobar" - would be an example of a structural action. To deduce a required structural action without an exhaustive or impossible search requires either (A) a known rule linking actions and results, simple enough to be reversible, or (B) deliberate analysis of the simpler elements making up the structure.

These definitions raise an army of fundamental issues - "time, causality, "subjunctivity, goals, searching, invention - but first, let's look at a concrete example. Imagine a "microworld" composed of Newtonian billiard balls - a world of spheres (or circles), each with a position, radius, mass, and velocity, interacting on some frictionless surface (or moving in a two-dimensional vacuum). (16).

The "world-model" for an AI living in that "microworld" consists of everything the AI knows about that world - the positions, velocities, radii, and masses of the billiard balls. More abstract perceptions, such as "a group of "three billiard balls", are also part of the world-model. The prediction that "billiard ball A and billiard ball B will collide" is part of the world-model. If the AI imagines a situation where four billiard balls are arranged in a square, then that imaginary world has its own, "subjunctive world-model. If the AI believes "imagining four billiard balls in a square" will prove useful in solving problem X", then that belief is part of the world-model. In short, the world-model is not *necessarily* a programmatic concept - a unified set of data structures with a common format and "API. (Although it would be wonderfully convenient, if we could pull it off.) The "world-model" is a cognitive concept; it refers to the content of all beliefs, the substance of all mental imagery.

Returning to the billiard-ball world, what is necessary for an AI to have a "model" of this world?

- A *sensory* binding occurs when there is covariance between internal data structures of the AI and external properties of the billiard-ball world. For example, when the floating-point number representing the position of the billiard ball varies with the actual position of the billiard ball. We would also require that the same mapping - the same rules of interpretation - suffice to establish a binding between the modeled positions and actual positions of all the other billiard balls. (17).
- A *predictive* binding occurs when the model is accurate enough to predict the future positions of billiard balls. Assume a sensory device that reveals the positions of billiard balls to the AI, with a sensory binding (correspondence) between the data output by the sensory device and the actual positions. When the AI can establish a sensory binding (correspondence) between predicted data and actual data, a predictive binding has occurred.

- A *decisive* binding requires that some limited set of actions be available to the AI - for example, choosing whether to subtract some fixed increment of momentum each time a billiard ball bounces off a wall. (This action has been chosen so as to introduce no quantitative elements.) It requires a goal state, such as "three balls halted on the north side of the board". It requires that the AI be able to project the results of actions - to predict the world-state given the current world-model plus the fact of the action. It requires that the AI be able to recognize, internally, whether a given imagined result meets the criteria of the goal-state. Given these cognitive capabilities in a perfect world (18), a blind search through possible actions, combined with the programmatic rule "When an imagined situation meets the goal criteria, implement the action-list leading to that situation", would create the "atomic" case of decisive binding. (Of course, in accordance with the Law of Pragmatism, simplifying the design down to the level where it's easy to visualize the code has stripped it of all useful intelligence. Real minds are vastly more complex.)
- A *manipulative* binding would occur if, for example, the AI could control a cue ball, and knew how to use this cue ball to "create two symmetrical groups of three billiard balls". In this particular example, a structural result (two groups of three) is obtained through a series of quantitative actions (forces applied to the cue ball at particular times).

In the last case, the AI may have been able to manipulate each of the six billiard balls as a separate object, or each action may have affected multiple balls simultaneously, requiring a more complex planning process. The important thing is that "creating two symmetrical groups of three billiard balls" is not something that would happen by chance, or be uncovered by a blind search. For the AI to create a structure of billiard balls, it will need 'heuristics - knowledge about rules - that not only link outcomes to actions, but reverse the process to link actions to outcomes.

Suppose that a cue ball travelling south at 4 meters/second, bumping into a billiard ball travelling south at 2 meters/second, results in the cue ball and the billiard ball travelling south at 3 meters/second. Suppose, furthermore, that these rules are contained within the AI's internal model of the environment, so that if the AI visualizes a cue ball at {8.2, 6} of radius 1 travelling south at 4 m/s, and a ball at {8.2, 10} of radius 1 going south at 2 m/s, the AI will visualize the balls bumping one second later at {8.2, 11}, and the two balls then travelling south at 3 m/s. It's a long way from there to knowing - consciously, 'declaratively - that two balls *in general* bumping at 4 m/s and 2 m/s while going in the same direction will travel on together at 3 m/s. It's an even longer way to knowing that "if billiard ball X bumps into billiard ball Y, then they will continue on together with the average of their velocities". And it's a still longer way to *reversing* the rule and knowing that "to get a group of two balls travelling together with velocity X, given billiard ball A with velocity Y, bump it with billiard ball B having velocity (2X - Y)". Finally, to close the loop, this last high-level rule must be *applied* to create a particular hypothesized action in the world-model, and the hypothesized action needs to be taken as a real action in external reality.

Without jumping too far ahead, there are a number of properties that a world-model needs to support high-level thought. It needs to support 'time - multiple frames or a temporal visualization - with accompanying extraction of temporal features. It needs to support predictions and expectations (and an expectation isn't real unless the AI notices when the expectation is fulfilled, and especially when it is violated). The world-model needs to support hypotheses, 'subjunctive frames of visualization, which are distinct from "real reality" and can be manipulated freely by high-level thought. (By "freely manipulated", I mean a *direct* manipulative binding; choosing to think about a billiard ball at position {2, 3} should cause a billiard ball to materialize directly within the representation at {2, 3}, with no careful sequence of actions required.) And for the visualization to be useful once it exists, the high-level thought which created the billiard-ball image must *refer* to the particular image visualized... and the reference must run both ways, a two-way linkage.

'Time, expectation, comparison, 'subjunctivity, visualization, introspection, and reference. I haven't defined any of these terms yet. (Most are discussed in 3: Cognition, although you can jump ahead to Appendix A: Glossary if you're impatient.) Nonetheless, these are some of the basic attributes that are present in human world-models, and which are Necessary (But Not Sufficient) for the existence of high-level features such as causality, intentionality, 'goals, memory, learning, association, focus, abstraction, categorization, and symbolization.

NOTE: I mention that list of features to illustrate what will probably be one of the major headaches for AI designers: If you design a system and forget to allow for the possibility of expectation, comparison, 'subjunctivity, visualization, or whatever, then you'll either have to go back and redesign every single component to open up space for the new possibilities, or start all over from scratch. *Actualities* can always be written in later, but the *potential* has to be there from the beginning, and that means a designer who knows the requirements spec in advance.

Interlude: The Consensus and the Veil of Maya

In a rainbow, the physical frequency of the light changes smoothly and linearly with distance (19). Yet, when you look at a rainbow, you see colors grouped into bands, with relatively sharp borders. And it's not just you. Everyone sees the bands.

It gets worse. Consider: The frequency of light is a linear, scalar, real number. The visible frequencies of light rise linearly from red to blue, bounded by infrared and ultraviolet. But if you look at a color wheel on your computer, you'll see that it's a *wheel*. Red to orange to yellow to green to blue to... purple? ... and back to red again. *Where does purple come from?* It's a color that doesn't exist, seemingly added on afterwards to turn a linear spectrum into a circle!

It turns out the color purple and the bands in a rainbow are both artifacts of the way humans perceive color space, which in turn is a result of the way our visual cortex has evolved to distinguish objects in the ancestral environment and maintain color constancy under natural lighting. (For more about this, see "The Perceptual Organization of Colors" in "The Adapted Mind". It's definitely a cool article.)

The color purple, and the bands in the rainbow, aren't *real*. But everyone sees them, so you can't just call them hallucinations. I prefer to strike a happy compromise and say that purple and rainbows exist in the *Consensus*. Nobody actually lives in external reality, and we couldn't understand it if we did; too many quarks flying around. When we walk through a hall, watching the floor and walls and ceiling moving around us, we're actually walking through our visual cortex. That's what we see, after all. We don't see the photons reflected by the walls, and we certainly don't see the walls themselves; every single detail of our perception is there because a neuron is firing somewhere in the visual system. If the wrong neuron fired, we'd see a spot of color that wasn't there; if a neuron failed to fire, we wouldn't see a spot of color that was there. From this perspective, the

actual photons are almost irrelevant. Furthermore, all the colors in the hall you're walking through are technically incorrect due to that old color-space thing. Heck, you might even walk past something purple.

This is the point where the philosopher usually goes off the solipsistic deep end. "It's all arbitrary! Nothing is real! Everything is true! I can say whatever I want and nobody can do a thing about it, bwahaha!" I *hate* this whole line of thinking. If I ever start sounding like this, check my forehead for lobotomy scars.

The Consensus usually has an extremely tight °sensory, °predictive, and °manipulative binding to external reality. No, it doesn't work 100% of the time, but it works 99.99% of the time, so the rules are just as strict. Just because you can't see external reality directly doesn't mean it isn't there.

Everything you see is illusion, the Veil of Maya. Where Eastern philosophy goes wrong is in assuming that the Veil of Maya is hiding something big and important. What lies behind the illusion of a brick is the actual brick. The vast majority of the time, you can forget the Veil of Maya is even there.

Nor does our residence in the Consensus grant the Consensus primacy over external reality. The Consensus itself is just another part of reality. That's how reality binds the Consensus; it's just one part of reality affecting another part, under the standard rules of interaction imposed by the laws of physics. External reality existed before the patterns in reality known as "humans" or "the Consensus". People who ignore external reality on the grounds that "all truth is subjective" tend to have their constituent quarks assimilated by the quark-patterns we call "tigers".

However, sometimes it's important to remember that tigers only exist in the Consensus. Suppose someone asks you for a definition of a "tiger", and you give them a definition that works 99.99% of the time - "big orange cat thingy with stripes". Then whoever it is paints a tiger green and says, "Ha, ha! Your definition is wrong!" What I would do in this case is give a more precise definition based on genetics, behavior patterns, and so on, but then you have cyborg tigers and mutant tigers. At that point, it becomes important to remember that it's "just" the Consensus. *You shouldn't expect things in the Consensus to have perfect mathematical definitions.* Evolution doesn't select for tigers, or tiger-perceiving minds, that have philosophically elegant definitions; evolution selects whatever works most of the time.

So why does the Consensus work? Because of a fundamental rule of °reducthologism: *Forget about definitions.* Anything true "by definition" is a tautology, and bears no relation to external reality - does not even *refer* to external reality.

Forget about definitions, and if you find that some cognitive perception is inherently °subjective or °observer-dependent - that the perception relies on qualities that exist only in the mind of the observer - then relax and accept it as being useful to intelligence most of the time, and don't go into philosophical fits. It isn't *real*, after all, so why should you worry?

Hey, that's life in the Consensus.

DEFN:	Consensus: The Consensus is the world of shared perceptions that humanity inhabits. Things in the Consensus aren't <i>really really</i> real, but they usually correspond tightly to reality - enough to make the rules about what you can and can't say just as strict. What distinguishes the Consensus from actual reality is that there is no <i>a priori</i> reason why things should be formalizable, philosophically coherent, or unambiguous.
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2.2: Sensory modalities

A human has a visual cortex, an auditory cortex, a sensorimotor cortex - areas of the brain specifically devoted to particular senses. Each such "cortex" is composed of neural modules which extract important mid-level and high-level features from the low-level data, in a way determined by the "laws of physics" of that domain. The visual cortex and associated areas (20) are by far the best-understood parts of the brain, so that's what we'll use for an example.

Visual information starts out as light hitting the retina; the resulting information can be thought of as being analogous to a two-dimensional array of pixels (although the neural "pixels" aren't rectangular). "Low-level" feature extraction starts right in the retina, with neurons that respond to edges, intensity changes, light spots, dark spots, et cetera. From this new representation - the 2D pixels, plus features like edges, light spots, and so on - the lateral geniculate nucleus and striate cortex extract mid-level features such as edge orientation, movement, direction of moving features, textures, the curvature of textured surfaces, shading, and binocular perception. This information yields °David Marr's two-and-a-half-dimensional world, which is composed of scattered facts about the three-dimensional properties of two-dimensional features - this is a continuous surface, this surface is curving away and to the left, these two surfaces meet to form an edge, these three edges meet to form a corner.

Finally, a 3D representation of moving objects is constructed from the 2.5D world. Constraint propagation: If the 3D interpretation of one corner requires an edge to be convex, then that edge cannot be concave in another corner. Object assembly: Multiple surfaces that move at the same speed, or that move in a fashion consistent with rotation, are part of a single object. Consistency: An object (or an edge, or a surface) cannot simultaneously be moving in two directions.

The resulting 3D representation, still bound to the 2.5D features and the 2D pixels, is sent to the temporal cortex for object recognition and to the parietal cortex for spatial visualization.

The visual cortex is the foundation of one of the seven senses. (Yes, at least seven. In addition to sight, sound, taste, smell, and touch, there's proprioception (the nerves that tell us where our arms and legs are) and the vestibular sense (the inner ear's inertial motion-detectors). (21).) The neural areas that are devoted *solely* to processing one sense or another account for a huge chunk of the human cortex. In the modular partitioning of the human brain, the single most common type of module is a sensory modality, or a piece of one. This demonstrates a fundamental lesson about minds in general.

Classical AI programs, particularly "expert systems", are often partitioned into microtheories. A microtheory is a body of knowledge, i.e. a big semantic net, e.g. propositional logic, a.k.a. suggestively named LISP tokens. A typical microtheory subject is a human specialty, such as "cars" or "childhood diseases" or "oil refineries". The content of knowledge typically consists of what would, in a human, be very high-level, heuristic statements: "A child that is sick on Saturday is more likely to be seriously ill than a child who's sick on a schoolday."

How do the microtheory-based modules of classical AI differ from the *sensory* modules that are common in the human mind? How does a "microtheory of vision" differ from a "visual cortex"? Why did the microtheory approach fail?

There are two fundamental clues that, in retrospect, should have alerted expert-system theorists ("knowledge engineers") that something was wrong. First, microtheories attempt to embody high-level rules of reasoning - heuristics that require a lot of pre-existing *content* in the world-model. The visual cortex doesn't know about butterflies; it knows about edge-detection. The visual cortex doesn't contain a preprogrammed picture of a butterfly; it contains the feature-extractors that let you look at a butterfly, parse it as a distinct object standing out against the background, remember that object apart from the background, and reconstruct a picture of that object from memory. We are not born with experience of butterflies; we are born with the visual cortex that gives us the capability to experience and remember butterflies. The visual cortex is not visual *knowledge*; it is the space in which visual knowledge exists.

The second, deeper problem follows from the first. All of an expert system's microtheories have the same underlying data structures (in this case, propositional logic), acted on by the same underlying procedures (in this case, a few rules of "Bayesian reasoning"). Why separate something into distinct *modules* if they all use the same data structures and the same functions? Shouldn't a real program have more than one real module?

I'm not suggesting that data formats and modules be proliferated because this will magically make the program work better. Any competent programmer knows not to use two data formats where one will do. But if the data and processes aren't complex enough to seize the programmer by the throat and *force* a modular architecture, then the program is too simple to give rise to real intelligence.

Besides, a single-module architecture *certainly* isn't the way the brain does it. Maybe there's some ingenious way to represent auditory and visual information using a single underlying data structure. If we can get away with it, great. But if no act of genius is required to solve the very deep problem of getting domain-specific representations to interact usefully, if the problem is "solved" because all the content of thought takes the form of propositional logic, if all the *behaviors* can fit comfortably into a single programmatic module - then the program doesn't have enough complexity to be a decent video game, much less an AI. (22).

We shouldn't be too harsh on the classical-AI researchers. Building an AI that operates on "pure logic" - no sensory modalities, no equivalent to the visual cortex - was worth trying. As Ed Regis would say, it had a certain hubristic appeal. Why does human thought use the visual cortex? Because it's there! After all, if you've already evolved a visual cortex, further adaptations will naturally take advantage of it. It doesn't mean that an engineer, working *ab initio*, must be bound by the human way of doing things.

But it didn't work. The recipe for intelligence presented by GISAI assumes an AI that possesses equivalents to the visual cortex, auditory cortex, and so on. Not necessarily these *particular* cortices; after all, Helen Keller (who was blind and deaf, and spoke in hand signs) learned to think intelligently. But even Helen Keller had proprioception, and thus a parietal lobe for spatial orientations; she had a sense of touch, which she could use to "listen" to sign language; she could use the sensory modalities she had to perceive signed symbols, and form symbols internally, and string those symbols together to form sentences, and think. (23) *Some* equivalent of *some* type of "cortex" is necessary to the GISAI design.

"Cortex" is a specifically neurological term referring to the surface area of the brain, and therefore I will use the term "sensory modality", or "modality", instead of cortex.

DEFN: **Modality:** Modalities in an AI are analogous to human cortices - visual cortex, auditory cortex, *et cetera* - enabling the AI to visualize processes in the target domain. Modalities capture, not high-level *knowledge*, but low-level *behaviors*. A modality has data structures suited to representing the target domain, and "codelets" or processing stages which extract higher-level features from raw data.

Why does an AI need a visual modality? Because the human visual cortex and associated neuroanatomy - our visual modality - is what makes our thoughts of 2D and 3D objects *real*. Drew McDermott, in *Artificial Intelligence Meets Natural Stupidity*, pointed out that, just because a LISP token is labeled with the character string "hamburger", it does not mean that the program understands hamburgers. The program has not even noticed hamburgers. If the symbol were called G0025 instead of hamburger, nobody would ever be able to figure out that the token was supposed to represent a hamburger.

When two objects collide, we don't just have a bit of propositional logic that says *collide(car, truck)*; we imagine two moving objects. We model 2D pixels and 3D features and visualize the objects crashing together. The edges touch, not as *touch(edge-of(car), edge-of(truck))*, but as two curves meeting and deforming at all the individual points along the edge. You could successfully look at a human brain and deduce that the neurons in question were modelling edges and colliding objects; this is, in fact, what visual neuroanatomists *do*. But if you did the same to a classical AI, if you stripped away the handy English variable names from the propositional logic, you'd be left with G0025(Q0423, U0111) and H0096(D0103(Q0423), D0103(U0111)). No amount of reasoning could bind those cryptic numbers to real-world cars or trucks.

Furthermore, our visual cortex is useful for more than vision. Philosophy in the Flesh (George Lakoff and Mark Johnson) talks about the Source-Path-Goal pattern (24) - a trajectory that moves, a starting point, a goal, a route; the position of the trajectory at a given time, the direction at that time, the actual final destination... Philosophy in the Flesh also talks about "internal spatial 'logic' and built-in inferences": If you traverse a route, you have been at all locations along the route; if you travel from A to B and B to C, you have traveled from A to C; if X and Y are traveling along a direct route from A to B and X passes Y, then X is further from A and closer to B than Y is.

These are all *behaviors* of spatial reality. Classical AI would attempt to capture *descriptions* of this behavior; i.e. "if *travel(X, A, B)* and *travel(X, B, C)* then *travel(X, A, C)*". The problem is that the low-level elements (pixels, trajectories, velocities) making up the model can yield a nearly infinite

number of high-level behaviors, all of which - under the classical-AI method - must be described independently. If A is-contained-in B, it can't get out - unless B has-a-hole. Unless A is-larger-than the hole. Unless A can-turn-on-its-side or the hole is-flexible. Trying to describe all the possible behaviors exhibited by the high-level characteristics, without directly simulating the underlying reality, is like trying to design a CPU that multiplies two 32-bit numbers using a doubly-indexed lookup table with 2^{64} (around eighteen billion billion) entries.

Real CPUs take advantage of the fact that 32-bit numbers are made of *bits*. This enables transistors to multiply using the wedding-cake method (or whatever it is modern CPU designs use). A 32-bit number is not a monolithic object. The numerical interpretation of 32 binary digits is not intrinsic, but rather a high-level characteristic, an observation, an abstraction. The individual bits interact, and yield a 32-bit (or 64-bit) result which can then be interpreted as the resulting number. The computer can multiply 9825 by 767 and get 7535775, not because someone *told* it that 9825 times 767 is 7535775, but because someone told it about how to multiply the individual bits.

A visual modality grants the power to observe, predict, decide, and manipulate objects moving in trajectories, not because the modality captures *knowledge* of high-level characteristics, but because the modality has elements which *behave* in the same way as the external reality. An AI with a visual modality has the potential to understand the concept of "closer", not because it has vast stores of propositional logic about closer(A, B), but because the model of A and B is composed of *actual pixels* which are *actually getting closer*. (25).

Source-Path-Goal is not just a *visual* pattern. It is a *metaphor* that applies to almost any effort. *Force* and *resistance* aren't just people pushing carts, they're companies pushing products. Source-Path-Goal applies not just to walking to Manhattan, but a programmer struggling to write an application that conforms to the requirements spec. It applies to the progress of these very words, moving across the screen as I type them, decreasing the distance to the goal of a publishable Web page. Furthermore, the visual metaphor is in many cases a *useful* metaphor, one which binds predictively. (26). A metaphor is useful when it involves, not just a similarity of high-level characteristics, but a similarity of low-level elements, or a single underlying cause. (See previous footnote.) The visual metaphor that maps the behavior of a programming task to the Source-Path-Goal pattern (a visual object moving along a visual line) is useful if some measure of "task completed" can be mapped to the quantitative position of the trajectory, and the perceived velocity used to (correctly!) predict the amount of time remaining on the task.

Of course, one must realize that having a visual modality is Necessary, But Not Sufficient, to pulling that kind of stunt. In such cases, *noticing* the analogy is ninety percent of the creativity. The atomic case of such noticing would consist of generating models at random, either by generating random data sets or by randomly mixing previously acquired models, until some covariance, some similarity, is noticed between the model and the reality. And then the AI says "Eureka!"

Of course, except for very simple metaphors, the search space is too large for blind constructs to ever match up with reality. It is more often necessary to deliberately construct a model - in this case, a visual model - whose behaviors correspond to reality. Discussion of such higher-level reasoning doesn't belong in the section on "sensory modalities", but being able to "deliberately construct" *anything* requires a way to manipulate the visual model. In addition to the hardware/code for taking the *external* action of "draw a square on the sheet of paper", a mind requires the hardware/code to take the *internal* action of "imagine a square". The consequence, in terms of how sensory modalities are programmed, is that feature extraction needs to be *reversible*. Not *all* of the features *all* of the time, of course, but for the cognitive act of *visualization* to be possible, there must be a mechanism whereby the perception that detects the "line" feature has an inverse function that constructs a line, or transforms something else into a line.

Feature reconstruction is much more difficult to program than feature extraction. More computationally intensive, too. It's the difference between multiplying the low-level elements of "7" and "17", and reconstructing two low-level elements which could have yielded the high-level feature of "119". This may be one of the reasons why thalamocortical sensory pathways are always reciprocated by corticothalamic projections of equal or greater size; for example, a cat has 10^6 neural fibers leading from the lateral geniculate nucleus to the visual cortex, but 10^7 fibers going in the reverse direction. (27).

Even a complete sensory modality, capable of perception and visualization, is useless without the rest of the AI. "Necessary, But Not Sufficient," the phrase goes. A modality provides some of the raw material that concepts are made of - the space in which visualizations exist, but nothing more. But, granting that the rest of the AI has been done properly, a visual modality will create the potential to understand the concept of "closer"; to use the concept of "closer", and heuristics derived from examining instances of the concept "closer", as a useful visual metaphor for other tasks; and to use deliberately constructed models, existing in the visual modality, to ground thinking about generic processes and interactions. (In other words, when considering a "fork" in chess or an "if" statement in code, it can be visualized as an object with a Y-shaped trajectory.)

Is a complete visual modality - pixels, edge detectors, surface-texture decoders, and all - really necessary to engage in spatial reasoning? Would a world of Newtonian billiard balls, with velocities and collision-detection, do as well? It would apparently suffice to represent concepts such as "fork", "if statement", "source-path-goal", "closer", and to create metaphors for most generic systems composed of discrete objects. The billiard-ball world has significantly less representative power; it's harder to understand a "curved trajectory" in spacetime if you can't visualize a curve in space. (28). But, considering the sheer programmatic difficulty of coding a visual modality, are metaphors with billiard balls composed of pixels *that* superior to metaphors with billiard balls implemented directly as low-level elements?

Well, yes. In a visual modality, you can switch from round billiard balls to square billiard balls, visualize them deforming as they touch, and otherwise "think outside the box". The potential for thinking outside the box, in this case, exists because the system being modeled has elements that are represented by high-level visual objects; these high-level visual objects in turn are composed of mid-level visual features which are composed of low-level visual elements. This provides wiggle room for creativity.

Consider the famous puzzle with nine dots arranged in a square, where you're supposed to draw four straight lines, without lifting pen from paper, to connect the dots. (29). To solve the puzzle one must "think outside the box" - that is, draw lines which extend beyond the confines of the square. A conventional computer program written to solve this problem would probably contain the "box" as an assumption built into the code, which is why computers have a reputation for lack of creativity. (30). A billiard-ball metaphor, even assuming that it could represent lines, might run into the same problem.

I suspect that many solvers of the nine-dot problem reach their insight because a particular configuration of tried-out lines suggests an incomplete triangle whose corners lie outside the box. "Seeing" an "incomplete triangle" is an *optical illusion*, which is to say that it's the result of high-level features being triggered and suggesting mid-level features - in this case, some extra lines that turn out to be the solution to the problem. Sure, you can make up ways that this could happen in a billiards modality, but then the billiards modality starts looking like a visual cortex. The point is that, for our particular human style of creativity, it is Necessary (But Not Sufficient) to have a modality with rich "extraneous" perceptions, and where high-level objects in the metaphor can be made to do unconventional things by mentally manipulating the low-level elements. (Even so, it would make development sense to start out with a billiards modality and work up to vision gradually.)

There are two final reasons for giving a seed AI sensory modalities: First, the possession of a codic modality may improve the AI's understanding of source code, at least until the AI is smart enough to make its own decisions about the balance between slow-conscious and fast-autonomic thought. Second, as will be discussed later, thoughts don't start out as abstract; they reach what we would consider the "abstract" level by climbing a layer cake of ideas. That layer cake starts with the non-abstract, autonomic intuitions and perceptions of the world described by modalities. The concrete world provided by modalities is what enables the AI to learn its way up to tackling abstract problems.

NOTE: One of the greatest advantages of seed AI - second only to recursive self-improvement - is going beyond the human sensory modalities. It's possible to create a sensory modality for source code. The converse is also true: Various processes that are autonomic in humans - memory storage, symbol formation - can become sensory modalities subject to deliberate manipulation.

In programmatic terms, *any program module with a coherent set of data structures and an API*, which could benefit from higher-level thinking, is a candidate for transformation into a modality with world-model-capable representations, feature extraction, reversible features to allow mental actions, and the other design characteristics required to support concept formation.

2.3: Concepts

2.3.1: Modality-level, concept-level, thought-level

Modalities in the human brain are mostly preprogrammed, as opposed to learned. (Human modalities require external stimuli to grow into their preprogrammed organization, but this is not the same as learning.) Individual neural signals can have meanings that are visible and understandable to an eavesdropper. Programmers may legitimately take the risk of creating modalities through deliberate programming, with low-level elements that correspond to data structures, and human-written procedures for feature extraction.

Within °GISAI, the term *concept* is used to refer to the kind of mental stuff that exists as a pattern in the modality. A learned sequence of instructions that reconstructs a generic, abstracted "light bulb" in the visual modality is a *concept*. Symbols, categories, and some memories are concepts. (Despite common usage, "concept" might technically refer to non-declarative mental stuff such as a human cognitive reflex or a human motor skill. However, in a seed AI, where everything is open to introspection, it makes sense to call the equivalents of human reflexes or skills "concepts".) Concepts are patterns, learned or preprogrammed, that exist in long-term storage and can be retrieved.

A structure of concepts creates a *thought*. The archetypal example, in humans, is words coming together to form sentences. Thoughts are visualized; they operate within the RAM of the mind, the "workspace" represented by available content capacity in the sensory modalities, commonly called "short-term memory" or "working memory". (The capacity of working memory in AIs is not determined by available RAM, but by available CPU capacity to perform feature extraction on the contents of memory. If you have the data structures without the feature extraction, the AI won't notice the information.) Thoughts manipulate the world-model.

In humans, at least, it's hard to draw clean boundaries between thoughts and concepts. (31). The experience of hearing the word for a single concept, such as "triangle", is not necessarily a mere concept; it may be more valid to view it as a thought composed of the single concept "triangle". And, although some concepts are formed by categorizing directly from sense perception, more abstract concepts such as "three" probably occur first as deliberate thoughts. We'll be discussing both types in this section.

2.3.2: Abstraction is information-loss; abstraction is not information-loss

In chemistry, *abstract* means *remove*; to "abstract" an atom from a molecule means to take it away. Use of the term "abstract" to describe the process of forming concepts implies two assumptions: First, to create a concept is to *generalize*; second, to generalize is to *lose information*. It implies that, to form the concept of "red", it is necessary to ignore other high-level features such as shape and size, and focus only on color.

This is the classical-AI view of abstraction, and we should therefore be suspicious of it. On the other hand, our mechanisms for abstraction *can* learn the concept for "red". In a being with a visual modality, this concept would consist of a piece of mindstuff that had learned to distinguish between red objects and non-red objects. Since redness is detected directly as a low-level feature, it shouldn't be very hard to train a piece of mindstuff to thus distinguish - whether the mindstuff is made of trainable neurons, evolving code, or whatever. A neural net needs to learn to fire when the "red" feature is present, and not otherwise; a piece of code only needs to evolve to test for the presence of the redness feature. At most, "red" might also require testing for solid-color or same-hue groupings. Given a visual modality, the concept of "red" lies very close to the surface.

Of course, to have a real concept for "red", it's not enough to distinguish between red and non-red. The concept has to be *applicable*; you have to be able to apply it to visualizations, as in "red dog". You also need a default exemplar (32) for "red"; and an extreme exemplar for "red"; and memories of experiences that are stereotypically red, such as stoplights and blood. (For all we know, leaving out any one of these would be enough to totally hose the flow of cognition.) Again, these features lie close to the surface of a visual modality. "Red" would be one of the easiest features to make reversible, with little additional computational cost involved; just set the hue of all colors to a red value. (Although hopefully in such a way as to preserve all detected edges, contrasts, and so on. Making everything *exactly* the same color would destroy non-color features.) The default exemplar for red can be a red blob, or a red light; the extreme exemplar for red may be the same as the default exemplar, or it may be a more

intensely red blob. And the stereotypically red objects, such as stoplights and blood, are the objects in which the redness is important, and much remarked upon.

(33).

For the moment, however, let's concentrate on the problem of forming categories. The conventional wisdom states that categorization consists of generalization, and that generalization consists of focusing on particular features at the expense of others.

We'll use the microdomain of letter-strings as an example. To generalize from the instances {"aaa", "bbb", "ccc"} to form the category "strings-of-three-equal-letters", the information about *which letter* must be abstracted, or lost, from the model. Actually, this misstates the problem. If you lose that information on a letter-by-letter basis, then "aaa" and "aab" both look like "****". What's needed is for the letter-string modality to *first* extract the features of "group-of-equal-letters", "number=3", and "letter=b", after which the concept can *lose* the last feature or *focus* on the first two. If the second feature, "number", is also lost, then the result is an even more general concept, "strings-of-equal-letters". Of course, this concept is precisely identical to the modality's built-in feature-detector for "group-of-equal-letters", which again points up that only very simple conceptual categories, lying very close to the surface of the modality's preprogrammed assumptions about which features are important, can be implemented by direct information-loss.

To examine a more complex concept, we'll look at the example of "three".

2.3.3: The concept of "three"

To a twenty-first-century human, trained in arithmetic and mathematics, the concept of "three" has enormous richness. It must therefore be emphasized that we are dealing solely with the concept of "three", and that a mind can understand "three" without understanding "two" or "four" or "number" or "addition" or "multiplication". A mind may have the concept "three" and the concept "two" without noticing any similarity between them, much less having the *aha!* that these concepts should go together under the heading "number". If a mind somehow manages to pick up the categories of groups-of-three-dogs and groups-of-three-cats, it doesn't follow that the mind will generalize to the category of "three".

To think about infant-level or child-level AIs, or for that matter to teach human children, it's necessary to slow down and forget about what seems "natural". It's necessary to make a conscious separation between ideas - ideas that, to humans, seem so close together that it takes a deliberate effort to see the distance.

Just because the AI exists on a machine performing billions of arithmetical operations per second doesn't mean that the AI itself must understand arithmetic or "three". (John Searle, take note!) Even if the AI has a codic modality which grants it direct access to numerical operations, it doesn't necessarily understand "three". If every modality were programmed with feature-extractors that counted up the number of objects in every grouping, and output the result as (say) the tag "number: three", the AI might *still* fail to really understand "three", since such an AI would be unable to count objects that weren't represented directly in some modality. An AI that *learns* the concept of "three" is more likely to notice not just three apples but that 've (the AI) is currently thinking three thoughts. A preprogrammed concept only notices what the programmer was thinking about when he or she wrote the program.

What is "three", then? How would the concept of "three" be learned by an AI whose modalities made no direct reference to numbers - whose modalities, in fact, were designed by a programmer who wasn't thinking about numbers at the time? How can such a simple concept be decomposed into something even simpler?

There's an AI called "Copycat", written by Melanie Mitchell and conceived by Douglas R. Hofstadter, that tries to solve analogy problems in the microdomain of letter-strings. If you tell Copycat: "'abc' goes to 'abd'; what does 'bcd' go to?", it will answer "'bce'". It can handle much harder problems, too. (See "Copycat" in the glossary.) Copycat is a really fascinating AI, and you can read about it in *Metamagical Themas*, or read the source code (it's a good read, and available as plain text online - no decompression required). If you do look at the source code, or even just browse the list of filenames, you'll see the names of some very fundamental cognitive entities. There are "bonds", "groups", and "correspondences". There are "descriptors" (and "distinguishing descriptors") and "mappings", and all sorts of interesting things.

Without going too far into the details of Copycat, I believe that some of the mental objects in Copycat are primitive enough to lie very close to the foundations of cognition. Copycat measures numbers directly (although it can only count up to five), but that's not the feature we're interested in. Copycat was designed to understand relations and invent analogies. It can notice when two letters occupy "the same position" in a letter-string, and can also notice when two letters occupy "the same role" in a higher-order mental construct. It can notice that "c" in "abc" and "d" in "abd" and "d" in "bcd" all occupy the same position. It can understand the concept of "the same role", if faced by an analogy problem which forces it to do so. For example: If "abc" goes to "abd", what does "pqrs" go to? Copycat sees that "c" and "s" occupy the same role, even though they no longer occupy the same numerical position in the string, and so replies "pqrt".

Correspondences and *roles* and *mappings* are probably autonomically-detected features on the modality-level (as well as being very advanced concepts in cognitive science). Intuitive, directly perceived correspondences allow two images in the same modality to be compared, and that is a basic part of what makes a modality go.

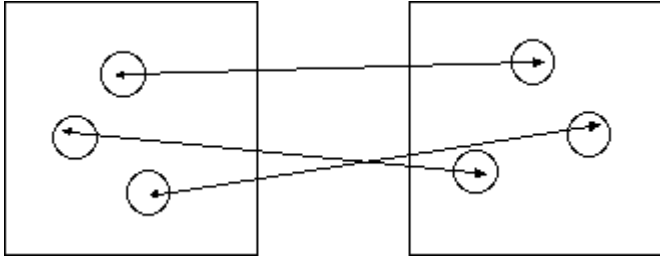
These intuitions obey certain underlying cognitive pressures (also modeled by the Copycat project): If two high-level structures are equal, then the low-level structures should be mapped to each other. *Symmetry*, which - very loosely defined - is the idea that each of these low-level mappings should be the same. If one is reflected, they should all be reflected, and so on. *Completeness*: You shouldn't map five elements to each other but leave the sixth elements dangling.

Copycat shows an example of how to implement this class of cognitive intuitions using conflict-detectors, equality-detectors, and a feature called a "computational temperature". Roughly speaking, conflicts raise the temperature and good structures lower the temperature. The higher the temperature, the more easily cognitive perceptions break - the more easily groups and bonds and mappings dissolve. Lower temperatures indicate

better answers, and thus answers are more persistent - perceived pieces of the answer in the cognitive workspace are harder to break. Copycat's intuitions may not have the same flexibility or insight as a human *consciously* trying to solve a "symmetry problem" or a "completeness problem", but they *do* arguably match a human's unconscious intuitions about analogy problems. Each low-level built-in cognitive ability has its analogue as a high-level thought-based skill, and it is dangerous to confuse the standards to which the two are held.

We now return to the concept of "three". We'll suppose for the moment that we're operating in a Newtonian billiard-ball modality, and that we want the AI to learn to recognize three billiard balls.

The first concept learned for "three" might look like this:



The mental image on the left is an "exemplar" (or "prototype"), attached to the *three* concept and stored in memory. The mental image on the right is the target, containing the objects actually being counted. The concept of "three" is satisfied when correspondences can be drawn between each object in the three-exemplar and each object in the target image. If the target image contains two objects, a dangling object will be detected in the three-exemplar image, and the concept will not be satisfied. If the target image contains four objects, then a dangling object will be detected in the target image. (34).

This isn't a full answer to the "problem of three", of course. A full answer would also consider the question of how to computationally implement a "unique correspondence" in a non-fragile way; how to distinguish each object from the background; how to *apply* the three-concept to a mental image formerly containing two or four objects to yield a new mental image containing three objects; how to retrieve the exemplar from memory; how to extend the intuition of "unique correspondence" across modalities. And the type of mindstuff needed to implement these instructions in a non-fragile way; and how the exemplar and concept were created or learned in the first place.

In fact, the problem of three is so complicated that it would probably be first solved by conscious thought, and compiled into a concept afterwards. This adds the problem of figuring out how the thoughts got started; what types of task would force a mind to notice "three" and evolve a definition like that above; and how the skill gets compiled into a pattern. Also, an understanding of *three* that generalizes from the concept "three billiard balls" to the concept "three groups of three billiard balls" means asking what kind of problem would force the generalization. It means asking how the generalization would take place inside the thought-based skill or mindstuff-based concept; how the need to generalize would translate into a cognitive pressure, and how that pressure would apply to a piece of the mindstuff-code, and how that piece would correctly shift under pressure. And then there are questions about moving towards the adult-human understanding of "three", such as noticing that it doesn't matter which *particular* billiard ball A corresponds to which billiard ball B.

However, the diagram above does constitute a major leap forward in solving the problem. It is a functional decomposition of three, one that invokes more basic forces such as unique correspondence and exemplar retrieval. It is a concept that could be learned even by an AI whose programmers had never heard of numbers, or whose programmers weren't thinking about numbers at the time. It is a concept that can mutate in useful ways. By relaxing the requirement of no dangling objects in the exemplar, we get "less than or equal to three". By relaxing the requirement of no dangling objects in the target image, we get "greater than or equal to three". By requiring a dangling object in the target image, we get "more than three". By comparing two images, instead of an exemplar and an image, we get "same number as" (35), and from there "less than" or "less than or equal to".

In fact, examining some of these mutations suggests a real-world path to threeness. The general rule is that concepts don't get invented until they're useful. Many physical tasks in our world require equal numbers of something; four pegs for four holes, and so on. The task of perceiving a particular number of "holes" and selecting, in advance, the correct number of pegs, might force the AI to develop the concept of corresponding sets, or sets that contain the same number of objects. The spatial fact that two pegs can't go in the same hole, and that one peg can't go in two holes, would be a force acting to create the perception of unique (one-to-one) correspondences. "Corresponding-sets" would probably be the first concept formed. After that, if it were useful to do so, would come a tendency to categorize sets into classes of corresponding sets, when it was useful to do so; after that would come the selection of a three-exemplar and the concept of three.

The decomposition of three in the above graphic is not the most efficient concept for three. It is simply the most easily evolved. After the formation of the exemplar-and-comparison concept for three would come a more efficient procedure: Counting.

To evolve the counting *concept* requires that the counting *skill* be developed, which occurs on the thought-level, which thought in turn requires a more sophisticated concept-level depiction of *three*. It requires that *one* and *two* have also been developed, and that *one* and *two* and *three* have been generalized into *number*. Once this occurs, and the AI has been playing around with numbers for a while, it may notice that any group of three objects contains a group of two objects. It may manage to form the concept of "one-more-than", an insight that would probably be triggered by watching the number of a group change as additional objects are added. It might even notice that physical processes which add one object at a time always result in the same sequence of numerical descriptions: "One, two, three, four..."

If multiple experiences of such physical processes can be generalized, and an exemplar experience of the process selected and applied, the result might be a counting procedure like that taught to human children: *Tag an object as counted and say the word 'one'; tag another object as counted and say 'two'; tag another object as counted and say the word that, in the learned auditory chanting sequence, comes after 'two'; and so on. Do not*

re-count any object that has already been tagged as "counted". The last word said aloud is the number of the group. This method is more efficient than checking unique correspondences, and the method also reflects a deeper understanding of numbers.

Finally, once "three" has been used long enough, it's likely that a human brain evolves some type of neural substrate for seeing threeness directly. That is, some piece of the human visual modality - probably the object-recognition system in the temporal lobe, but that's just a wild guess - learns to respond to groups of three objects. (Larger numbers like "five" or "six" are harder to recognize directly - that is, without counting - unless the objects are arranged in stereotypical five-patterns and six-patterns, like those on the sides of dice.) The analogue for an AI might be a piece of code (or assembly language, or a neural net - you know, mindstuff) that counts items directly.

However, even if the AI eventually creates a highly-optimized counting method, implemented directly, the previous definitions of the concept will still exist. When new situations are encountered, new situations that force the extension of the concept, the mind can switch from the optimized method to the methods that reflect underlying causes and underlying substrate. If necessary, the problem can rise all the way to the level of conscious perception, so that the deliberate, thought-level methods - the thoughts from which the concepts first arose - are used. The experiences that underlie the original definition, the experience of noticing the definition, the experience of using the definition - all can be reviewed. This is why a concept is so much richer, so much more powerful, if it's *learned* instead of preprogrammed. It's why learned, rich concepts are so much more *flexible*, so much likelier to mutate and evolve and spin off interesting specializations and generalizations and variations. It's why learned concepts are more useful when a mind encounters special cases and has to resort to high-level reasoning. It's why high-level cognitive objects are vastly more powerful, more *real*, than the flat, naked "predicate calculus" of classical AI.

Thus the idea of "information-loss" or "focus" is cast in a different light. Sure, calling something a three-group, or placing it into the three-category, can be said to "lose" a lot of information - in information-theoretical terms, you've moved from specifying the distinct and individual object to specifying a member of the class of things that can be described by "three". In classical-AI terms, you've decided to focus on the feature called "number" and not any of the other features of the object. But to label a rich, complex, multi-step act of perception "information loss" borders on perversion. Seeing the "threeness" of a group doesn't *destroy* information, it *adds* information. One perceives everything that was previously known about the object, and its threeness as well; nor could that threeness be "focused" on, until the methods for perceiving threeness were learned.

2.3.4: Concept combination and application

"When you hear the phrase "triangular light bulb", you visualize a triangular light bulb... How do these two symbols *combine*? You know that light bulbs are fragile; you have a built-in comprehension of real-world physics - sometimes called "naive" physics - that enables you to understand fragility. You understand that the bulb and the filament are made of different materials; you can somehow attribute non-visual properties to pieces of the three-dimensional shape hanging in your visual cortex. If you try to design a triangular light bulb, you'll design a fluorescent triangular loop, or a pyramid-shaped incandescent bulb; in either case, unlike the default visualization of "triangle", the result will not have sharp edges. You know that sharp edges, on glass, will cut the hand that holds it."

-- 1.2: Thinking About AI

How do the concepts of "triangular" and "light-bulb" combine? My current hypothesis involves what might be called "reductionist energy minimization" or "holistic network relaxation", a conflict-resolution method that takes cues from both the "potential energy surface" of chemistry and the "computational temperature" of Copycat.

Neural networks, when perturbed, are known to seek out what might be called "minimal-energy states". A network-relaxation model of concept combination could be computationally realistic - an operation that neurons can accomplish in the 200 operations-per-second timescale. My current hypothesis for the basic neural operation in concept-combination is the *resonance*. A neural resonance circuit - perhaps not a physical, synaptic circuit, but a virtual message-passing circuit, established by one of the higher-level neural communication methods (binding by neural synchrony, maybe) - can either resonate positively, reinforcing that part of the concept-combination, or resonate negatively, generating a conflict. My guess at the network-relaxation method resembles the "potential energy surface" of chemistry in that multiple, superposed alternatives are tried out simultaneously, so that the minima-seeking resembles a flowing liquid rather than a rolling ball.

The high-level, salient facets of the concepts being combined are combined first. These high-level features then visualize the mid-level features; if no conflict is detected, the mid-level features visualize the low-level features. If a conflict is detected at any level, the conflict propagates back up to the conflicting high-level or mid-level features causing the problem. Who wins the conflict? The more salient, more important, or more useful feature - remember, we're talking about combining two concepts, each with its own set of features along various dimensions - is selected as dominant, and the network relaxation algorithm proceeds. When one concept modifies another, the "more salient" feature is the one specified by the concept doing the modifying. (Note also that, in casual reading, not all the facets of a concept may be important, just as you don't fully visualize every word in a sentence. Only the facets that *resonate* with the subject of discussion, with the paragraph, will be visualized.)

In the case of "triangular light bulbs", "triangular" is an adjective. The concept for "triangle" or "triangular" is modifying the concept of "light bulb", rather than vice versa. The default exemplar for "light bulb" - that is, an image of the generic light bulb - is loaded into the mental workspace, including the visual facet of the exemplar being loaded into the visual cortex. Next, the concept for "triangular" is *applied* to this mental image.

The concept of "triangular", as it refers to physical objects, has a single facet: It alters the physical *shape* of the target image. Note that I say "physical shape", not "visual shape". The default exemplar for "light bulb" is a mental image - not a mental *picture*, but a mental *image*; in GISAI, an "image" means a representation in any modality or modalities, not just the visual cortex. The "light bulb" exemplar is an image of a three-dimensional bulb-shaped object, made of glass, having a metal plug at the bottom, whose purpose is to emit light. It is this multimodal mental image that "triangular" modifies, not just the visual component of the image. In particular, the "shape" facet of the light-bulb concept, the facet being modified, is a high-level feature describing the shape of the three-dimensional physical object, not the shape of the visual image. Thus, modifying the light-bulb shape will modify the mental image of the physical shape, rather than manipulating the 2-D visual shape in the visual cortex.

The "triangular" concept, when applied along the dimension of "shape", manipulates the mental image of the light bulb, changing the 3D model to be triangle-shaped. However, since the image of a flat light bulb fails to resonate, "triangle" automatically slips to "pyramid".

(I'm not sure whether this conflict is detected at the mid-level feature of "flat light bulb", or whether a flat light bulb actually begins to visualize before the conflict is detected. The slippage happens too fast for me to be sure. I suspect that "triangular" has slipped to "pyramidal" before, when applied to three-dimensional mental images; for neural entities, anything that happens once is likely to happen again. Neurons learn, and neural thinking wears channels in the neurons. It could be that the non-flatness of light bulbs is salient because of their bulbous shape, and that this resonance with non-flatness causes "triangular" to slip to "pyramidal" before the concept is even applied.)

Pyramids are sharp. I know, from introspection, that the "sharp pyramidal light-bulb" got all the way down to the visual level before the conflict was noticed. (The conflict rose to the level of conscious perception, but was resolved more or less intuitively; I didn't have to "stop and think". So this is probably still a valid example of concept-level processes.) The particular conflict: Sharp glass cuts the person who holds it. We've all had visual experience of sharp glass, and the associated need for visual recognition and avoidance; thus, the mental image of sharp glass would trigger this recognition and create a conflict. This conflict, once detected, was also visualized all the way down to the visual cortex; I briefly saw the mental image of a thumb sliding along the edge of the pyramid.

The problem of sharp edges is one that is caused by sharpness and can be solved by rounding, and I've had visual experience of glass with rounded edges, so the sharp edges on the mental image slipped to rounded edges. The result was a complete mental image of a pyramidal light bulb, having four triangular sides, rounded edges and corners, and a square bottom with a plug in it. (36)

Every sentence in the last five paragraphs, of course, is just begging the question: "Why? Why? Why?" A full answer is really beyond the scope of the section on "Mind"; I just want to remind my readers that often the real answer is "Because it happened that way at least once before in your lifetime." A human mind is not necessarily capable of simultaneously inventing all the reflexes, salient pathways, and slippages necessary to visualize a triangular lightbulb. Neurons learn, and thoughts wear channels in the network. The *first* time I ever had to select which level triangle-imposition should apply to - visual, spatial, or physical - I may have made a comical mistake. A seed AI may be able to avoid or shorten this period of infancy by using deliberate, thought-level reasoning about how concepts should combine; if so, this is functionality over and above that exhibited by humans.

You'll note that, throughout the entire discussion of concept combination, I've been talking about humans and even making appeals to specific properties of neurally based mindstuff, without talking about the problem of implementation in AIs. Most of the time, the associational, similarity-based architecture of biological neural structures is a terrible inconvenience. Human evolution always works with neural structures - no other type of computational substrate is available - but some computational tasks are so ill-suited to the architecture that one must turn incredible hoops to encode them neurally. (This is why I tend to be instinctively suspicious of someone who says, "Let's solve this problem with a neural net!" When the human mind comes up with a solution, it tends to phrase it as code, not a neural network. "If you really understood the problem," I think to myself, "you wouldn't be using neural nets.")

Concept combination is one of the few places where neurons really shine. It's one of the very rare occasions when the associational, similarity-based, channel-wearing architecture of biological neural structures is so appropriate that a programmer might reinvent naked neurons, with no features added or removed, as the correct computational elements for solving the problem. Neural structures are just very well-suited to "reductionist energy minimization" or "holistic network relaxation" or whatever you want to call it.

Even so, neural networks are very hard to understand, or debug, or sensibly modify. I believe in the ideal of mindstuff that both human programmers and the AI can understand and manipulate. To expect direct human readability may be a little too much; that goal, if taken literally, tends to promote fragile, crystalline, simplistic code, like that of a classical AI. Still, even if concept-level mindstuff doesn't have the direct semantics of code, we can expect better than the naked incomprehensibility of assembly language. We can expect the programmer to be able to see and manipulate what's going on, at least in general terms, perhaps with the aid of some type of "decompiler". I currently tend to lean towards code for the final mindstuff, while acknowledging that this code may tend to organize itself in neural-like patterns which will require additional tools to decode.

2.3.5: Thoughts are created by concept structures

Thoughts are created by structures of concept-level patterns. The archetypal example is a grammatical sentence: a linear sequence of words parsed by the brain's linguistic centers into a more-or-less hierarchical structure, in which the referents of targetable words and phrases (an adjective needs a target image, for example) have been found, either inside the sentence or in the most salient part of the current mental image. The inverse of this process is when a fact is noticed, turned into a concept structure, translated into a sentence, and articulated out loud within the mind. (A possible reason for the stream-of-consciousness phenomenon is discussed in 2.4.3: Thoughts about thoughts.)

The current section has discussed concepts as mindstuff-based patterns in sensory modalities - that is, the mindstuff is assumed to pay attention to, or issue instructions to, the sensory modalities and the features therein. That concepts interact with other concepts, and are influenced by the higher-level context in which they are invoked, has been largely ignored. This was deliberate. The farther you go from the mindstuff level, and the more "abstract" you get, the closer you are to the levels that are easily accessible to human introspection. These are the introspective perceptions that come out in words; the qualities that modern culture associates with above-average intelligence; the levels enormously overemphasized by classical AI.

Still, there are some thoughts that are so abstract as to appear distant from any sensory grounding. In that last sentence, for example, only the term "distant" has an obvious grounding, and since the sentence wasn't interpreted in a spatial context, it's unlikely that even that term had any direct visualizational effect. Metaphors do show up more often than you might think, even in abstract thought (see "Lakoff and Johnson, *Metaphors We Live By* or *Philosophy in the Flesh*). Still, there are concepts whose definition and grounding is primarily their effect on other concepts - "abstract concepts". Why doesn't the classical-AI method work for abstract concepts?

Even abstract concepts, mental images composed entirely of concepts referring to other concepts, exist within a "reductoholistic" system. Abstract concepts may not have reductionist definitions that ground directly in sensory experience, but they have reductionist definitions that ground in other concepts. What are apparently high-level object-to-object interactions between two abstract concepts can, if conflicts appear, be modeled as mid-

level structure-to-structure interactions between two definitions. Abstract concepts still have lower-level structure, mid-level interactions, and higher-level context.

Still, defining concepts in terms of other concepts is what classical AIs do. I can't actually recall, offhand, any (failed!) classical AIs with explicit holistic structure - I can't recall any classical AIs that constructed explicitly multilevel models to ground reasoning using semantic networks - but it seems likely that someone would have tried it at some point. (Eurisko and Copycat don't count for reasons that will be discussed in future sections. Besides, they didn't *fail*.) So, why doesn't the classical method work for abstract concepts?

Many classical AIs lack even basic quantitative interactions (such as fuzzy logic), rendering them incapable of using methods such as holistic network relaxation, and lending all interactions an even more crystalline feeling. Still, there are classical AIs that use fuzzy logic.

What's missing is flexibility, mutability, and above all *richness*; what's missing is the complexity that comes from *learning* a concept. Perhaps it would be theoretically possible to select a piece of abstract reasoning in an adult AI in which the complexity of sensory modalities played no part at all. Perhaps it would even be possible to remove all the grounding concepts below a certain level, and most of the modality-level complexity, without destroying the causal process of the reasoning. Even so - even if the mind were deprived of its ultimate grounding and left floating - the result wouldn't be a classical AI. Abstract concepts are learned, are *grown* in a world that's almost as rich as a sensory modality - because the grounding definitions are composed of slightly less abstract concepts with rich interactions, and those less-abstract concepts are rich because they grew up in a rich world composed of interactions between even-less-abstract concepts, and so on, until you reach the level of sensory modalities. Richness isn't automatic. Once a concept is created, you have to play around with it for a while before it's rich enough to support another layer. You can't start from the top and build down.

Another factor that's missing from classical AIs is the ability to attach experience to concepts, to gain experience in *thinking*, to wear a channel in the mind. Even a concept-combination like "triangular light bulb" has a dynamic pattern, a flow of cause and effect on the concept level, that relies on the thinker having done most of the thinking in advance. That complexity is also absent from classical AIs. (And of course, most classical AIs just don't support all the other dimensions of cognition - attention, focus, causality, goals, subjunctivity, et cetera.)

I think this provides an adequate explanation of why classical AI failed. This is why classical AIs can't support thought-level reasoning or a stream of consciousness; why sensory modalities are necessary to learn abstract thought; and why concepts must be *learned* in order to be rich enough to support coherent thought.

Interlude: Represent, Notice, Understand, Invent

Rational reasoning is very large, and very complicated. In trying to duplicate the functionality of a line of rational reasoning, it's easy to bite off too much, and despair - or worse, oversimplify. The remedy is an understanding of precedence, a sequence that tells you when you're getting ahead of yourself and building the roof before you've laid the foundations; °heuristics that tell you when to slow down and build the tools to build the tools. Before you can create a thing, there must be the potential for that thing to exist, and sometimes you have to recurse on creating the potential.

Drew McDermott, in the classic article "Artificial Intelligence Meets Natural Stupidity", pointed out that the first task, in AI, is to get the AI to notice its subject. Not "understand". *Notice*. If a classical AI has a LISP token named "hamburger", that doesn't mean the token is a symbol, or that there's any hamburgerness about it. For an AI to notice something, its internal behavior must change because of what is noticed. A LISP token named "hamburger" has no attached hamburgerness. A philosopher of classical AI would say that the LISP token has semantics because it refers to hamburgers in external reality, but the AI has no way of noticing this alleged reference. The "reference" does not influence the AI's behavior - neither external behavior, nor the internal flow of program causality.

I've extended McDermott's heuristic to describe a sequence called **RNUI**, which stands for Represent, Notice, Understand, and Invent. *Represent* comes before *Notice*; before you can write feature-detectors in a modality, you need data structures (or non-°crystalline equivalents thereof) for the data being examined and the features being perceived. *Understand* comes before *Invent*; before an AI can design a good bicycle, it needs to be able to tell good bicycles from bad bicycles - perceive the structure of goals and subgoals, understand a human designer's explanation of why a bicycle was designed a particular way, be capable of *Representing* the explanation and *Noticing* the difference between explanations and random babbling. Only then can the AI independently invent a bicycle and explain it to someone else.

Represent is when the skeleton of a cognitive structure, or the input and output of a function, or a flat description of a real thought, can be represented within the AI. *Represent* is about static data, what remains after dynamic aspects and behaviors have been subtracted. *Represent* can't tell the difference between data constituting a thought, and data that was provided by a random-number generator.

Notice provides the behaviors that enforce internal relations and internal coherence. *Notice* adds the dynamic aspect to the data. Applied to the modality-level, *Notice* describes the feature-extractors that annotate the data with simple facts about relations, simple bits of causal links, obvious similarities, temporal progressions, small predictions and expectations, and other features created by the "laws of physics" of that domain. The converse of modality-level *Notice* perception is *Notice* manipulation, the availability of choices and actions that manipulate the cognitive representations in direct ways. The RNUI sequence also applies to higher levels, and to the AI as a whole; it's possible to be capable of *Representing* and *Noticing* threeness without *Understanding* it, or being able to do anything useful with it.

Understand is about °intentionality and external relations. *Understand* is about coherence with respect to other cognitive structures, and coherence with respect to both upper context and underlying substance (the upper and lower levels of the °reductholistic representation). *Understanding* means knowledge and behaviors that reflect the goal-oriented aspects of a cognitive structure, and the purpose of a design feature. *Understanding* reflects the use of heuristics that can bind high-level characteristics to low-level characteristics. *Understanding* means being able to distinguish a

good design from a bad one. *Understanding* is the ability to fully represent the cognitive structures that would be created in the course of designing a bicycle or inventing an explanation, and to verify that these cognitive structures represent a good design or a good explanation.

Invent is the ability to design a bicycle, to invent a heuristic, to analyze a phenomenon, to create a plan for a chess game - in short, to think.

If you have trouble getting an AI to design a bicycle, ask yourself: "Could this AI understand a design for a bicycle if it had one? Could it tell a good design for a bad design?" If you have trouble getting an AI to understand the design for a bicycle, ask yourself: "Can this AI notice the pieces of a bicycle? Could it tell the difference between a bicycle and random static?" If you have trouble getting an AI to notice the pieces, ask yourself: "Can this AI represent the pieces of the bicycle? Can it represent what is being noticed about them?"

2.4: Thoughts

NOTE:	This section is about what thoughts <i>do</i> . For an explanation of what thoughts <i>are</i> - how they work, where they come from, and so on - see the previous sections.
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2.4.1: Building the world-model

Before the AI can act, it needs to learn. "Learning" can be divided into knowledge-formation and skill-formation. Skill formation happens when mindstuff, reflexes, or other unconscious processes are modified. In humans, the modification is autonomic; in seed AIs, it can be either autonomic or deliberate; but skills are always *executed* autonomically. (Note that "skill", as used here, includes not only motor reflexes but cognitive reflexes, and that "skill" does not include conscious skills like knowing (in theory!) how to disassemble a motorcycle.) The usual term for the dichotomy between skill and knowledge is "procedural vs. declarative", although this involves an assumption about the underlying representation that isn't necessarily true. In general, "knowledge" is the world-model, the contents of the mind, and "skill" is the stuff the mind is made of. Because skills tend to be located at the concept-level or modality-level, this section focuses on knowledge.

The world-model is holistic or reductionist, depending on whether you're looking up or looking down. We live in a Universe where complex objects are built from simpler structures, and stochastic regularities in the interactions between simple elements become complex elements that can develop their own interactions.

Thus, broadly speaking, there are at least three kinds of knowledge problems. You can look for a regularity in the way an object interacts with another object. You can take an object, an event, or an interaction, and try to *analyze* it; explain how the visible complexity is embodied in the constituent elements and their interactions. Or you can take elements and interactions that you already know something about, and try to understand the high-level behavior of the system. Starting from what you know, you can look sideways, down, or up.

Actually, this is speaking *too* broadly. Where, for example, do you fit "taking an object that you know something about, and suddenly understanding its purpose within a higher system"? I suppose you could explain this as a variant of analysis - when the "Aha!" is done, the result is a better understanding of a system in terms of its constituents. But then there are other knowledge problems, like guessing the properties of an element by taking the intentional stance towards the system and assuming the object is well-designed for its purpose. Where does that fit in? The moral, I suppose, is that "reductholism" has its uses as a paradigm, but there are limits.

Maybe we should generalize to generic causal models, regardless of level? Then you could divide activities into noticing a property or interaction, deducing the cause of a property or interaction, or projecting from known causes to the expected results. This model is a little more useful, since it sounds like the three problem types may correspond to three problem-solving methods: (A) Examine the model for unexpected regularities, correspondences, covariances, and so on. (B) Generate and test possible models to explain an effect. (C) Use existing knowledge to fill in the blanks (and, if you're a scientific mind, test the predictions thus created).

Still, even that view has its limitations. For example, asking *Why?* or looking for an explanation isn't strictly a matter of generate-and-test. In fact, generate-and-test is simply a genteel, thought-level version of that old bugaboo of AI, the search algorithm. It seems likely that some type of "genteel search algorithm" - not "blind", but not really deliberate either, and with a definite random component - is responsible for sudden insights and intuitive leaps and a lot of the go-juice of intelligence on the concept level. On the thought level, however, it's often more efficient to take a step back and think *about* the problem. One implementation for thinking about the problem is "abstraction is information-loss" classical-AI-type "abstract thought", running the problem through with Unknown Variables substituted in for everything you don't know, to see if there are places where the Unknowns cancel out to yield partial results that would hold true of every possible solution, thus constraining the search space. A more accurate implementation would be "applying heuristics that operate on the general information you have, to build up general information about the answer".

The thought-level is a genuine layer of the mind. There isn't any simple way to characterize it. There's a complex way to characterize it, which would consist of watching people solve problems while thinking out loud ("protocol analysis"), then figuring out a set of generalizations that corresponded to underlying neurology or underlying functional modules of the problem-solving method, and which categorized all the individual thoughts in the experimental observations. This problem is large, but finite; the set of underlying abilities and mental actions is limited. Still, such a project is beyond the scope of this particular section. (What I will attempt to do, in later topics, is describe *enough* of the underlying abilities - enough that implementing them would give rise to sustainable thought. Remember, seed AI isn't about perfectly describing the complete functionality of humans, it's about building minds with sufficient functionality to work.)

The thought-level is a genuine layer of the mind, and has around the same amount of internal complexity as might be associated with the modality-level or the concept-level. The difference is that thoughts are open to introspection, and thus, when I make sweeping generalizations, my readers can catch me at it. Nonetheless, I hope that the generalizations that have been offered here are sufficient to convey a vague general image of what

goes on in a mind searching for knowledge. Noticing interesting coincidences and covariances and similarities (looking sideways), building and testing and thinking about the reason why something happens (analysis, looking down in the holistic model, looking backwards in the causal model), trying to fill in the blanks from the knowledge you already have (prediction, looking up in the holistic model, looking forwards in the causal model). The goal is a holistic model with good high-level/low-level *bindings*, or a causal model where the consequences and preconditions of a perturbation are well-understood, or a goal-and-subgoal model with plans and convergences and intentionality. The goal is a model that holds together, on all levels, when you think about changing it; a model rich enough to support what we think of as intelligent thought.

2.4.2: Creativity and invention

It is literally impossible to draw a sharp line between understanding and creativity. Sometimes the solution to a difficult knowledge question must be *invented*, almost *ab initio*. Sometimes the creation of a new entity is not a matter of searching through possibilities but of seeing the *one* possibility by looking deeper into the information that you already have. But, *usually*, when building the world-model, you're trying to find a single, unique solution; *the* answer to the question. When trying to design something new, you're looking for *any* answer to the question. Understanding is more strongly constrained, but this actually makes the problem easier, since a solution exists and the problem is finding it... the constraints might rather be called *clues*.

In invention, each constraint eliminates options and makes it less likely that a solution exists. The distinction between understanding and invention is something like the difference between P and NP, between verifying a solution and finding it. Returning to the quadrivium of Sensory, Predictive, Decisive, and Manipulative binding, and to Manipulation's sub-trinity of qualitative, quantitative, and structural bindings, then invention, or high-level manipulation, adds a fourth binding, the *holic* binding. It's the ability to take a desired high-level characteristic and specify the low-level structure that creates it. It's the ability to engage in hierarchical design, to start from the goal of rapid travel and move to a complete physical design for a bicycle.

The methods of invention are even less clear-cut than the methods of understanding. Unless the problem is one of qualitative manipulation (choice from among a limited number of alternatives), the design space is essentially infinite. An intelligent mind reduces the effective search space through possession of a holistic model that ultimately grounds in heuristics capable of direct backwards manipulation. In other words, if you can choose any real number to specify the width of the wheel, what's needed is a heuristic that binds it - reversibly - to a higher-level design feature, such as desired stability on turns. If desired stability on turns is itself a design variable, a heuristic is needed that binds it to a known quantity, such as the weight range of the rider. And so on.

Such reasoning acts to reduce the search space from the space of all possible low-level specifications of a design, to the space of cognitive objects constituting reasonable high-level designs. If there are enough heuristics left to constrain the design further, or to specify design features from high-level goals, then the task can be completed without special inspiration. If there's a gap, a high-level feature with no heuristics that directly determine how it might be implemented, then there sometimes comes that special event known as an "insight", an intuitive leap.

Sometimes you try to invent the bicycle without knowing about the wheel. The crucial insight may consist of remembering logs rolling down a hill. It may consist of just suddenly seeing the answer. Or it may lie in finding the right heuristic to attack the problem. The key point is that a wide search space is crossed to find the single right answer, apparently without any guide or heuristic that simplifies the problem. (If the *aha!* is finding the right heuristic, then the act of creativity lies in crossing the search space of possible *heuristics*.)

What is creativity? Creativity is the name we assign to the mental shock that occurs when a large and novel load of high-quality mental *material* is delivered to our perceptions. I would say that it's the perception of "unexpected" material, meaning "unexpected" not in the sense that the delivery comes as a surprise, but in the sense that our mental model can't predict the specific content of the material being delivered. We perceive a thought as "creative", in ourselves or others, on one of two occasions: First, seeing someone thinking outside the box; second, on perceiving a single good solution selected from a nearly infinite search space. In the first case, a concept is redefined, or what was thought to be a constraint is broken; the answer is *unexpected*, which creates - to the viewer - the mental shock that we name "creativity". The second case consists of seeing the very large gap between "high-speed travel" and "bicycle" crossed; the viewer - unless he or herself has designed a bicycle - has no single heuristic that can cross a gap of that size, that can anticipate the content of the material presented. There's a nearly infinite space of possible paintings, so when we see any single painting of reasonable quality, a large quantity of unexpected cognitive material is delivered to our eyes and we call it "creativity".

It seems likely to me that the experience of creative insight happens when the mind decides to brute-force, or rather intelligent-force, the search problem. The *aha!* of wheels comes because, somewhere in the back of your mind, possible memories were tested at random for applicability to the problem until the memory of logs rolling down a hill resonated with the problem and rose to conscious attention. This unconscious "blind" search may employ some of the tricks of deliberation, such as searching through memories of objects that were seen traveling very fast. (Or not. It seems likely to me that only deliberate thought produces that kind of constraint.) Even so, it remains in essence a try-at-random algorithm. If there's anything more to subconscious creative insights than that, I don't know what it is.

2.4.3: Thoughts about thoughts

Since thoughts are reasonably accessible to the human mind, there's a good deal of existing research on how they work. The specific methods are important, but what's more important is getting a working *system* of thoughts, enough methods that work well enough that the AI can continue further.

Most important to the *system* of thoughts is introspection. Introspection is the glue that holds the thought-level together. Coherent thoughts don't happen at random. They happen because we know how to think, and because we have the right reflexes for thinking. The problem of what to think next is itself a problem domain. To prevent an infinite-recursion error, our solution to this problem on the moment-to-moment level is dictated entirely by reflex, the channels worn into our neural minds. Even when we deliberately stop and say to ourselves, "Now, what topic should I think about next?", the thinking about thinking proceeds by reflex. These reflexes are formed during infancy, and before they exist, coherent thought doesn't happen. To get past that barrier you'd have to be a seed AI, capable of watching a replay of your own source code in action, or halting and storing the current state of high-level thought to recurse on examining the stuff the thought is made of.

The self is a domain fully as complex as any in external reality. It consists not just of perceiving the self but of manipulating the self. The experience you remember of introspection consists of the occasions when the problems became large enough to require conscious thought. Beneath that remembered, introspection-accessible experience lies perceptions and reflexes that have become so invisible we don't even notice them. The *intuitions* of introspection are far more basic to thought than Hamlet's soliloquy. The problem of introspection should be approached with the same respect, and the same attention to the RNUI method, that would be given to the problem of designing a bicycle.

Introspection requires introspective senses, perhaps even an introspective modality. But the idea of an introspective modality is a subtle and perhaps useless one. The obvious implementation is to have an introspective modality that reports on all the cognitive elements inside the AI, but what does this add? The AI has already noticed that the cognitive elements are there. How does "the introspective modality" differ from "a useless and static additional copy of all the information inside the AI"? What can you do with the detected feature of "the feature of redness" that you can't do with the feature of redness itself?

To answer this question, it is necessary to step back and consider the problem in context. Sensory modalities don't exist in a vacuum. They are useful because concepts lie on top. The question, then, is not how to build an introspective sensory modality, but how to insure that concepts about introspection can form. This may involve creating a new introspective modality, or it may involve attaching a new dimension to the old modalities and to the other modules of cognition.

Concepts *manipulate* their referents, as well as extracting information from them. How would you go about tweaking the visual modality so that you could imagine "thinking about redness"? How do you get the AI to notice, declaratively, that a concept has been activated, and how is this perception reversed to give rise to visualizing the consequences of activating a concept?

This design problem may go a bit towards explaining that peculiar phenomenon called "stream of consciousness". You notice a fact, the fact gets turned into a conceptual structure, the conceptual structure gets turned into a sentence by your language centers, and then you speak the sentence "out loud" within your mind. The fascinating thing is this: If you try to skip the step of "speaking the sentence out loud" within your mind, *even after you know exactly what the words will be*, you can't go on thinking. Why? What new information is added by this act?

One possible explanation is that the human mind notices concepts by noticing the auditory cortex. Humans have no built-in introspective modality, so concepts become "visible" to our mental reflexes when they add recognizable content - words - to the auditory cortex. This closes the loop. Concept activation becomes detectable, and we can form concepts about concepts. I don't think this is the entire explanation, but it's a good start.

What about thoughts? On the thought-level, human introspection is fairly primitive. There's this tendency to lump everything together under the term "I". When we attribute causality, we say "I remembered" instead of "the long-term memory-retrieval subsystem reports..." Perhaps this is because, historically speaking, we didn't *know* anything about what was inside the mind until yesterday afternoon. Perhaps it's because fine-grained introspection doesn't contribute useful complexity to self-modeling unless you're, oh, writing a paper on AI or something. There's plenty of useful heuristics about the self that can be learned by looking at cause and effect, even when all the causal chains start at a monolithic self-object. A seed AI may have uses for more fine-grained self-models, but with both design and source code freely accessible, it shouldn't be too hard for such a self-model to develop.

2.4.4: The legitimate use of the word "I"

When can an AI legitimately use the word "I"?

Understand that we are asking about a very limited and purely technical aspect of self-awareness. We are not talking about the kind of self-awareness that will cause an ethical system to treat you as a person. We are not talking about "qualia", the hard problem of conscious experience, what it means to be a bat, or anything of that sort. These are different puzzles.

The question being asked is: When can an AI legitimately use the word "I" in a sentence, such as "I want ice cream", without Drew McDermott popping up and accusing us of using a word that might as well be translated as "shmeerp" or G0025?

Consider the SPDM distinction: Sensory, Predictive, Decisive, Manipulative. A *binding* between a model and reality starts when the model "maps" in some way to reality (although this is ultimately arbitrary), becomes testable when the model can predict experiences, and becomes useful when the model can be used to decide between alternatives, with the acid test being manipulation of reality in quantitative or structural ways. Consider also the distinction between modality-level, concept-level, and thought-level.

Self-modeling begins when the AI - let's call it Aisa, for "AI, self-aware" - starts to notice information about itself. Introspective sensations of sensations are hard to distinguish from the sensations themselves, so this ball doesn't really get rolling until Aisa forms introspective concepts. The self-model doesn't begin to generate novel information, information that can impose a coherent view of internal events, until it can make predictions - for example: "Skipping from topic to topic, instead of spending a lot of time on one topic, will result in conceptual structures that are connected primarily through association." Likewise, this information doesn't become useful until it plays a part in goal-oriented decisions - a *decisive* binding.

When Aisa can create introspective concepts and formulate thought-level heuristics about the self, it will be able to reason about itself in the same fashion that it reasons about anything else. Aisa will be able to manipulate internal reality in the same way that it manipulates external reality. If Aisa is impressively good at understanding and manipulating motorcycles, it might be equally impressive when it comes to understanding and manipulating Aisa.

But to say that "Aisa understands Aisa" is not the same as saying "Aisa understands itself". Douglas Lenat once said of Cyc that it knows that there is such a thing as Cyc, and it knows that Cyc is a computer, but it doesn't know that *it* is Cyc. That is the key distinction. A thought-level SPDM binding for the self-model is more than enough to let Aisa legitimately say "Aisa wants ice cream" - to make use of the term "Aisa" materially different from use of the term "shmeerp" or "G0025". There's still one more step required before Aisa can say: "I want ice cream." But what?

Interestingly, assuming the problem is real is enough to solve the problem. If another step is required before Aisa can say "I want ice cream", then there must be a material difference between saying "Aisa wants ice cream" and "I want ice cream". So that's the answer: You can say "I" when the behavior generated by modeling yourself is materially different - because of the self-reference - from the behavior that would be generated by modeling another AI that happened to look like yourself.

This will never happen with any individual thought - not in humans, not in AIs - but iterated versions of Aisa-referential thoughts may begin to exhibit materially different behavior. Any individual thought will always be a case of A modifying B, but if B then goes on to modify A, the system-as-a-whole may exhibit behavior that is fundamentally characteristic of *self-awareness*. And then Aisa can legitimately say of herself: "I want an ice-cream cone."

Humans also throw a few extras into the pot. We have observer-biased social beliefs, a whole view of the world that's skewed toward the mind at the center, which tends to anchor the perception of the self. We attribute internal causality to a monolithic object called the "self", which generates a lot of *perceived* self-reference because you don't notice the difference between the thought doing the modifying and the cognitive object being modified - the source of the thought is the "self", and the item being modified is part of the "self".

A seed AI will probably be better off without these features. I mention them because they constitute much of what a human means by "self".

3: Cognition

- 3.1: Time and Linearity
 - 3.1.1: The dangers of the system clock
 - 3.1.2: Synchronization
 - 3.1.3: Linear metaphors: Time, quantity, trajectory
 - 3.1.4: Linear intuitions: Reflection, simultaneity, interval, precedence
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3.1: Time and Linearity

- 3.1.1: The dangers of the system clock
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- 3.1.6: Trajectories
 - 3.1.6.1: Identification of single objects across temporal experiences
 - 3.1.6.2: Defining attributes of sources, trajectories, and destinations
 - 3.1.6.3: Source, path, target; impulse, correction, resistance, and forcefulness

Time in a digital computer is °discrete and has a single °space of simultaneity, so anyone who's ever played °Conway's Game of Life knows everything they need to know about the True Ultimate Nature of time in the AI. With each tick of the clock, each frame is derived from the preceding frame by the "laws of physics" of that °ontology. (Higher-level regularities in the sequence of frames form what we call causality; more about this in *Unimplemented section: Causality*.)

A general intelligence needs to be able to perceive and visualize when two events occur at the same time; when one event precedes or follows another event; when two sequences of events are identical or opposite-symmetrical; and when two intervals are equal, lesser, or greater. Most of this comes under the general heading of having a feel for time as a quantity and time as a trajectory, which requires both concept-level and modality-level support.

3.1.1: The dangers of the system clock

To support temporal metaphors and temporal concepts - to provide an °API with sufficient complexity for the °mindstuff to hook into - the AI needs modality-level support. The most obvious method would be to tag all events with a 64-bit number indicating the nanoseconds since 1970 - a plain good-old-fashioned system clock. The problem is that then the AI can't think about anything that happened before 1970. Or about °picoseconds.

If we humans have a built-in system clock - there are several candidates, ranging from the heartbeat to a 40-°hertz electrical pulse in the brain - we don't have conscious, abstract access to it. What we remember is the *relative* times; that event A came before event B, that event C was between A and B, that a lot of stuff happened between A and B, that D seemed to take a long time, that E seemed to go by very quickly, that E and F happened at the same time, and so on. If I know that a particular event happened at 4:58 PM on July 23rd 2000, it's because I looked at my watch and associated the visual or auditory label "4:58" with the event. That's why I can think - at least abstractly - about the age of the Universe or picosecond time frames. Our abstract concepts for quantitative time aren't really built on our internal modality-level clocks, but on the external clocks we built. Or rather, the internal modality-level clocks are used for immediate perceptions only, and the abstract concepts create the modality level through a layer of abstraction that can handle millennia as easily as minutes.

Because it's very easy to derive all the relative perceptions of time by comparing absolute quantitative times, we'll almost certainly wind up tagging every event with a 64-bit system-clock time (or equivalent interpreter token), and building any other modality functions on top of that. It's just important to remember that the really important *concepts* about time should not be founded directly on the underlying, absolute numbers, because then the AI really *can't* think about picoseconds or pre-1970 events; the mindstuff making up the concepts will crash. Concepts about time, if they refer to quantitative numbers at all, should be founded on the *relative* times of the cognitive events that occur while thinking about a temporal problem. Thus the AI can imagine a process that takes place on picosecond timescales, and because the visualization itself takes place on nanosecond timescales (or whatever speed the AI's system clock runs at), there's no crash. It's a kind of automatic scaling.

To put it another way: Generality requires that there be at least one layer of complete abstraction between temporal concepts and temporal modalities. Even if stored memories also store the attached system-clock time, a replay of those memories obviously won't take place at the recorded time! If all remembered times are purely abstract characteristics, and only concretely visualized times give rise to temporal intuitions, then the AI can freely manipulate temporal aspects of a visualized process. Symbols such as *slow* and *fast* (37) can be abstracted from temporal intuitions and applied to aspects of any visualized temporal process.

Of course, because we aren't slavishly following human limitations, a seed AI should probably have some mode of direct access to the system clock. We've all been in situations where we've wanted to know exactly what time it is, or exactly what time it was when we had breakfast. That's why God gave us wristwatches (38). This should be safe as long as the direct access occurs through the same conceptual filter, the same layer of abstraction, so that the modality-level system clock time 203840928340 comes out as the abstract characteristic "System-clock time 203840928340".

3.1.2: Synchronization

Another subtlety of human temporal understanding is that our senses are synchronized even though different senses presumably have different processing delays. It takes time for the visual cortex to process an image, and time for the auditory cortex to process a sound - not necessarily the *same* amount of time. But a physical sound and a physical sight that *arrive* simultaneously should be *perceived* as simultaneous. Since a seed AI should be able to tag *sensory* events as distinct from the derivative *perceptual* events, this should be relatively easy to handle on the modality level... although it's possible to imagine problems popping up if there are °heuristics or concepts that act on the derivative and possibly unsynchronized high-level features of multiple modalities.

For some cases, this problem can be solved by only allowing multimodality concepts to act on events that have been completely processed by all targeted modalities. If a vision and a sound arrive at t=10, the sound finishes feature-extraction at t=20, and vision finishes extraction at t=30, then no audiovisual concept can begin acting until t=31, with both the sound and the vision having a perceived time of t=10. In other words, rather than skimming the cream off the modalities, the perceived *now* of the AI will lag a few seconds behind real time.

This introduces two new problems: One, it may introduce severe delays into the system. Modalities don't just apply to external sensory information; modalities are where all the internal thoughts take place as well. To some extent this problem may be solvable by not requiring *complete* processing before concepts can activate, but only that level of processing which is necessary to the concept. After all, a concept can't act on information it doesn't have. But this may still lose some efficiency; there may be cases where concepts don't need synchronization.

The second problem is synchronization of subjective time. If the AI's *now* lags a few seconds behind, when are thoughts perceived to have taken place? If the AI thinks "fool!" at a time that *looks* to the AI like t=10 but is actually t=40, is the concept "fool!" labeled as having taken place at t=10 or t=40? And what difference does it make? I can't see that using t=40 makes any difference, so I'm strongly in favor of labeling all events as occurring when they actually occur. Still, the AI may eventually find useful °heuristics that act on "subjective time".

All these modality-level and concept-level problems are simply echoes of the far more difficult problem of change propagation on the thought-level - how to ensure that "Aha!" experiences and "Oops" experiences propagate to all the corners of the mind, so that beliefs remain in a reasonably consistent state. The issue of Consistency doesn't belong in this section. However, it seems likely that issues of concept-level (and thought-level) synchronization are not problems that should be solved by autonomic processes; concept synchronization may need to be decided on a case-by-case basis. It may be that, in the process of learning thought-level reflexes, and finding concepts that work well, the AI will be forced to invent whatever forms of synchronization are necessary for each concept. If a multimodal concept must act on modality-images that began processing at the "same time" (39), and will otherwise fail (not generate useful results), it should be a relatively simple tweak/mutation, of the sort that even °Eurisko could have performed easily enough. The same goes for whatever concepts are specified by the programmer during the initial stages.

As a general rule: All derivative perceptual events should be tagged with their true cognitive time as well as the external-world time of the derivative event. Human-programmed concepts should enable the programmer to decide which time should be used; learned concepts won't even be noticed unless the proper timeframe is used. Try to maintain the regularities in reality that all intelligence is supposed to represent; figure out whether the useful regularities represented by a temporal concept are perceptual/external or cognitive/internal.

3.1.3: Linear metaphors: Time, quantity, trajectory

"A general intelligence needs to be able to perceive and visualize when two events occur at the same time; when one event precedes or follows another event; when two sequences of events are identical or opposite-symmetrical; and when two intervals are equal, lesser, or greater. Most of

this comes under the general heading of having a feel for time as a quantity and time as a trajectory..."
-- above

Several of the most fundamental domains of cognition are one-dimensional or monotonically increasing, and thus share certain linear characteristics. In a sense, any possible use of the word "close" or "far" invokes a kind of linear intuition. So do the words "more" and "less". Time, because it is both monotonically increasing and one-dimensional (40), is one of the linear domains. The linear domains tend to relate very closely to each other - you can have "more" time or "less" time, treating time as a quantity; you can be "close" to a given time, treating time as a trajectory. We freely mix-and-match the words because the target domains share behaviors and underlying properties. In some sense, the relation between time and quantity and trajectory is not, as Lakoff and Johnson would call it, a "metaphor"; it is a real identity.

When you consider that time is almost always mathematically described as a real number (41); that one of the words for real number is "quantity"; that in most trajectories the spatial distance to the target decreases monotonically with time; and that time "moves forward" at constant velocity; then, the identity seems so perfect that there is no complexity to be gained by the metaphor. Lakoff and Johnson kindly remind us that "quantity" applies not just to mathematics, but to piles of bricks and stacks of coins; that "trajectories" are not just simple flights from source to target, but complex spatial maneuvers, with huge chunks of the visual subsystems dedicated to their visualization.

By observing that piles of two bricks plus piles of three bricks equal piles of five bricks, it is possible to guess that two hours plus three hours will equal five hours. Using the underlying numerical concept described in 2.3.3: The concept of "three", it can be seen that this "metaphor" requires the ability to treat temporal intervals as distinct objects, so that unique correspondences can be drawn between each of three hours and each of three bricks. To learn (concept-level) to treat time as a quantity requires that the AI encounter a task with a uniqueness constraint; one in which it can't do two things in the same minute (42). This leads to treating time as a limited resource, which leads to an even stronger analogy with time-as-material-substance.

Lakoff and Johnson describe the time-is-movement metaphor in terms of the motion of an observer. The "location" of the observer is the present, the "space" in front of the observer is the future, the "space" behind the observer is the past. "Objects" are events or times, "located" at various "points" along the "line". The time-is-motion metaphor has two (incompatible) interpretations: The observer can be thought of as moving forward at a constant speed, passing the events; or the events can be thought of as moving towards the observer. (L&J note that this is why "Let's move the meeting ahead a week" is ambiguous.) Lakoff and Johnson note that we also map time onto body image; in almost all languages, the observer "faces" the future - although a few languages (presumably noting that one can see the past, but not the future) have the observer facing the past. However, this is getting away from the primary topic - the *utility* of describing time as a trajectory.

One primary use of time-as-space is to visualize multiple events simultaneously. That is, by conceptualizing time as a line, we can simultaneously consider three points/events along the line, where a true temporal visualization would force us to consider the events sequentially. *But* this only applies to *humans*, with our single and indivisible stream of consciousness. A seed AI might be able to simultaneously visualize the dynamic qualities of three different events; in effect, placing three different moving observers at three different points along the timeline! Likewise, visualizing time as space makes it easier for humans to perceive certain types of qualitative relations. Visualizing a quantity plotted against time - you know, an ordinary 2D graph - enables us to perceive properties of the *curve* that would not be visible to a human observer watching the 1D variable change with time. Humans have one set of intuitions for static spatial properties, allowing us to stand back and look at the graph and form compounded perceptions and connected thoughts; we have another set for dynamic systems in which the sensory images change at the same rate as our stream of consciousness.

For an AI, the benefit of spatial metaphors might be provided *directly* by rewriting the spatial-modality perceptions directly for the temporal modality - rewriting a visual curve-detector so that it operates on data in the temporal modality, so that an AI watching a single quantity change over time has the same set of "smooth curve" or "sharp curve" or "global maximum" perceptions as a human contemplating a 2D graph.

In conclusion: Time, quantity, and trajectory share certain basic underlying properties. The primary driver for high-level metaphors between time and quantity is a task in which time is a limited resource. In humans, the primary driver for metaphors between time and trajectory is the greater sophistication of our static visual intuitions, but this may not apply to seed AIs.

3.1.4: Linear intuitions: Reflection, simultaneity, interval, precedence

Hofstadter, writing about Copycat - an AI that performs analogies in the domain of letter-strings, such as "abc->abd::pqrs->?" - notes that, despite the simplicity of Copycat's domain, the domain can contain analogy problems so complex as to embrace a significant chunk of human thought. A few years back, when I was only beginning to think about AI, I set out to brainstorm a list of a few hundred perceptions relating to analogies - "before, next, grow, quantity, add, distance, speed, blockage, symmetry, interval..." - and noticed that most of them could be represented on a linear strip of Xs and Os. These perceptions I collectively name to myself the *linear intuitions* - the perceptions that apply to straight lines.

3.1.4.1: Reflection

One such perception is reflection: "XXOX" is the reflection of "XOXX", and the image "XXOXOXX" is bilaterally symmetric. (Note that it may take you more time to verify that "XXOXOXXO" is the reflection of "OXXOXOXX", or that "OXOXOXOXOXO" is bilaterally symmetric, and you may need to do so consciously rather than intuitively; our perceptions have horizons, limits to the amount of processing power expended. Of course, your perceptions are analyzing huge collections of two-dimensional pixels, not just the on-off "pixels" of a linear image.) Writing a computational procedure to verify reflection is trivial, but this would leave out some of the most important design features. On seeing the letter-strings "ooabao", "cxcdcxc", and "rauabauar", the letter-string "oomemool" would come as rather a surprise, and the "l" would stick out like a sore thumb. Even without precedents to establish the expectation, the image "WHMMOW" has something wrong about it (43).

The perception of reflection is not simply a binary, yes-or-no verification; once a partial reflection is visible, it establishes an *expectation* of complete reflection - a mental image of how the structure "ought" to look, if the reflection were complete - and if the expectation is violated, if the actual image conflicts with the imagined, then the violation is detected, and the violating object becomes more salient ("sticks out like a sore thumb"). If there is some way to look at the violating object that preserves perfect reflection, it will resonate strongly with the expectation. (A more complete discussion of expectation, especially on the concept-level rather than modality-level, is in *Unimplemented section: Causality*.) The

point is that the perception of reflection, like most perceptions, has complex internal structure. In particular, it is possible to expect reflection, and for the property of "reflection" to be *applied* to a previously asymmetric object.

And the usual caveats: It is possible to notice reflection *within* an image, or to notice reflection of two structures in two *different* images; and it is easier to see reflection if you're looking for it in advance.

Since it would be computationally expensive to compare every possible set of pixels for reflection, and yet we notice even unexpected reflections within an image - implying that the detectors are always on - the human brain probably detects for prerequisites to reflection first, and tries to perceive reflection *per se* only if the prerequisites trigger. If two visual images are related by the property of reflection, they are likely to have very similar high-level properties, so that the simultaneous perception of an image and its reflection would lead to perceptual structures that, in the human neuron-based brain, would resonate very strongly with each other, suggesting that tests should be performed for both identity and reflection. If the object is recognizable, then both the object and its mirror image would usually be classified identically by the temporal lobe (44) - a bird and its mirror image are both classified as "bird" - so that the visual signals from object and mirror image would rendezvous at that point, and could be backtraced to their origins, and the test for symmetry then applied.

That's how humans detect visual symmetry, anyway. It is possible that the human brain uses its underlying electrical properties to detect neural synchronies on a global scale, a physically based method that it would be computationally extravagant to match on a von-Neumann-architecture digital computer. It could be that a Monte Carlo method would do as well; a million random samplings and comparisons of parts of the global state might often find local similarities between sufficiently large similar structures - if not always, then often enough to give perception a humanlike flavor of spontaneity. A Monte Carlo method that randomly tried to detect a million possible resonances might do to duplicate almost all the functionality of neural resonance, without the combinatorial explosion that would defeat a perfect implementation.

But that sort of thing is a major, fundamental, and underlying design issue, and somewhat beyond the scope of this section, or even 3: Cognition. The perception of 1D temporal reflection is much simpler than the perception of true 2D or 3D spatial reflection. The modality-level design requirement is that the AI should be able to independently notice blatantly obvious temporal reflections; detecting anything more subtle can be left to heuristics, concepts, and the full weight of deliberate intelligence. The AI needs to be able to verify temporal reflections suggested by concept-level or thought-level considerations, but this, as said, is relatively simple.

Scenario 1

A glass drops, and grapes explode in the microwave, and the computer turns itself on - and then, a few minutes later, the computer turns itself on, grapes explode in the microwave, and a glass drops.

The reactivation of the infrequently-used exploding-grape concept (or perceptual structure, if it doesn't rate a concept) should be enough to suggest that events are being repeated; enough to draw correspondences between each unusual pair of events. The computational procedure for detecting reflection is simple enough that it could conceivably be run on every consciously perceived event-line where correspondences are drawn between events - at least, with respect to the events salient enough to have correspondences drawn between them.

Perhaps this example is a bit *outré*, but then it's hard to come up with examples of *useful* temporal reflections. The only example that springs to mind would be disassembling and reassembling a motorcycle (45). A stock-trading AI might find a temporal-reflection intuition useful, or an AI watching a light bob up and down and trying to deduce a pattern. "Run the process backwards" is an incredibly useful heuristic in a wide variety of circumstances, but such a high-level idea is a thought-level process; even the concept "backwards" properly belongs under *Unimplemented section: Symmetry*.

There are still some subtleties remaining in Scenario 1 (the exploding-grape scenario). First, the correspondences drawn are between high-level events. The concept of "exploding grape" is not represented directly in a sensory modality; at most, the sound and sight of the exploding grape are represented, and no two real-world sights and sounds will ever be precisely equal. The similarities between the first and second events that lead both of them to be classified as "exploding grape" are higher-level - either low-level conceptual or very high-level modality.

However, the *modality*-level intuition for temporal reflection can operate on *concept*-level cognitive events. In humans, for example, the thought *exploding grape* results in the visualization of the syllables "exploding grape" in the auditory cortex, which - in theory - could have a time-tag attached. In practice, it seems likely that the AI architecture will be such as to locate concept-level cognitive events and label them as objects - so that, among other things, thoughts can be tagged with the system-clock-time that's used for modality-level temporal intuitions. In general, thinking about thinking - introspection - obviously requires some way of observing the temporal sequence of thoughts, knowing *when* you thought something. Either the architecture needs to explicitly represent the activation of concepts and thoughts (the likely solution (46)); or, if it's all a big puddle of mindstuff with higher levels being emergent (47), the thoughts need to spill over into modalities in some way that allows evolved concepts and thought-level reflexes to do things like identify the time of a thought.

The second subtlety is that the temporal reflection is not likely to be *perfect*. The intervals between the dropped glass and the exploding grape are not likely to be exactly 20 seconds apiece. Only the comparative precedences - which event came first - are tested for reflection. That said, a reflection which preserves intervals constitutes a much stronger binding, although human temporal perceptions are too approximate for us to notice that sort of thing without a stopwatch. (Our spatial intuitions for reflection *do* require the preservation of distances.)

3.1.4.2: Simultaneity

Simultaneity is when two events occur at the same time. Perfect simultaneity is when two events are tagged as occurring at exactly the same time, to the limits of the resolution of the modality-level system clock. Even in AIs that totally avoid parallel processing, sensory modalities will tag all the components of an incoming image as having *arrived* at the same time, so any mind is full of insignificant simultaneities. Significant simultaneities are those that are unexpected and that occur in high-level, salient objects. For example, two objects simultaneously disappearing from a sensory input.

Because a seed AI's system clock will probably run much much faster than our own, it may be necessary to define intuitions that detect imperfect simultaneities - for example, any sensory coincidence within 1/40th of a second, or any internal coincidence within 1/1000th of a second (or some other time scale chosen to match the speed of the AI's stream of consciousness). (48).

Aside from that, take all the caveats I listed in 3.1.4.1: Reflection and apply them to simultaneity. For example, if simultaneity is repeated often enough to be expected, then the expectation of simultaneity is applied to sensory inputs to create an image, a violated expectation should be noticed as a conflict of the real image with the expectation, the violating stimulus should become salient, and so on. (And if stimulus A appears without the expected simultaneous stimulus B... and stimulus B still hasn't appeared after the AI gets over the shock... then both stimulus A and the *absence* of B become salient.)

3.1.4.3: Interval

The human perception of intervals is approximate rather than quantitative. We divide how long something *feels* into "less than a second", "a second", "ten seconds", "a minute", "ten minutes", "an hour", "a few hours", "a day", "a few days", "a few weeks", "a few months", "a few years", "a lifetime", and "longer than a lifetime". (That's a guess. I don't know the actual categories or their boundaries. It would be an interesting thing to know, if someone has already done the research.)

The human perception of temporal intervals is also at least partially °subjective, dependent on how much thinking is going on. A process relatively empty of events, in which our mind processes incoming data much faster than it becomes available, is paradoxically perceived as being longer - it is "boring" (49). A process packed full of emotionally significant events may appear as being longer; when it's over, "it feels much longer than it was". (Again, with the time-as-pathway metaphor, passing a lot of events may appear to make the intervals longer.) There's also the proverb "time flies when you're having fun"; if events happen so fast that "there's no time to think" or pay attention to underlying intervals, time may appear to move by much more quickly. (50).

However, it appears to me that human subjective intervals implement no important functionality. If the AI uses system-clock intervals to control the actual subjective perception, so that perceived intervals are precise, then the perception of exact intervals is more likely to be useful - that is, when two processes unexpectedly have the same intervals, it is more likely to signal a useful underlying correlation. The AI does need a perception for "approximately the same amount of time", since this is a useful human perception. (Such a perception might have a quantitative as well as a qualitative component; in other words, the perception of "approximately the same amount of time" might be strongly true or weakly true.)

It may be that we humans have no modality-level "equal interval detectors" at all - after all, we have to count heartbeats or glance at a watch when we want to even *verify* the equality of two intervals. If so, an AI with a modality-level appreciation for intervals might spot surprises that a human would miss.

"Temporal Reasoning" in °MITECS notes that comparative operations on intervals can be more complex than the simple precedence or simultaneity of instantaneous events: "There are thirteen primitive possible relationships between a pair of intervals: for example, before (<) meets (m) (the end of the first corresponds to the beginning of the second), overlaps (o) and so on." Since these thirteen possible relationships can be built up from the relationships of the "start" and "end" events, I don't *think* they would require architecture-level support. Overlapping intervals should be intuitively noticed because salient intervals should be perceived as solid, filling in every point between the two events, and collisions should be detected in the same way as collisions of solid objects. Computationally, this can be implemented either by using a 1D collision-detection algorithm, or by creating an internally perceived "timeline", with temporal pixels that can be occupied by multiple events, with a computationally tractable resolution (the system clock might be too fast) that is nonetheless fine enough to detect overlap. (52).

Finally, intervals have the same caveats as 3.1.4.1: Reflection. For example, intervals are perceived only for salient events; they aren't computed for every pair of cognitive events in the mind. (This is, in fact, impossible, since the perception of an interval is itself a cognitive event.)

3.1.4.4: Precedence

Temporal precedence is which of two events - A or B - came first. Precedence is the most often-used and most useful temporal perception; it is the one by which humans order reality. We don't care about the exact intervals in milliseconds (although an AI might - see above); we care whether event A or event B came first. Precedence is the most useful temporal intuition because it is the most deeply intertwined with causality - effects follow causes. (See *Unimplemented section: Causality*.)

Mathematically, transitivity of precedence is the defining characteristic of a linear ordering. If $A < B$ and $B < C$, then $A < C$; if this relation holds true for all events A, B, and C in a group, then that defines a linear ordering of the group (53). The set of precedence relations defines a linear string of events. It is this definition that we humans use, most of the time. Without access to an actual calendar, we will almost never reconstruct a series of events by trying to remember the actual temporal labels and performing a sort(). Rather, we try to reconstruct the series by remembering that B came after A and before C, that D came after B, and so on

It is also noteworthy that we tend to remember precedences that have *reasons* behind them - such as the precedence of cause and effect. If the series is a causal chain, we may be able to rattle off the whole series without effort. If we're trying to describe the ordering of events that belong to multiple different causal series, we often have to consciously reconstruct the complete ordering from intersections in the partial orderings we remember; from remembering whether something was "a short time ago" or "a long time ago"; and so on. We do not remember an internal calendar or timeline, and we do not remember - on the modality level - the times of events. We remember precedences, and it is from these precedences that the timeline of our lives is constructed.

A seed AI should probably use a modality-level clock or a modality-level timeline, but it will still need to understand precedence.

Precedence in general is ubiquitous; we invoke it every time we say *before* or *after*. Precedence can be spatial as well as temporal. Precedence applies to priorities, not just in terms of what must be done *first*, but the *first choice*. In this sense, we invoke precedence every time we say *better*

or worse. The metaphors for precedence apply to every comparator that operates on a linear ordering: This is why linear and temporal metaphors are ubiquitous in human language.

What all the metaphors have in common is that the comparative operation on the quantity or trajectory usually reflects an actual temporal precedence - the *first* choice is usually the one that is considered first; the cognitive events associated with extrapolating that choice will take place earlier. If a simpler theorem comes before a more complex one, it's because the complex theorems are constructed from simple ones; the simple ones are *learned first* or *invented first*, and the cognitive event of that learning or invention will have an earlier clock-time attached.

Comparison is as ubiquitous in modalities as it is in ordinary source code. The modality-level intuitions for temporal precedence are a single case of this general rule.

Usual caveats about expecting precedence and broken expectations and so on.

3.1.5: Quantity in perceptions

"Quantity" is invoked with every perception containing a real number, as ubiquitous as floating-point numbers in ordinary source code. When I say "quantity", I do not just refer to a continuously divisible material substance, like water or time; I generalize to the internal use of floating-point numbers in representations and intuitions - all the perceptions that can be "stronger" or "weaker".

3.1.5.1: Zeroth, first, and second derivatives

Given two quantities, we can notice which is more or less; given two quantitative properties, such as height, we can notice which is higher or lower; given two quantitative perceptions, we can tell which is stronger or weaker. This perception can operate statically, in the absence of a temporal component.

As discussed in *Unimplemented section: whenextract*, quantities and comparators are too ubiquitous to initiate thoughts directly, unless the quantities and comparators are properties of very high-level objects; thus, low-level quantities and comparisons would be computed either as preludes to feature extraction, or only when demanded by the context of a higher thought. Comparisons computed for feature extraction are also generally local. A human visual pixel is compared with nearby pixels for edge detection, but not with every other pixel in the image, using $O(N)$ instead of $O(N^2)$ comparisons. A seed AI should be able to compare arbitrary pixels in arbitrary modalities - but only on demand. For more about the differences between on-demand and automatically-computed perceptions, the difference between low-level and high-level perceptions, and the difference between thought-initiating and guess-verifying perceptions, see *Unimplemented section: whenextract*.

The list of basic operations that can be performed on static quantities is basically the set of useful arithmetical operations: Subtraction (in other words, interval calculation), comparison, equality testing. It would also be possible to include addition, multiplication, division, bit shifting, bitwise & and |, remainder calculations, exponentiation, and all the other operations that can be performed on integers and floating-point numbers; however, these operations are less likely to be useful - less likely to pick out some interesting facet of reality.

3.1.5.2: Patterns and broken patterns

A field of quantities, extended across time or space or both, can give rise to the mid-level features called patterns; patterns are higher-level than quantities, and richer, and rarer as a perception (a hundred pixels give rise to one pattern); thus, patterns are more meaningful. Patterns can be broken, and the high-level feature that constitutes the breaking of a pattern is rarer, and *far* more meaningful, than either the patterns themselves or the low-level quantities. (I speak here of modality-level patterns; the problem of seeing thought-level patterns is nearly identical with the problem of intelligence itself.)

One example of a pattern is a rising quantity - "rising" implying either a single quantity changing with time, or a field of quantities changing continuously with with some spatial dimension.

- A: 27, 28, 29, 30, 31, 32.
- B: 29, 31, 33, 35, 37, 39.
- C: 8, 19, 22, 36, 45, 71.

A and B are not only monotonically increasing, but steadily increasing. The only pattern in C is that the numbers are always rising; each number, when compared to the previous number, is greater than that previous number. In each case, a pattern at a lower level becomes a constant feature at a higher level. The first derivative - "increase by 1", "increase by 2" - is a constant in A and B. In C, the feature "previous number is less than next number" is a constant.

A modality observing D: 8, 16, 32, 64, 128, 256 should notice that the numbers are constantly increasing, and that the rate of the increase is constantly increasing. A human modality would *not* notice that the numbers formed a doubling sequence - and neither, in all probability, should an AI's modality, unless the sequence is examined by a thought-level process. I say this to emphasize that the problem of modality-level pattern detection is *limited*, in contrast to the problem of understanding patterns in general - if the AI's modality can understand a simple, limited set of patterns, it should be enough.

To notice a pattern is to form an expectation. When this expectation is violated, the pattern is broken. Observing a single quantity changing, as in sequence C, the feature "increasing" remains constant. If C continues but suddenly starts decreasing - 8, 19, 22, 36, 45, 71, 62, 21, 7, 6, 1 - an "edge" has been detected. On a higher level, this is what is observed: "...greater than, greater than, greater than, less than, less than, less than..." Thus the presence of the low-level feature detector for "greater than" or "less than" enables the AI to notice a pattern it could not otherwise

notice, and to detect an edge it could not otherwise see. That is the function of modality-level feature detectors: To enable the discovery of regularities in reality that would otherwise remain hidden.

As a general rule, notice equality, continued equality, and broken equality in the quantity, in the first derivative, and in the second derivative. We notice when a constant quantity changes and when a constant rate of change changes, but we humans do not directly perceive changes in acceleration. We compute the quantity and the quantitative first derivative, but not the quantitative second derivative. Since the second derivative - for humans - is not quantitative but qualitative, we can notice it crossing the zero line, or notice large (order-of-magnitude) changes, but not notice small internal variances. An AI might find it useful to perceive the second derivative quantitatively, but computing a quantitative third derivative (and thus a qualitative fourth derivative) would probably not contribute significantly to intelligence outside of specialized applications.

(54).

3.1.5.3: Saliency of noticed changes

There is still a question of saliency. We would wish a financial AI, or a human accountant, to notice and wonder if a bank account customarily showing transactions measured in hundreds of dollars suddenly began showing transactions measured in millions - the mid-level feature "magnitude", formerly constant at "hundreds", suddenly jumps to "millions". But we wouldn't want to notice a change from the mid-level feature "magnitude: 150-155" to the mid-level feature "magnitude: 153-160", even though - on the surface - both look like equally sharp inequalities. (As a "crystalline" compare operation, "hundreds" != "millions" is neither more nor less unequal than "150-155" != "153-160".) Similarly, we would not notice a change from the mid-level feature "frequency of numbers ending in 5: 20%" to "frequency of numbers ending in 5: 25%"; or, if we did somehow notice, we wouldn't attach as much significance.

We have learned from experience, or from our cultural surroundings, that money is extremely significant, that people often try to tamper with it, and that the order-of-magnitude of monetary quantities should be paid attention to; we have not learned a similar heuristic for shifts in a few dollars, or shifts in percentage frequency of digits, which is why monitoring either quantity is a specialized technique used only by auditors.

Learning which patterns and broken patterns to pay attention to is a concept-level problem; it's not trivial, but "Eurisko-oid" techniques should suffice.

3.1.5.4: Feature extractors for general quantities

These are the feature extractors that can operate on quantities in general:

- Identity
 - Equality and inequality
- Comparators
 - Greater-than and less-than
- Perception of qualitative changes
 - Expected change (within range of observed variance)
 - Fractional change (plus or minus a few percentage points)
 - Significant change (plus or minus a few dozen percentage points)
 - Order-of-magnitude change
- Quantitative computation of first derivative, which is then another quantitative perception
 - Computing a second derivative may sometimes work
 - Computing a third derivative is probably useful only for specialized applications

The *lack* of these simple intuitions is one of the reasons why computer programs look so *stupid* to humans. We always notice when salient quantities change; most programs are incapable of noticing anything at all, unless specifically programmed, and they certainly aren't programmed to notice the general properties of the things they notice. A bank account won't notice if you make one deposit a day, then suddenly make ten deposits in one day, then go back to one deposit a day; it's programmed to handle financial transactions, but not notice patterns in them. Since knowing about a deposit is a high-level perception to a human - one which rises all the way to the level of conscious attention - we automatically compute the basic quantitative perceptions and notice any unexpected equalities or unexpected changes.

On the concept-level, all these features should be computed for all salient high-level quantities, and for all higher-level features rare enough that computing all the features is computationally tractable. Figuring out which features to compute for a quantity, and which features to pay attention to, is a major learning problem for the AI; learning in this area contributes significantly to qualitative intelligence as well as efficiency, since compounding extractors can lead to the computation of entirely new features.

On the modality level, these feature extractors can be composed to yield some basic mid-level features, such as edge detection in pixels, although anything more than that is probably a domain-specific problem. For example, a problem as simple as computing changes in velocity will not fit strictly within the domain of quantitative perceptions, unless the velocity is broken up by domain-specific perceptions into quantitative components of speed and direction.

3.1.6: Trajectories

"Lakoff and Johnson, arguing that our understanding of trajectories is fundamentally based on motor functions, offer this list of the basic elements of a trajectory (quoted from "Philosophy in the Flesh"):

- A trajectory that moves
- A source location (the starting point)
- A target (L&J call it a "goal"), an intended destination of the trajectory
- A route from the source to the target
- The actual trajectory of motion
- The position of the trajectory at a given time
- The direction of the trajectory at that time
- The actual final location of the trajectory, which may or may not be the intended destination

"Trajectory" can also be generalized to any series of changes to a single object, any series of modulations to a state, that takes place over time and has a definite beginning and end; any perception that changes continuously, and smoothly or monotonically enough to be perceived as a trajectory rather than a series of unrelated change-events. (55). The trajectory behaviors - especially trajectories with definite beginnings and ends and directions - intersect planning, which intersects goals, which is a different topic. However, we will discuss intuitions that have "intentional aspects - goal-oriented characteristics - such as force and resistance.

3.1.6.1: Identification of single objects across temporal experiences

The concept of a trajectory can be represented in the temporal XO modality. Zooming out from the following frame, "00000X0000000X000X00000", it could be described as "three points on a line". Given a temporal sequence of XO frames, the points on the line can "move"; they can have position, speed, direction, and velocity.

The XO modality suffices to represent an example of a trajectory, e.g.: "XX000X", "XOX00X", "X00X0X", "X000XX"; an observing human would say that the middle X has moved from the starting point defined by the first X to the endpoint defined by the third X. (Note that I do not yet use the word "target".)

For the sake of form, we should name all the intuitions giving rise to the start-move-endpoint perception. The largest hurdle is the perception of each middle X as an instance of the same continuous object - that is, that the X at position 2 in t_1 , the X at 3 in t_2 , the X at 4 in t_3 , and the X at 5 in t_4 , are all instances of a single object with a continuous existence. A human makes this interpretation immediately because we have built-in assumptions about the continued existence of discrete objects - domain-specific instincts that become visible within a few months after birth.

An AI could probably make the same interpretation, but it would be more difficult. To establish a strongly bound perception of each X as a discrete object and the middle X as a continuous object, it would probably take a trajectory lasting, say, ten frames, instead of four. Assume for the moment that the sequence is expanded to encompass ten frames and ten one-unit steps for the middle X. In this case, the following facts are visible immediately: First, that there are the same number of Xs in each frame. (I will not say "three Xs in each frame", since this implies an understanding of "three".) Second, that each frame has an X in position 1 and an X in position 12. To a human, it is "obvious" that the constant number of Xs implies a constant number of discrete objects; to a human, it is obvious that the three Xs are each different objects; to a human, it is obvious that an X maintaining an identical position in each frame is the same object in each frame; therefore, since the first and last Xs are accounted for, the leftover middle X in each frame must be the third object. And indeed, the "movement" of the third object (or "shift in the positional attribute", as an AI might see it) is incremental and constant.

A tremendous amount of cognition has just flashed by. Getting the AI to perceive two experiences as belonging to the same object is almost as deep a problem as that of getting the AI to perceive two objects as belonging to the same category. Some of the underlying forces are visible in the source code of Hofstadter's "Copycat"; Copycat can see two different letters in two different strings as occupying the same role. (Copycat can also see bonds formed by "movements" in letterspace; it knows that "c" follows "b".) The general rule, however, goes much deeper than this.

Rules of Identification

1. Equality of attributes across experiences, particularly those attributes that remain constant for constant objects, implies equality of identity.
2. Continuous change in an attribute, particularly those attributes that can change without changing the underlying object - such as "position" or "speed" - implies equality of identity.

Rule of Improbability Binding

When two images are equal or very similar, the probability that there is a shared underlying cause behind the equality is proportional to the improbability of a coincidental equality.

The Rule of Improbability implies that, the wider the range of *possible* values for an attribute, the more strongly equality of values *implies* equality of underlying objects. "XOX" binds to "XOX" much more weakly than "roj" binds to "roj". "3" binds to "3" much more weakly than "23,083" binds to "23,083".

Thus, even so basic a task as knowing when two experiences are the "same" object requires that the AI have previously have learned which attributes are good indicators of identity, which in turn requires that the AI have watched over objects known to be identical so that it can observe which attributes remain constant. If this were a seminar on logic we'd be in trouble, but since we're pragmatists we can break the circularity by cheating, just as the human mind does - it seems highly likely that equality of visual signatures and continuous change in position are hardwired into the brain as signals of identity. Similarly, we can start by identifying a few good attributes to begin with, and giving some sample sets with pre-identified objects, and letting the seed AI work it out from there.

What are the consequences of identifying an object?

Rules of Objectification

1. Objects constitute a major source of regularities in reality, and many heuristics - perhaps even modality-level feature extractors - will operate on objects rather than experiences.
2. Objects often continue to exist even when they are not directly experienced, and may require continuous modeling.
3. Objects will often have internal attributes and complex, dynamic internal structure.
4. All nonvisible attributes of an object remain constant across experiences, unless there is a reason to expect them to change. (If the object has intrinsic variability, then the description of the variability remains constant.)

(Author's note: The discussion of objects should probably be somewhere other than 3.1.6: Trajectories, probably the section on categorization, and should have a much longer discussion.)

3.1.6.2: Defining attributes of sources, trajectors, and destinations

In what sense does labeling objects as "sources", "trajectors", and "destinations" - we will not use the term *target* just yet - differ from identifying them as "Object 1", "Object 2", and "Object 3"? In what sense is a "path" different from a "trajectory"? What expectations are implied by the labels, and what experiences are preconditions for using the labels?

Conceptually, a path can exist apart from the traversing objects. If, on multiple occasions, one or more objects is observed to precisely traverse the same path - perhaps at the same speed - then a generalization can be made; an observed feature can be extracted from the single experience and verified to apply across a set of different experiences. To observe the existence of a path is useful only if the observation is reflected in external reality - for example, if the *reason* a rolling ball follows a path down a mountain is because someone dug a trench. A seed AI is unlikely to need to deal with physical trajectories of the type we are familiar with, but the metaphor of "trajectory" extends to the more important modality of source code - a piece of data can follow a path through multiple functions.

Similarly, the conditions that lead us to identify some object or position as "source" is that one or more observed trajectories originate from that source; what leads us to identify a position as "endpoint" is that one or more observed trajectories terminate at that endpoint. What makes the perception of "source" *useful* is if there is a causal *reason* why the position is the source of the trajectory, especially if the object or position is actually *generating* the trajectors - if a pitcher throws a ball, for example; or, in AI terms, if a function outputs pieces of data that then travel through the system. Similarly, the perception of "endpoint" is especially useful if the endpoint actually halts the trajector, or consumes it.

One cue that a real cause may exist - that the perception of a position/object as "source"/"path"/"endpoint" is useful - is if multiple, *varying* paths/trajectories have the same source or endpoint. Imagine that a randomly moving point darts over a screen, and then the movie is played back three times; the fact that the sources and endpoints were identical may not mean that the sources and endpoints have any particular significance; the rest of the path was identical too.

Rule of Variance Binding

Multiple, variant experiences sharing a single higher-level characteristic, but not others, means that the shared characteristic is likely to be significant. Multiple identical experiences can have any number of possible sources; only if at least some properties differ is there a reason to focus on a particular shared characteristic as opposed to others.

Thus, the perception of "source" or "endpoint" exists whenever multiple trajectories share an starting position or ending position, and exists more strongly when multiple *different* trajectories share a source or endpoint but not other characteristics. The perception of "source" and "endpoint" is *useful* when the perception reflects the underlying cause of the initiation or termination of the trajectory.

A "source" or "endpoint" can be any characteristic shared by multiple origins or terminating points, not just position. If the trajectory of a grenade always ends at the location of the blue car, regardless of where the blue car goes, then it's a good guess that someone is trying to blow up the blue car - that the blue car is the endpoint. The greater the variance, the less probability that the covariance is coincidence, and the stronger the binding. The more unique the description of the endpoints - e.g., the blue car was the only car which shared a location with all endpoints, and the green car and the purple car were elsewhere - the stronger the binding. This binding is *predictive* if it can be used to predict the position of the next trajectory termination by reference to the position of the perceived "endpoint", and *manipulative* if moving the perceived "endpoint" can change the trajectories - that is, if you can guess where the grenade will fall by looking at the blue car, and make the grenade fall in a particular place by driving the blue car there. If the binding is strong enough, the endpoint may deserve the name of "target" (see below).

Finally, it is noteworthy that "source" and "endpoint" do not necessarily imply that the trajector goes into and out of existence. Any interval which bounds the trajectory, or any conditions which bound the trajectory, or any sharp changes within the trajectory, may make salient the location of the trajector during the boundary change. (To perform the computational operations which check multiple trajectories for binding of sources or endpoints, it is necessary that the source and endpoint be salient - salient enough that the additional processing is performed which discovers the binding.)

3.1.6.3: Source, path, target; impulse, correction, resistance, and forcefulness

When defining what it means to take the *intentional stance* with respect to a system, the archetypal example given is usually that of the thermostat. A thermostat turns on a cooling system when the temperature rises above a certain point, and turns on a heating system when the temperature falls below a certain point. A thermostat behaves as though it "wants" the temperature to stay within a certain range; as if the thermostat had a goal state and deliberately resisted alterations to that goal state. In reality, a thermostat possesses no model of reality whatsoever, but we may still find it convenient to speak of the thermostat's *behavior* as goal-oriented or "intentional".

To describe a trajectory using the terms source, path, and *target*, the trajector's arrival at the target must be non-coincidental. If the trajector is continuously propelled, then use of the word "target" usually implies that the trajector's path is self-correcting - that if an *impulse* is applied which causes the trajector to depart from the path, a *correction* (originating from inside or outside the trajector) will correct the trajectory so that the trajector continues to approach the goal state. A trajector typically approaches the target such that the distance between trajector and target tends to decrease continuously, in spite of any interfering impulses. (This is not always true, particularly in cases where the "trajector" actually *is* an intelligent or semi-intelligent entity capable of taking the long way around, but you get the idea.) In a slightly different usage of the word "target", the trajector moves at a constant and unalterable velocity, but tends to hit the target - or at least come close to it - because the trajector was *aimed*. (Which is how "aiming is defined".) (Author's note: Expand this area.)

Resistance is the name given to an "obstacle" on the way to the target or goal state. The perception of "resistance" arises when we observe a trajector hit some type of barrier and bounce, or slow down, or be pushed back. The implication is that the trajector has not merely encountered some random impulse, but that there are specific forces preventing the achievement of a specific goal state or subgoal state.

Forcefulness is the ability to overcome resistance. The perception of "forcefulness" - force that, to humans, is viscerally impressive - arises when we see the trajector applying additional forces to overcome resistance.

All of this applies, not just to actual moving objects, but to goals in general; to the higher-level metaphor *similarity is closeness*. The idea of "closeness" does not apply only to two °quantitative attributes, but also to two structures built from a number of °qualitative attributes. If, over time, the qualitative attributes of the first structure are one by one adjusted so that they match the corresponding attributes of the second structure, then the first structure is "approaching" the second.

Mathematically, we might say that one point is approaching a second in the multi-dimensional °phase space defined by the qualitative attributes, but this is being overly literal. The perception of similarity is *useful* when two objects being more similar means that the two objects are more likely to behave similarly. The similarity-is-closeness metaphor is useful and *manipulative* when two objects being "closer" means that less additional work is required to make them match completely - one object has become closer to the *target* represented by the other.

Use of the term "close" to mean "similar" is an astonishingly general metaphor. "Close" is used to describe almost any object, event, or situation that can "approach" a goal state. "Approach" is used as a metaphor to describe goals in general.

The ultimate underpinning of this metaphor, in humans, may actually be the human *emotional* state of *tension*. We feel tension as we watch something approach a goal; tension rises as the goal comes closer and closer... The same rising tension applies when we watch a trajector approach a target. The closer the approach, the sharper our attention, the more we're on the lookout for something that might go wrong at the last second. The metaphor between spatial closeness and generalized similarity is probably a shadow of the much stronger metaphor between approaching a target and approaching a goal.

Generally speaking, it's a bad idea to weigh down an AI with slavish imitations of human emotions. It may not even be necessary to duplicate the metaphor; I'm not all that sure that the space-to-similarity metaphor contributes to intelligence. It does seem likely that the AI will either experience (or learn) some type of heightened attention as events approach a goal state.

For we humans, who inhabit a physical world, trying to make an object achieve a certain *position* is one of the most common goal states; position is one of the attributes that is most commonly manipulated to reach a goal state. Indeed, we might be said to instinctively apply the metaphor *state is position*. Perhaps the AI will learn a similar set of extensive metaphors for source code.

There should probably be some type of modality-level support that indicates the feeling of approaching a goal, so that the concept of "approaching a goal" lies very close to the surface, and generalizations across tasks and modalities are easy to notice. The idea of "approach" is an opening wedge, a way to split reality along lines that reveal important regularities; the behavior of the "trajectory" towards the goal in one task is often usefully similar to the behavior of trajectories in other tasks.

Version History

May 18, 2001: GISAI 2.3.02. Split the original document, "Coding a Transhuman AI", into General Intelligence and Seed AI and Creating Friendly AI. Minor assorted bugfixes. GISAI now 349K.

Apr 24, 2001: GISAI 2.3.01. Uploaded printable version. Some minor suggested bugfixes. Removed most mentions of the phrase "Eliezer Yudkowsky" to make it clearer that GISAI is a publication of the Singularity Institute.

Apr 18, 2001: GISAI 2.3.0. (This version number previously reflected the addition of Creating Friendly AI, which later became a separate document.) Changed copyright to "2001" and "Singularity Institute" instead of legacy "2000" and "Eliezer Yudkowsky". Uploaded multi-page version.

Sep 7, 2000: GISAI 2.2.0. Added 3.1: Time and Linearity and Interlude: The Consensus and the Veil of Maya. Uploaded old bugfixes. 358K.

Jun 25, 2000: GISAI 2.1.0. Added Appendix A: Glossary and Version History. Much editing, rewriting, and wordsmithing. 220K. Not published.

May 18, 2000: GISAI 2.0a. *General Intelligence and Seed AI* was originally known as *Coding a Transhuman AI*. As the Singularity Institute did not yet exist at that time, CaTAI was then copyrighted by Eliezer S. Yudkowsky. 180K.

Appendix A: Glossary

NOTE: If a referenced item does not appear in this glossary, it may be defined in *Creating Friendly AI*.

- affector
- API
- atto
- Bayesian binding
- Bayesian Probability Theorem
- cache
- CFAI
- codelet
- computational horizon
- computational temperature
- Consensus
- continuous
- Copycat
- counterfactual
- crystalline
- cytoarchitecture
- declarative
- Deep Blue
- discrete
- e.g.
- Eurisko
- exa
- femto
- GEB
- gender-neutral pronouns
- giga
- GISAI
- granularity horizon
- hertz
- heuristic
- holism
- horizon
- i.e.
- iff
- instantiate
- intelligence
- intentionality
- Kasparov
- kilo
- Lakoff and Johnson
- latency
- Law of Pragmatism
- Life
- LISP tokens
- Marr
- mega
- micro
- microworld
- milli
- mindstuff
- MITECS
- nano
- Necessary, But Not Sufficient
- ontology

- ontotechnology
- orthogonal
- past light cone
- peta
- Physicist's Paradigm
- pico
- predictive horizon
- procedural
- Q.E.D.
- qualia
- qualitative
- quantitative
- reductholism
- reductionism
- reflection
- relevance horizon
- RNUI
- salience
- scalar
- search trees
- seed AI
- sensory modality
- SIAI
- space of simultaneity
- SPDM
- stochastic
- structural
- subjunctive
- tera
- three
- time
- Turing-computability
- ve
- ver
- verself
- vis
- world-model

- **affector:**

To be defined in *Unimplemented section: Causality*. An "affector" is a factor, something that affects things, something that has effects. Subtly different from describing something as a *cause* or as a *factor*; somewhere between the two. The term is useful, in that I often find no other term has the exact connotation I want to use. To say that A *causes* B is to say that A *completely* accounts for B. A can *affect* B without completely accounting for B. Also, to describe something as a "cause" is generally to describe it as an *intermediate* point on a causal chain; that is, a cause is a combination of effect and affector; to describe something as an "affector" is to look only forward, rather than looking both forward and backward. "Cause" is often overloaded with other meanings and does not have the precision of "affector". Finally, "cause" is a noun and a verb where "affector" is clearly a noun, a difference of terminology which subtly affects the way we think about causality. So please excuse the jargon.

- **API:**

Application Programming Interface. The membrane that separates libraries from the programmer. A set of functions, objects, methods, properties, formats, and data structures that a programmer can use to communicate with an operating system, a commercial code library, an Internet browser, et cetera. Essentially, an API is the structure of the inputs and outputs of a system.

- **atto:**

10⁻¹⁸. One-quintillionth; a thousandth of a "femto". See the Hacker's Dictionary for the full list of quantifiers.

- **Bayesian**

The strength of the binding between a piece of information and the conclusion derived from it, under the Bayesian Probability Theorem. The more improbable a given pattern is, the less likely that the pattern was produced by a pure coincidence. Suppose that you flip coin A thirty times, and then flip coin B thirty times, and the pattern of heads and tails matches exactly. Since this is a billion-to-one improbability for fair, independent coins, there is a very strong *Bayesian binding* between the two patterns, leading an observer to conclude that there is almost certainly some cause tying the two coins together.

In discussions of *meta-rationality*, the Bayesian Probability Theorem is used to estimate the strength of the binding between your beliefs and reality - i.e., the extent to which the fact that you believe X licenses you to conclude that X is true. Suppose that I observe myself to say, "The sky is green." If I know that I believe in the greenness of the sky so strongly that I would declare the sky green even though it were purest sapphire blue, then my observing myself to say "The sky is green" says nothing - according to the BPT - about the actual

binding:

likelihood that the sky is green. This effect - that I will see myself believing "The sky is green" - is predicted in both the groups where the sky is green and the groups where the sky is blue; thus the observation does nothing to indicate which group the actual sky falls into. If, on the other hand, I don't care much about the color of the sky, then I am only likely to say that the sky is green if it's actually green, and my observing myself to make this statement is strong evidence in favor of the greenness of the sky.

What was that about faith?

- **Bayesian Probability Theorem:**
The Bayesian Probability Theorem relates observed effects to the *a priori* probabilities of those effects in order to estimate the probabilities of underlying causes. For example, suppose you know the following: 1% of the population has cancer. The probability of a false negative, on a cancer test, is 2%. The probability of a false positive, on a cancer test, is 10%. Your test comes up positive. What is the probability that you have cancer?
The instinctive human reaction is terror. After all, the probability of a false positive is only 10%; isn't the probability that you have cancer therefore 90%?
The Bayesian Probability Theorem demonstrates why this reasoning is flawed. In a group of 10,000 people, 100 will have cancer and 9,900 will not have cancer. If cancer tests are administered to the 10,000 people, four groups will result. First, a group of 8,910 people who do not have cancer and who have a negative test result. Second, a group of 990 who do not have cancer and who have a positive test result. Third, a group of 2 who have cancer and who have a negative test result. Fourth, a group of 98 who have cancer and who have a positive test result.
Before you take the test, you might belong to any of the four groups; the Bayesian Probability Theorem says that your probability of having cancer is equal to $(2 + 98)/(8,910 + 990 + 2 + 98)$, $1/100$ or 1%. If your test comes up positive, it is now known that you belong to either group 2 or group 4. Your probability of having cancer is $(98)/(990 + 98)$, $49/544$ or approximately 9%. If your test comes up negative, it is known that you belong to either group 1 or group 3; your probability of having cancer is $2/8,912$ or around .02%.
Colloquially, the good Reverend Bayes is invoked wherever prior probabilities have a major influence on the outcome of a question. However, the Bayesian Probability Theorem has a *much* wider range of application - in normative reasoning, the BPT controls the binding between *all* sensory information and *all* beliefs. Normative reasoners are often called "Bayesian reasoners" for this reason. The Bayesian Probability Theorem is so ubiquitous and so intricate that I would cite it as one of the very, very few counterexamples to the "Law of Pragmatism" - in CFAI 3.1.4: Bayesian reinforcement, for example, I discuss how some of the functionality of pain and pleasure, though implemented in a separate hardwired system in humans, could emerge directly from the BPT in normative reasoners (i.e., in "Friendly AIs").
- **cache:**
To "cache" a result is to store it for later use. For example, your Web browser has a "cache folder" containing files that you have already downloaded from the Internet; when your browser encounters a URL that it's already followed, it can retrieve the file from the cache folder instead of downloading it again.
- **CFAI:**
An abbreviation used for Creating Friendly AI, a publication of the Singularity Institute for Artificial Intelligence. Located at <http://singinst.org/CFAI/index.html>
- **codelet:**
A free-standing piece of code - a piece of code that can be detached or reattached, or a piece of code that can float independently. Hofstadter and Mitchell's "Copycat" uses a codelet-based architecture to implement perception; bond-forming codelets are dumped into the Workspace to form bonds. If the "computational temperature rises, bond-breaker codelets are dumped into the Workspace.
The concept of a codelet doesn't necessarily imply the free-floating status of a "daemon", or the programmatic individualism of an "agent" (two (flawed) concepts from traditional AI). A codelet is to a function what an object is to a data structure. A codelet is a function regarded as a thing in its own right, with its own properties and characteristics and behaviors, with the freedom to move around or be moved around.
- **computational horizon:**
The scope of a problem. The amount of computing power that needs to be devoted to a task. Outside the "horizon" lie all the facts that are not relevant to the task, all the details that are too fine to be processed with the available computing power, all the consequences that are too hazy or too unimportant to predict, and so on. Deciding where the computational horizon lies often has a major impact on the quality and speed of a cognitive task. See "predictive horizon", "granularity horizon", and "relevance horizon".
- **computational temperature:**
A technique used in Hofstadter and Mitchell's "Copycat" (an AI that solves analogy problems) to control the degree of randomness and flexibility in the system. When the computational temperature is high, bonds break and form easily; avenues of exploration are selected on a more random basis. When the computational temperature is low, bonds don't break easily and only the better possibilities are explored. In Copycat, the computational temperature is linked to the goodness of the current cognitive structures. Elegant correspondences lower the temperature; conflicts raise the computational temperature.
Thus, for example, Copycat may begin by seeing a plausible set of bonds. This drops the computational temperature. More perceptions are built up, and the computational temperature keeps dropping, until a conflict is detected - for example, an extra correspondence, or lack of correspondence, that breaks a one-to-one mapping. This raises the computational temperature; the offending perceptual structures - bonds, groups, correspondences - dissolve; and a new set of structures, hopefully more elegant, is given the opportunity to form.
An interesting side effect - besides the fact that Copycat can successfully solve analogy problems - is that the final computational temperature provides a good measure of the *elegance* of Copycat's answer. Low computational temperatures in Copycat correspond fairly well to the analogy-problem answers that human observers would regard as elegant.
In some ways, Copycat's computational temperature may be the first true artificial emotion.
- **Consensus:** Defined in Interlude: The Consensus and the Veil of Maya.
A term for the perceptions that are shared by all of humanity but which are not identical with external reality. The color purple, the bands in a rainbow, and social/political perceptions are extreme examples, but none of our perceptions are precisely identical with external reality. Nobody actually lives in external reality, and we couldn't understand it if we did; too many quarks flying around. When we walk down a hall, watch the floor and walls and ceiling moving past us, we're actually walking around inside our visual cortex. Despite this, the Consensus usually has an extremely tight "sensory", "predictive", and "manipulative binding to external reality, so the rules are just as strict.
If an object or description is part of the Consensus rather than external reality, that doesn't mean the description is arbitrary. It does mean that you're likely have trouble defining the description mathematically, or coming up with a firm philosophical grounding, or otherwise inventing a definition that works 100% of the time. 99.99% is almost always good enough.

- **continuous:**
Can be divided and subdivided indefinitely. Contrast to °discrete.
- **Copycat:** Defined in 2.3: Concepts.
An AI, written by Melanie Mitchell and conceived by Douglas R. Hofstadter, which tries to solve analogy problems in the microdomain of letter-strings.

	If	'abc'	goes	to	'abd'	For	what	does	'bcd'	go	to?
	If	'abc'	goes	to	'abd'		what	does	'pqrs'	go	to?
	If	'abc'	goes	to	'abd'		what	does	'vuts'	go	to?
	If	'abc'	goes	to	'bcd'		what	does	'pqrs'	go	to?
	If	'abc'	goes	to	'bcd'		what	does	'ace'	go	to?
	If	'abc'	goes	to	'bcd'		what	does	'lwmb'	go	to?

If 'abc' goes to 'abd', what does 'xyz' go to? (Bear in mind that Copycat has no concept for "circularity".)
Copycat solves these problems through a *perceptual* architecture. It mentally builds up a structure of bonds, groups, correspondences, concept-mappings, until it has a "rule" for the first transition that can be applied to the second transition. If the rule can't apply, or can't apply well, then new cognitive pressures come into play, breaking down previously perceived perceptions and allowing new groups and correspondences and rules to form.
Note that Copycat doesn't select the correct answer from a list of alternatives; it actually °invents the answer, which is very impressive and very rare. Copycat is a really fascinating AI, and you can read about it in *Metamagical Themas*, or read the source code (it's a good read, and available as plain text online - no decompression required).
- **counterfactual:**
A what-if scenario deliberately contrary to reality; e.g. "What if I *hadn't* dropped that glass of milk?" See also °subjunctive.
- **crystalline:** Defined in 1.1: Seed AI.
Loosely speaking, "crystalline" is the opposite of "rich" or "organic". If vast loads of meaning rest on the shoulders of individual computational tokens, so that a single error can break the system, it's crystalline. "Crystalline" systems are the opposite of "rich" or "organic" error-tolerant systems, such as biological neural networks or seed-AI mindstuff. Error-tolerance leads to the ability to mutate; mutation leads to evolution; evolution leads to rich complexity - networks or mindstuff with lots of tentacles and connections, computational methods with multiple pathways to success.
- **cytoarchitecture:**
"Cytoarchitecture" refers to the general way neurons connect up in a given lump of neuroanatomy - many-to-many, many-to-one, and so on.
- **declarative:**
The distinction between "procedural" and "declarative" knowledge/skill/information is one of the hallowed dichotomies of traditional AI. Although the boundary isn't as sharp as it's usually held to be - especially in seed AI - the distinction is often worth making.
Your knowledge of how to walk is "procedural" and is stored in procedural form. You don't say to yourself: "Right leg, left leg, right leg, left leg." All the knowledge about how to balance and maintain momentum isn't stored as conscious, abstract, *declarative* thought in your frontal lobes; it's stored as unconscious *procedural* thought in your cerebellum and spinal cord. (Neurological stereotyping included for irony.)
Inside source code, the procedural/declarative distinction is even sharper. A piece of code that turns on the heat when the temperature reaches 72 and turns on the air conditioning when the temperature reaches 74, where 72 and 74 are hard-coded constants, has *procedurally* stored the "correct" temperature of 73. The number "73" may not even appear in the program.
A piece of code that looks up the "target temperature" (which happens to be 73) and twiddles heat or A/C to maintain that temperature has *declaratively* stored the number 73, but has still *procedurally* stored the method for maintaining a temperature. It will be easy for the programmer - or for internally generated programs - to refer to, and modify, the "target temperature". However, the program still doesn't necessarily know how it's maintaining the temperature. It may not be able to predict that the heat will go on if the temperature reaches 72. It may not even know that there's such a thing as "heat" or "air conditioning". All that knowledge is stored in procedural form - as code which maintains a temperature.
In general, procedural data is data that's opaque to the program, and declarative data is data that the program can focus on and reason about and modify. Seed AIs blur the boundary by analyzing their own source code, but this doesn't change the basic programmatic truth that declarative=good and procedural=bad.
- **Deep** **Blue:**
The chess-playing device that finally beat the human champion, °Kasparov. A great, glorified search tree that beat Kasparov essentially through brute force, examining two billion moves per second. Built by IBM.
- **discrete:**
Composed of a finite number of parts which cannot be further divided. Contrast to °continuous.
- **e.g.:**
Exempli gratia; Latin, "for example".
- **Eurisko:**
Eurisko was the first truly self-enhancing AI, created by Douglas B. Lenat. Eurisko's mindstuff - in fact, most of the AI - was composed of °heuristics. Heuristics could modify heuristics, including the heuristics which modified heuristics.
I've never been able to find a copy of Eurisko's source code, but, by grace of Jakob Mentor, I have obtained a copy of Lenat's original papers. It turns out that Eurisko's "heuristics" were arbitrary pieces of LISP code. Eurisko could modify heuristics because it possessed "heuristics" which acted by splicing, modifying, or composing - in short, mutating - pieces of LISP code. Many times this would result in a new "heuristic" which caused a LISP exception, but Eurisko would simply discard the failed heuristic and continue. In a sense, Eurisko was the first attempt at a seed AI - although it was far from truly self-swallowing, possessed no general intelligence, and was created from °crystalline components.
Engines of Creation (by K. Eric Drexler) contains some discussion of Eurisko's accomplishments.
- **exa:**
10¹⁸. One billion billion; a thousand "°peta". See the Hacker's Dictionary for the full list of quantifiers.
- **femto:**
10⁻¹⁵. One-quadrillionth; a thousandth of a "°pico". See the Hacker's Dictionary for the full list of quantifiers.
- **GEB:**
Gödel, Escher, Bach: An Eternal Golden Braid by Douglas R. Hofstadter. This book is *mandatory* reading for all members of the human species.

- gender-neutral**

°Ve, °vis, °ver, °verself. I was forced to start using gender-neutral pronouns when referring to intelligent AIs, since to use "he" or "she" would imply cognitive hardware that such an AI would very specifically *not* have.

I realize that these pronouns strike people as annoying the first time around. I'm sorry for that, and I truly regret having to annoy my readers, but "it" is simply inadequate to refer to AIs. Not only is "it" used as a pronoun for inanimate matter, but "it" is also a general anaphor, like "this" or "that". "It" can refer to anything at all in a sentence, not just the AI, so complex sentences - especially ones that use "it" for other purposes - become impossible to parse syntactically. Sometimes a sentence can be rewritten so that no pronoun is necessary, but for sentences with multiple pronoun references, this rapidly becomes either impossible, or too tangled. I would rather use unusual words than tangled syntax. At least "ve" gets easier to parse with time.

At one point I was using "ve" to refer to a human of indefinite gender, but I have since realized that this is just as inaccurate as referring to an AI as "he" or "she". I now keep a coin near my computer that I flip to decide whether a human is "he" or "she". (No, I haven't fallen into the bottomless pit of political correctness. Everyone has the right to use whatever language they like, and I can flip a coin if I want to. Your right to use "he" implies my right to flip a coin. Right?)
- giga:**

10^9 . One billion; a thousand "mega". See the Hacker's Dictionary for the full list of quantifiers.
- GISAI:**

"General Intelligence and Seed AI". An abbreviation for this document. Permanent location: <http://singinst.org/GISAI/index.html>
- granularity**

horizon:

The fineness of the detail that needs to be modeled. How much °reductionism needs to be applied to capture all the relevant details. The tradeoff between expenditure of computing power, and the benefits to be gained from finer modeling.

Every time your eye moves, the amount of processing power being devoted to each part of the visual field changes dramatically. ("The cortical magnification factor in primates is approximately inversely linear, at least for the central twenty degrees of field." "It has been estimated that a constant-resolution version of visual cortex, were it to retain the full human visual field and maximal human visual resolution, would require roughly 10^4 as many cells as our actual cortex (and would weigh, by inference, roughly 15,000 pounds).") °MITECS, "Computational Neuroanatomy".) And yet we can watch an object rotating, so that different parts move all over the visual cortex, and it doesn't appear to distort.

Every time you change scale or the level of detail at the °modality level of representation, the data may wind up going into essentially a different format - at least, from the perspective of someone trying to detect identity by bitwise comparison. Even adding or subtracting pieces of the puzzle, without changing scale, can be a problem if the AI has painstakingly built up a perceptual tower that doesn't take well to tampering with the foundations. Cognitive methods need to be able to take these kinds of random pushes and shoves. "Error-tolerance" isn't just important because of actual *errors*, but because all kinds of little flaws naturally build up in a cognitive task, as the result of cognition.

See °computational horizon.
- hertz:**

A measure of frequency, equal to 1 cycle per second. A neuron that fires 200 times per second is operating at 200 Hz. CPU clock speeds are currently measured in °megahertz (MHz) or °gigahertz (GHz).
- heuristic:**

I use the term to refer to any piece of knowledge which provides a rule of thumb - anything from "Don't rest your hand on a hot stove" to "Try to control the center of the chessboard".

Some other definitions: Douglas Lenat once wrote an AI called °Eurisko in which the mindstuff - in fact, practically the entire AI - was composed of "heuristics" which could modify other heuristics, including the heuristics doing the modifying. For example, "investigate extreme cases" was modified by a heuristic to yield "investigate cases *close* to extremes". (Douglas Lenat went on to state that "Heuristics are compiled hindsight; they are judgemental rules which, if only we'd had them earlier, would have enabled us to reach our present state of achievement more rapidly.") In classical AI, a "heuristic" is usually a function used to prune °search trees by indicating branches which are likely or unlikely to be desirable.
- holism:**

Holism: The attitude that the whole is greater than the sum of the parts. To take the holistic view is to look upward, focus on the high-level properties. See °reductionism and °reductholism. See also Gödel, Escher, Bach: An Eternal Golden Braid, particularly the dialogues "Prelude" and "Ant Fugue".

"No one in his right mind could deny holism." -- °GEB
- horizon:**

See °computational horizon.
- i.e.:**

Id est; Latin for "that is"; usually used to mean "in other words".
- iff:**

"Iff" is shorthand for "if-and-only-if".
- instantiate:**

Loosely, program A "instantiates" program B if it can perfectly simulate program B. An "instantiation" of a program is a running copy of that program.

This issue actually gets waaay more complicated, but I'm not going to inflict that on you now. Maybe later. Nobody has ever come up with a mathematical definition of "instantiation", but it's a useful concept.
- intelligence:**

Defined in 2.1: World-model.

What is intelligence? In the case of humans, intelligence *is* a brain with around 40 billion neurons, and 104 °cytoarchitecturally distinct areas in the cerebral cortex alone. What intelligence *is* is the subject of this whole web page.

The *cause* of intelligence can be more succinctly described: Evolution is the cause of intelligence, and intelligence is an evolutionary advantage because it enables us to model, predict, and manipulate reality. Or rather, it enables us to model, predict, and manipulate *regularities* in reality.
- intentionality:**

Defined in 3.1: Time and Linearity.

Behaving in such a way as to give rise to the appearance of deliberate, goal-oriented behavior. The classic example is the corrective action of a thermostat; by switching on the heat when the temperature drops below a certain point, or switching on the air-conditioning when the temperature rises above a certain point, a thermostat gives the appearance of "wanting" the temperature to stay in a certain range. We can take the *design stance* and talk about cause and effect in sensors and circuits, or take the *physical stance* and talk about

the underlying atoms, but it's usually most convenient to take the *intentional stance* and say that the thermostat maintains a certain temperature.

- **Kasparov:**
Gary Kasparov, the human who finally lost the world chess championship to °Deep Blue.
- **kilo:**
10³. One thousand. See the Hacker's Dictionary for the full list of quantifiers.
- **Lakoff** and **Johnson:**
George Lakoff and Mark Johnson, coauthors of *Metaphors We Live By* and *Philosophy in the Flesh*. George Lakoff is also the author of *Women, Fire, and Dangerous Things*, a book about cognitive categories.
- **latency:**
Latency describes delays, specifically delays introduced by *communication* rather than local processing, and *irreducible* delays rather than delays caused by sending a large amount of data. Most commonly used in discussion of computer networks and hardware systems. The *latency* on a motherboard is, for example, the time it takes a message from the CPU to reach a video card. The *latency* between nodes of a network is the time it takes for a message from node A to reach node B. For a fine explanation of why "latency" is entirely distinct from "bandwidth" - adding an identical second channel doubles bandwidth but does not affect latency at all - see "It's the Latency, Stupid".
Note that our Universe specifies that the minimum latency between two nodes will be at least one second for every 186,000 miles. The latency between two nodes 300 microns apart must be at least one picosecond.
Anders Sandberg, in "The Physics of Information Processing Superobjects: Daily Life Among the Jupiter Brains", suggests the measure *S* of the "diameter" of a single mind, where *S* is the ratio of the clock speed to the latency between computing elements. (*S* = distance / (clock speed * message speed)). Anders goes on to note that the human brain has *S* ~ 1 (56). An oft-voiced conjecture is that the subjective "feel" of having a single, unified mind may require *S* <= 1. (As far as I know, this conjecture is only applied to °superintelligences, and nobody has suggested that *S* played a role in shaping human neurology.)
- **Law of Pragmatism:** Defined in 1.2: Thinking About AI.
Any form of cognition which can be mathematically formalized, or which has a provably correct implementation, is too simple to contribute materially to intelligence.
- **Life:**
So, you want to know the meaning of life, eh?
When capitalized, "Life" usually refers to Conway's Game of Life, a two-dimensional cellular automaton. Cells are laid out in a square grid, and cells can either be alive or dead. Each cell is affected only by the eight cells around it. With each tick, these rules are applied to each cell:
1: A cell with fewer than two living partners becomes or remains dead.
2: A cell with two living partners maintains its current state.
3: A cell with three living partners becomes or remains alive.
4: A cell with four or more living partners becomes or remains dead.
These rules are enough to generate almost endless variance.
As for the age-old controversy among biologists about how to define "life", I suggest the following: "Life is anything designed primarily by evolution, plus anything that counts as a person." Note that this definition includes mules, excludes computer viruses, includes biological viruses, and (by special clause) includes designed minds smart enough to count as people.
- **LISP** tokens:
LISP is a programming language, the traditional language of AI.
"LISP" stands for "List Processor". In LISP, everything is made from lists, including the code. For example, a piece of code that adds 2 and 2 would be (plus 2 2). This code is a list composed of three tokens: plus, 2, and 2. If "num1" was a LISP token that contained the value of 2, the code could be written (plus num1 2) and would return 4.
When I say that classical AI is built from suggestively-named LISP tokens, I mean that the classical AI contains a data structure reading ((is(food hamburger)) (is(eater human)) (can-eat (eater food))); the classical AI then deduces that a "human" can "eat" a "hamburger", and this is supposed to be actual knowledge about hamburgers and eating. What the AI really knows is that a G0122 can H8911 a G8733. Drew McDermott pointed this out in a famous article called "Artificial Intelligence Meets Natural Stupidity".
(LISP does have one real property which is important to AI, however; the code and the data structures follow the same format, making LISP the single premier language for self-modifying code. A true AI would probably read C++ as easily as LISP, since the amount of complexity needed to parse code is comparatively trivial relative to the amount of cognitive complexity needed to *understand* code. Even so, using a language well-suited to self-modification may simplify the initial stages of the AI where self-improvement is mostly blind. Since LISP is getting ancient as programming languages go, and linked lists are awkward by today's standards, I've proposed a replacement for LISP called "Flare", which (among many other improvements) would use XML instead of linked lists. Even so, of course, putting faith in the token level of Flare would be no better than putting faith in the token level of LISP. At most, Flare might be well-suited to programming sensory modalities and mindstuff. It would be nice to have such a language, since none of the existing languages are really suited to AI, but it's more likely that we'll just hack something up out of Python - during the initial stages, at least. For more about Flare, see the obsolete document *The Plan to Singularity*.)
- **Marr:**
David Marr pioneered the field of computational neurology - in particular, the computational theory of vision. It is almost impossible to convey the magnitude of Marr's contribution to AI. Marr was the first person who *did the work* and *wrote the code* necessary to embody a piece of cognition. He was the first person to propose a theory of AI that wasn't oversimplified to the point of caricature - the first theory to correctly identify the token level of processing. Reading a document like *General Intelligence and Seed AI*, which is essentially *built* on top of David Marr's paradigms, it's hard to convey what the field was like before him.
David Marr died of leukemia in 1980, at the age of thirty-five.
- **mega:**
10⁶. One million; a thousand ""kilo". See the Hacker's Dictionary for the full list of quantifiers.
- **micro:**
10⁻⁶. One-millionth; a thousandth of a ""milli". See the Hacker's Dictionary for the full list of quantifiers.
- **microworld:** Defined in 2.1: World-model.
A microworld is a virtual environment in which the AI can learn and grow. Programmatically, such a world would consist of a simulated external environment, plus an AI watching the simulated external environment through a simulated camera or some other kind of simulated sensory system, and possibly altering the simulated external environment through some kind of simulated fingers, simulated cue sticks (for a billiard-ball world), or other simulated manipulators. Henceforth I will leave out the word "simulated" - the AI is living in a

very real external environment, just one that happens to be implemented on transistors instead of quarks. (Actually, "implemented on transistors *as well as* quarks" might be a better way of putting it.) The environment, to us, is "inside the computer", but it is still outside the AI.

- **milli:**
10⁻³. One-thousandth. See the Hacker's Dictionary for the full list of quantifiers.
- **mindstuff:** Defined in Executive Summary and Introduction.
Mindstuff is the basic substrate from which the AI's permanently stored cognitive objects (and particularly the AI's concepts) are constructed. If a cognitive architecture is a structure of pipes, then mindstuff is the liquid flowing through the pipes.
The mindstuff of classical AI is °suggestively named LISP tokens. The mindstuff of connectionist AI is neurons (neuroids, rather) plus the neuroid learning behaviors created by the training algorithms. Of course, the comparison is deceptive, since neither classical nor connectionist AI have a lower modality layer or a higher thought layer.
Insofar as seed AI has a "stuff that concepts are made of", it might be described as °reducthologic multilevel descriptions subject to conscious and autonomic manipulation", with the ultimate substrate probably being interpreted source code (in the early stages) or AI-coded assembly language (in the later stages).
- **MITECS:**
The MIT Encyclopedia of the Cognitive Sciences. Wilson and Keil, 1999. A truly excellent book containing 471 short articles about topics in the cognitive sciences. See also my review in the Bookshelf.
- **nano:**
10⁻⁹. One-billionth; a thousandth of a °micro". See the Hacker's Dictionary for the full list of quantifiers.
- **Necessary, But Not Sufficient:** Defined in 1.2: Thinking About AI.
A design feature may be Necessary for intelligence, but that does not mean it is Sufficient. The latest fad AI may use "the same parallel architecture found in the human brain", but it will also use the same parallel architecture found in an earthworm's brain.
- **ontology:**
The basic level of reality. Our ontology involves quarks, spacetime, and probability amplitudes. The ontology of a °Life game consists of dead cells, live cells, and the cellular-automaton rules. The ontology of a Turing machine is the state transition diagram, the read/write head, and an infinitely long tape with ones and zeroes written on it.
- **ontotechnology:**
Ontotechnology is what you move on to when you're bored with nanotechnology and megascale spacetime engineering. "Ontotechnology" is the art of meddling with underlying reality, and refers to technologies that change the laws of physics, create new Universes, or - this is the best part - *change the kind of things that can be real*. Quarks are real, which we understand. If you believe that °qualia are °objectively real - I do, but in my personal capacity, not my °SAI capacity - then thoughts and experiences and perceptions can also apparently be made °objectively real. If we can create qualia, why not make *other* things real too? Why not turn the laws of physics into material substances that can be manipulated directly? Heck, you could even tamper with the First Cause and make the whole of Reality go out like a candle!
I invented the idea of ontotechnology. Did you guess?
- **orthogonal:**
A mathematical term; in geometry, it means perpendicular. Colloquially, two variables that can change independently of each other; not necessarily mutually irrelevant, but decoupled. See also the entry in the Hacker's Dictionary.
- **past light cone:**
The set of all events in causal contact with a given spacetime point. The past light cone is the space of all events from which a ray of light could have reached the current event. The future light cone is the space of all events that can be reached by a ray of light from the current event. Any event occurring outside our past light cone can have no causal impact on this moment.
- **peta:**
10¹⁵. One million billion; a thousand °tera". See the Hacker's Dictionary for the full list of quantifiers.
- **Physicist's Paradigm:** Defined in 1.2: Thinking About AI.
The idea that advances in AI will be characterized by the discovery of a single bright idea that explains everything yet fits on a T-Shirt; e.g. "Physical Symbol Systems", "expert systems", "parallelism", "neural networks". Confusing the task of a physicist, who "explains" a skyscraper by reference to molecular dynamics, with the task of the engineer who must design that skyscraper.
- **pico:**
10⁻¹². One-trillionth; a thousandth of a °nano". See the Hacker's Dictionary for the full list of quantifiers.
- **predictive horizon:**
How far into the future the consequences of an event need to be projected. The amount of computing power devoted to projecting the consequences of an action. In °Friendly AI, the °computational horizon for disaster-checking.
Humans seem to do very well at recognizing the need to check for *global* consequences by perceiving *local* features of an action. It remains to be seen whether this characteristic of 10¹⁴ x 200Hz synapses can be duplicated in N 2GHz CPUs. See CFAI 3.2.2: Layered mistake detection.
- **procedural:**
See °declarative.
- **Q.E.D.:**
Quod erat demonstrandum; Latin for "So there!"
- **qualia:**
The substance of conscious experience. "Qualia" is the technical term that describes the *redness of red*, the mysterious, indescribable, apparently irreducible quality of *redness* that exists above and beyond a particular frequency of light. If a JPEG viewer stores a set of red pixels, pixels with color 0xFF0000, does it see red the way we do? No. Even if a program simulated all the feature-extraction of the human visual °modality, would it actually *see red*?
I first "got" the concept of qualia on reading the sentence "You are not the person who speaks your thoughts; you are the person who hears your thoughts."
See "Facing Up to the Problem of Consciousness" by David Chalmers for a more extensive definition.
- **qualitative:** Defined in 2.1: World-model.
Qualitative properties are selected from a finite set; for example, the binary set of {on, off}, or the eighty-eight member set of "piano keys". A *qualitative match* is when two qualities have identical values. A *qualitative binding* is when the two qualities are hypothesized to be bound together - if, for example, the same item of the 88-member set "piano keys" occurs in two instances. A qualitative binding is the

weakest type of binding, since it can often occur by sheer coincidence. However, the larger the set, the less likely a coincidence. Small integers are usually qualitative properties; large integers should be treated as quantitative. See also °SPDM, °quantitative binding, and °structural binding.

- **quantitative:** Defined in 2.1: World-model.
Quantitative characteristics occupy a continuous range; they are selected from a range of real (i.e., floating-point) numbers. A *quantitative match* is when two quantities are identical. A *quantitative binding* occurs when two or more quantitative variables are equal to sufficient precision that coincidence is effectively impossible. Quantitative bindings can also be established by covariance or other quantitative relations. See also °SPDM, °qualitative, °structural.
- **reductholism:**
A word appearing in Gödel, Escher, Bach. "Reductholism" is a synthesis of °reductionism and °holism; I use it to indicate the general theory of systems with multiple levels, including both the holistic disciplines of looking up and the reductionist disciplines of looking down. See also °reductionism and °holism.
- **reductionism:**
Reductionism: The attitude that the whole is the sum of the parts. To take the reductionist view is to look downward, focus on the low-level elements and the rules governing their interactions. See °holism and °reductholism. See also Gödel, Escher, Bach: An Eternal Golden Braid, particularly the dialogues "Prelude" and "Ant Fugue".
"No one in his left brain could deny reductionism." -- °GEB
- **reflection:** Defined in 2.4: Thoughts.
The ability of a thinking system to think about itself. A reflective mind is one that has an image of itself; a self-model. In °Friendly AI, a reflective goal system is one that can regard its own components and content as desirable or undesirable. In mundane programming, a "reflective" programming language is one in which code can access information about code (for example, obtaining a list of all the methods or properties of an object).
- **relevance** **horizon:**
The Universe goes on forever, and what we can say about it goes on even longer. Outside the "relevance horizon" lie all the knowledge and °heuristics and skills that are not relevant to the task. See °computational horizon.
Humans and AIs probably have *very different* relevance horizons.
- **RNUI:** Defined in Interlude: Represent, Notice, Understand, Invent.
Represent, Notice, Understand, Invent. First your AI has to *represent* something, then it has to *notice* it, then it has to *understand* it, then *invent* it. You can't take these items out of sequence.
Representing means having the static data structures to hold the information.
Noticing means being able to see simple relations, to perceive internal coherence; to tell the difference between a representation that makes sense, and a representation composed of random numbers.
Understanding means being able to see goal-oriented properties, and how the thing understood fits into the larger structure of the Universe - the thing's functionality, the causes of that thing's characteristics, and so on.
Inventing means being able to start with a high-level goal - "rapid transportation" - and design a bicycle.
- **salience:**
How much attention we're paying to something. The salient event/object is the focus of attention (or *a* focus of attention); the salient event/object occupies the foreground of the mind, rather than the background.
Deciding what to pay attention to is often a significant part of the problem.
- **scalar:**
A single number, as opposed to two or more numbers. The *speed* of an airplane is a scalar quantity; it can be described by a single number. The *velocity* of an airplane, which includes the direction as well as the speed, is a vector - it must be described by two numbers. (Three numbers, if it's a three-dimensional airplane.)
There's a timeworn joke about mosquitoes and mountain climbers that is usually mentioned at this point, but forget it.
- **search** **trees:**
One of the most venerable tools of AI. In a game of tic-tac-toe, you can make any of nine possible moves, then I can make any of eight possible moves, then you can make any of seven possible moves... The computational representation of the game - in a classical AI - would look like a tree; a single node representing the start of the game, with nine branches leading to nine first-move nodes; each first-move node would have eight branches leading to a total of seventy-two possible second-move nodes, and so on. By searching through the entire tree, a classical AI could play a perfect game of tic-tac-toe.
It is possible, even likely, that human cognition involves the use of similar (although much messier) search trees. Or not.
- **seed AI:** Defined in 1.1: Seed AI.
An AI designed for self-understanding, self-improvement, and recursive self-improvement. See 1.1: Seed AI, or the introductory article "What is Seed AI?" on the Singularity Institute's website.
- **sensory modality:** Defined in 2.2: Sensory modalities.
A sensory modality, in an AI, is a module analogous to the human visual cortex, the human auditory cortex, or some other chunk of neurology underlying one of the senses. A modality contains the data structures needed to represent the target domain; the active processing which enables the perception of higher-level features and coherence in that domain; and the interface to the concept level which enables the abstraction of, and visualization of, patterns and objects in that domain.
- **SIAI:**
An abbreviation for "Singularity Institute for Artificial Intelligence".
- **space** **of** **simultaneity:**
A cross-section of the Universe consisting of every event that is happening "right now" according to your reference frame. In Special Relativity, everyone has a different space of simultaneity depending on how fast they're going, and there is no "correct" space of simultaneity. °Turing machines and digital computers have a single, correct space of simultaneity. (For a Turing machine, the space of simultaneity is the state of the tape during any given tick.)
- **SPDM:** Defined in 2.1: World-model.
Sensory, Predictive, Decisive, Manipulative.
Intelligence is an evolutionary advantage because it enables us to model, predict, and manipulate reality. This idea can be refined into describing four levels of *binding* between a model and reality.
A *sensory* binding is simply a surface correspondence between data structures in the model and whatever high-level properties of reality are being modeled.
A *predictive* binding is one that can be used to correctly predict future sensory inputs.

A *decisive* binding is one that can be used to decide between limited sets of possible actions based on the utility of the predicted results.

A *manipulative* binding is one that can be used to start from a specified result and plan a sequence of actions that will bring about the desired result.

See also °qualitative, °quantitative, °structural.

- **stochastic:**

Describing a statistical feature of a population. Since stochastic processes use multiple parallel elements or averaged populations of elements to carry information, they are much more tolerant of errors than more °crystalline constructs such as most modern-day software.

Evolution tends to result in systems that are tolerant of errors and mutation; not necessarily because of a °selection pressure for error-tolerance, but because every evolved system is *ipso facto* one that mutated without dying. Systems with redundant low-level elements - stochastic processes - are the simplest way to get error-tolerance, and generally the first method that evolution hits on. This is a reason why evolved organisms tend to use multiply layered stochastic systems for everything, even though each extra layer of stochastic abstraction considerably decreases efficiency.

- **structural:** Defined in 2.1: World-model.
Structural characteristics are made up of multiple °qualitative or °quantitative components. A *structural match* is when two complex patterns are identical. A *structural binding* occurs when two complex patterns are identical, or bound together to such a degree that coincidence is effectively impossible - only a pattern copy of some kind could have generated the identity. A structural binding is usually the strongest form. See also °SPDM, °qualitative, °quantitative.

- **subjunctive:**

A what-if scenario, e.g. "What if you were to drop that glass of milk?" Something imagined or visualized that *isn't* supposed to describe present or past reality, although a sufficiently attractive what-if scenario might later be turned into a plan, and thence into reality. See also °counterfactual.

- **tera:**

10¹². One thousand billion; a thousand °°giga". See the Hacker's Dictionary for the full list of quantifiers.

- **three:** Defined in 2.3: Concepts.
The concept of "three" is decomposed in some detail in 2.3.3: The concept of "three".

- **time:** Defined in 3.1: Time and Linearity.
Time in a digital computer is °discrete and has a single °space of simultaneity, so anyone who's ever played °Conway's Game of Life knows everything they need to know about the True Ultimate Nature of time in the AI. With each tick of the clock, each frame is derived from the preceeding frame by the "laws of physics" of that °ontology. (Higher-level regularities in the sequence of frames form what we call causality; more about this in *Unimplemented section: Causality*.)

- **Turing-computability:**

If you really haven't heard the term "Turing-computable" before, the first thing you need to do is read Douglas R. Hofstadter's Gödel, Escher, Bach: An Eternal Golden Braid. Drop whatever you're doing, get the book, and read it. It's no substitute, but there's also a nice definition of "Turing machine" in the Stanford Encyclopedia of Philosophy; also, I give a sample visualization of a °Turing machine in *Unimplemented section: Causality*.

Any modern digital computer can, in theory, be simulated by a Turing machine. Any modern computer can also simulate a Turing machine, at least until it runs out of memory (Turing machines, as mathematical concepts, have an infinite amount of memory). In essence, Turing demonstrated that a very wide class of computers, including modern Pentiums and PowerPC chips, are all fundamentally equivalent - they can all simulate each other, given enough time and memory.

There is a task known as the *halting problem* - in essence, to determine whether Turing machine X, acting on input Y, will halt or continue forever. Since the actions of a Turing machine are clearly defined and unambiguous, the halting problem obviously has a true, unique, mathematically correct yes-or-no answer for any specific question. Turing, using a diagonalization argument, demonstrated that no Turing machine can solve the general halting problem. Since any modern digital computer can be simulated by a Turing machine, it follows that no digital computer can solve the halting problem. The halting problem is *noncomputable*. (This doesn't necessarily demonstrate a fundamental "inferiority" of computers, since there's no reason to suppose that *humans* can solve the halting problem. In fact, we can't do so simply by virtue of the fact that we have limited memories.)

A controversial question is whether *our Universe* is Turing-computable - that is, can the laws of physics be simulated to arbitrary accuracy by a digital computer with sufficient speed and memory? And if not, do the uncomputabilities carry over to (a) °qualia and (b) °intelligence? I don't wish to speculate here about °qualia; I don't understand them and neither do you. However, I do understand intelligence, so I'm fairly sure that even if qualia are noncomputable, that noncomputability doesn't carry over into human general intelligence.

I happen to believe that our physical Universe is noncomputable, mostly because I don't trust the Turing formalism. The concept of causality involved strikes me as fundamentally subjective; there's no tail-end recursion explaining why anything exists in the first place; I've never seen a good mathematical definition of °°instantiation"; plus a lot of other reasons that are *way* beyond the scope of this document. If the physical Universe is noncomputable, I believe that qualia are probably noncomputable as well. However, this belief is strictly in my personal capacity and is not defended (or attacked, or discussed) in °GISAI. I am in the extreme minority in so believing, and in an even smaller minority (possibly a minority of one) in believing that *qualia* are noncomputable but that this says nothing about the impossibility, or even the difficulty, of achieving *real intelligence* on computable hardware. I don't believe that semantics require special causal powers, that Godel's Theorem is at all difficult to explain to a computable intelligence, that human mathematical ability is noncomputable, that humans are superior to mere computers, or any of the other °memes that customarily go along with the "noncomputability" meme. Anyway, back to GISAI.

- **ve:**

A °gender-neutral pronoun. The equivalent of "he" or "she".

- **ver:**

A °gender-neutral pronoun. The equivalent of "him" or "her".

- **verself:**

A °gender-neutral pronoun. The equivalent of "himself" or "herself".

- **vis:**

A °gender-neutral pronoun. The equivalent of "his" or "her" (or "hers").

- **world-model:** Defined in 2.1: World-model.

The AI's mental image of the world. Includes °declaratively stored memories containing knowledge about the world, and the contents of sensory modalities - whether sensed, or imagined - constituting the mental workspace. The latter includes °subjunctive or even °counterfactual mental imagery, since even a subjunctive chain of reasoning has internal consistency.

- [1](#): Initial versions of the AI will almost certainly run on interpreted code, and self-modification will take place at or above that level. Eventually, however, a sufficiently intelligent AI should be able to dispense with the interpreted code and rewrite itself in assembly language.
- [2](#): This is actually part of the procedure for building gcc, the GNU C Compiler. Given the gcc source, you build the first version of the gcc binaries using the compiler that came with the system, then build the second version of gcc using the first version, then build a third version using the second version. The idea is that any idiosyncrasies in the included compiler might show up in the first gcc binaries, but won't break them; the second version should be fairly true to the original source, and can be safely used to compile a third version.
- [3](#): The nanotechnology described in Nanosystems, which is basically the nanotechnological equivalent of a vacuum tube - acoustic computing, diamondoid rod logics (4) - describes a one-kilogram computer, running on 100 kW of power, which performs 10^{21} ops/sec using 10^{12} CPUs running at 10^9 ops/sec. The human brain is composed of approximately 100 billion neurons and 100 trillion synapses, firing 200 times per second, for approximately 10^{17} ops/sec total. Thus a seed AI on a nanocomputer would run at ten thousand times the raw power and a million times the linear speed of a human, even before superior software was taken into account. (And if an AI with that much brainpower can't write awesomely superior software, the project has failed.)
- [4](#): That is, the transistor-equivalents consist of kiloatom structures physically moving at the speed of sound in diamond, rather than photons or electrons moving at an appreciable fraction of the speed of light
- [5](#): It might be interesting to learn, just for the record, whether a chess master using a chess program could beat (a) Deep Blue and (b) Kasparov.
- [6](#): Note that I say "instincts". It doesn't mean that AIs are automatically socially stupid. AIs might run human instincts in emulation, or they might develop cognitively-informed rules that make them far savvier than humans. What I mean is that the social heuristics they learn will be matters of conscious thought, not self-delusion and built-in reflexes. They will not make politically-caused *mistakes*.
- [7](#): Whether this process continues indefinitely or dies out will depend on the behavior of the power/intelligence/efficiency curve. That is, what is the software efficiency as a function of the intelligence of the programmer-AI? What is the intelligence of the programmer-AI, as a function of the efficiency of the software on a given amount of hardware?

Since this curve folds in on itself, most "reasonable" images of the local curves for intelligence and efficiency, when combined, are likely to result in a breakthrough-and-bottleneck series at the global level. At least, this is what's likely to happen in the prehuman areas of the curve. Once a breakthrough carries the seed AI past the human level, I would expect the "nanotechnology-to-Transition Guide" curve to take over.

- [8](#): "Hardware" intelligence doesn't necessarily refer to raw computing power as such. I use "hardware" intelligence to refer to that component of intelligence which is determined by genetics and particularly the variance between species, as opposed to the variance between humans. (See "complex functional adaptation".)
- [9](#): At the same time, there's *less* going on than might appear to "naive" introspection. If asked why the metal crumples and the glass shatters, you would say that metal is ductile and glass is fragile. But you didn't need to perform that reasoning process, consciously or unconsciously, to visualize the anvil hitting the car. You've seen car crashes on TV and movies, and your visualization "borrowed" the "cached outcome". You didn't need to know that anvils are heavy metal objects and that gravity accelerates them downwards with enough force to damage cars; you've seen anvils falling in cartoons.

Introspection, like evolutionary reasoning, is an incredibly powerful tool. Like evolutionary reasoning, it takes practice, talent, and self-awareness to use it on a professional level - to reliably distinguish between *post facto* and "pre facto" (10) reasoning, or between original thought and "cached thought".

Some people, maybe even a majority of readers, may not have needed to visualize the car smashing before deducing that it would break - or, rather, accepting that the sentence "Dropping an anvil on a car will break it" is true - or, rather, continuing to read without noticing that the sentence was false.

- [10](#): Carl Feynman points out that the correct term is "ante facto", which surprises me, because I didn't think there'd be a term for it at all.
- [11](#): Every rule has an exception. The exception to the "Law of Pragmatism" is the "Bayesian Probability Theorem".
- [12](#): It's actually rather surprising that the vast body of knowledge about human neuroscience and cognition has not yet been reflected in proposed designs for AIs. It makes you wonder if there's some kind of rule that says that AI researchers don't study cognitive science. This wouldn't make any sense, and is almost certainly false, but you do get that impression.
- [13](#): After some soul-searching, I decided to use "his" instead of "vis", since (a) hunter-gatherer societies are often blatantly sexist; (b1) I'd have no qualms about using "his" or "her" if we were talking about Alice and Bob in cryptography; (b2) from a cosmic perspective, one occupation has no greater significance than the other.
- [14](#): It is very likely that human intelligence derives not from the need to outwit tigers, but the need to outwit other humans. (See "conspecifics, and "sexual selection in the glossary.") Hopefully, none of this will hold true of AIs. It's just an important thing to know about humans.
- [15](#): Philosophers have been wrestling with this problem, "the meaning of meaning", for ages. Attempts to create a mathematical definition are probably doomed; there are no selection pressures in favor of reasoning processes which are precisely definable and provably correct. Evolution favors the creation of *useful* models - that is, models whose use promotes inclusive reproductive fitness. In some cases, such as tribal politics, selection pressures may have favored inaccurate, observer-biased models, with consequent problems for modern-day humanity. See also Interlude: The Consensus and the Veil of Maya.
- [16](#): This may sound like the setup for one of those jokes that ends with the physicist saying "First, assume a spherical chicken...", but the billiard-ball domain is complex enough to pose nearly every problem that would be faced by a real-world AI, including uncertainty. Even if sensory information is perfect and complete, the *internal* model is still uncertain - will spending 30 CPU-seconds on a problem-solving strategy yield results, or just another blind alley?

- [17](#): Since a mapping inherently requires a mapper, I do not believe that there is any way to mathematically define a sensory binding in an "observer-independent" fashion. In fact, I do not believe there is any way to define *any* binding in an observer-independent way. I do not believe there is any mathematical way to define when "Turing-computable process A" instantiates Turing-computable process B. I've tried.
- [18](#): In an uncertain world, the AI would need to be able to recognize if a plan had worked, and re-plan if the actions did not have the predicted results. Smart minds design plans that bear in mind the possibility of error.
- [19](#): I could be wrong about this. Ask your local physicist.
- [20](#): Vision in the brain involves an enormous amount of cortex, not just the occipital lobe. In neuroanatomical terms, the processes I'll be talking about take place in the retina, the lateral geniculate nucleus, the striate cortex, and the higher cortical visual areas.
- [21](#): Proprioception and the sense of touch are sometimes lumped together as the "haptic" modality. However, proprioception is served by a separate set of nerves - position sensors embedded in muscle tissue and bone. I think that proprioceptive sensory information is routed to a distinct parietal area (not just sensorimotor cortex), but I can't find a reference that says so one way or the other. (I do know that there's a separate spinal pathway.)

If proprioception does have a separate area of cortex (with distinct representations and extractable features), then it's a distinct sensory modality and should be known as such.

- [22](#): It may occur to some readers, at this point, to suggest that "XML" or "LISP tokens" or even "ones and zeroes" represent a common, underlying data format. But remember the "Law of Pragmatism". The level at which everything is composed of XML does not contribute materially to intelligence. It's the particular *content* of the XML/tokens/binary digits, maybe even the content at a level above that, which is generating interesting behaviors. If you can take the level on which pixels can be viewed as colors in a 2D array, or auditory "pixels" viewed as pitches in a 1D array (both levels which are quite a bit above "ones and zeroes", or even "XML"), and all the high-level features extracted, from "edges" to "descending tones", and make all of *that* obey a common format - create a universal format that allows direct interchange *on the level where* domain-specialized representations are necessary - well, you'd have done something *really cool*.

Otherwise, it's like suggesting that translating between Microsoft Word and HTML should be programmatically trivial because both files are really just magnetic patterns in the atoms of the hard disk. What matters is the level where they're *different* - that's where the Law of Pragmatism says the intelligence is. And if they aren't different *anywhere* - why, then, there's probably no intelligence.

- [23](#): There's even a dramatic play, "Helen Keller", about the necessity of symbol tags to intelligent cognition.
- [24](#): Lakoff and Johnson call it a 'schema'.
- [25](#): And don't tell me that they're just simulations. How do you know that all the quarks in this Universe aren't being simulated on some big honkin' computer somewhere? Unless the fact that it's a simulation makes an observable, experimentally detectable difference, who cares? I don't believe in zombies.
- [26](#): The necessity for a strongly bound metaphor was something that Plato, for example, never understood. If you make a metaphor between, say, human death and the setting of the sun, it doesn't prove that "Because the sun rises tomorrow, death is impermanent." For a metaphor to bind predictively, it is necessary that the metaphor result from a single underlying *cause* which produces both sets of effects. It is not enough to say that there is a shared high-level characteristic. Mere similarity, on a high level, is not enough to produce high-level predictions. However, similarity on a *low* level - a metaphor which analogizes between the *elements* of A and the *elements* of B - is often enough to predict high-level similarities.
- [27](#): "MITECS, "Thalamus":

It [the thalamus] has a simple position in the overall architecture; virtually all information arriving at the cerebral cortex comes from the thalamus, which receives it from subcortical structures... In particular, all visual, auditory, tactile, and proprioceptive information passes through the thalamus on its way to cortex...

These facts give rise to the classic view that the thalamus is a passive relay station which generates virtually all the information bearing input to the cortex...

BUT the above picture has omitted one fundamental fact: *all projections from thalamus to cortex are reciprocated by feedback projections from cortex to thalamus of the same or even larger size*. For instance, Sherman and Koch (1986) estimate that in cat there are roughly 10^6 fibers from the lateral geniculate nucleus in the thalamus to the visual cortex, but 10^7 fibers in the reverse direction! (Italics in original.)

The most popular hypothesis is that these fibers play a gatekeeping role, assisting in focus of attention (why do you need *more* fibers to do that?); or, more plausibly, top-down constraints in feature extraction. And since this particular statistic is for cats, the latter hypothesis may be mostly correct. Visualization - *imagination* - is stereotypically associated with minds directed by general intelligence. While cats may need a memory, and thus the ability to reconstruct images from remembered high-level features, they probably don't need the detailed, fine-grained imagination of a human. So I wouldn't be surprised to find an even *greater* discrepancy in humans!

Or perhaps, even for cats, more fibers go from cortex to thalamus than vice versa because even mnemonic sensory manipulation is just computationally *harder* than sensory perception.

- [28](#): A "curved trajectory" exists only in spacetime; in any given frame, any instant in time, the trajectory is in exactly one place. We understand the concept of a "curved trajectory" by mapping the 4D trajectory onto a 3D (or 2D) spatial curve, which allows us to use our visual feature-extractors to determine whether the curve is tightly curved, sharp-edged, et cetera. A billiard-ball model can represent the 4D trajectory, but not the 3D spatial curve.
- [29](#): It's also possible to connect nine dots with *three* straight lines without lifting pen from paper. But you have to think even farther outside the box.

- [30](#): As I've said elsewhere, a real AI isn't a computer program any more than a human is an amoeba. To say it more formally (which I can do, now that we're in *this* Webpage), an AI's useful complexity is as far from the program level as a human's useful complexity is distant from the cellular level. A frustrated Windows user assuming that AIs would make the same kind of mistakes as Windows applications is being as foolish as an extraterrestrial assuming that humans would have characteristics stereotypical of amoebas.
- [31](#): Also, some of the things that I'm lumping together as "concepts" in the AI are probably stored and retrieved by distinct subsystems in humans.
- [32](#): See George Lakoff, "Women, Fire, and Dangerous Things".
- [33](#): Also, "red" is a *basic-level category*. (See George Lakoff, in "Women, Fire, and Dangerous Things".) The basic level includes categories such as "red", "dog", and "chair", but not "color", "animal", or "furniture", nor "scarlet", "Irish setter", and "rocker". The basic level is the highest level at which you can summon up a mental image for the category. The basic level is cognitively privileged in a number of ways. For example, basic-level words are the lexically short, the first words learned by children, and the first words to enter a language. Since basic-level categories are more salient, a cognitive scientist looking for an example of a "category" will almost always select a basic-level category. But what holds true of basic-level categories may not hold of higher or lower categories. It's important to bear this in mind whenever you see a discussion of categories that offers a basic-level category as an example.
- [34](#): Note that, for this to work, each correspondence has to be unique. Each object must have only one slot for a unique correspondence in each particular mapping, and once that slot is detected as being filled, it must be impossible to form a correspondence. It's noteworthy that, in spatial modalities modeled on macroscopic physics, the same object cannot occupy two positions, and two objects cannot occupy the same position. In other words, "unique correspondence" involves not so much counting to one, or vetoing an object with "two" correspondences, but simply noticing when objects are bumping into each other, or when the *correspondence* slot is "full" or "empty".
- [35](#): Or rather, "one-to-one correspondence between group membership". "Same number as" implies counting, then comparing the numerical descriptions. The object-by-object implementation described would directly compare objects within the two images. "Same number as" is a much more powerful concept requiring much deeper prerequisites.
- [36](#): "Triangle" slipped to "pyramid" instead of "tetrahedron", even though I know in theory that a tetrahedron is more triangular. I suppose that I've seen more pyramids, or that pyramids are easier to visualize, or that I learned about pyramids at an earlier age, or that I've seen more pyramidal physical objects, or that pyramids resonate more strongly with the bulbish shape. If I'd been constructing a "triangular light bulb" via a deliberate thought-level reasoning process, it probably would have come out as a tetrahedron.
- [37](#): *Slow* and *fast* are not the first concepts abstracted. First comes *slower* and *faster*. Then comes *slower-than-expected* and *faster-than-expected*. Finally comes the concept of *slow* in the absolute sense of a process that occurs slowly relative to the general stream of consciousness (with lots of extra space for thought) and *fast* as describing a process that flits by almost too quickly to be noticed. Since a seed AI should be able to replay cognitive events at will, slowly or quickly to taste, such observer-relative speeds are likely to be a far less important feature of life than for us environmentally-bound humans.
- [38](#): Haven't you ever heard the phrase "The blind watchmaker"?
- [39](#): Say, "within 20000 ticks of each other", or a fuzzy-boundaries ("within 10 ticks is even better") equivalent thereof.
- [40](#): In digital computers! I'm not talking about Relativity!
- [41](#): Again, ducking the question of whether we're considering the "t" dimension or the Minkowskian interval that describes subjective time.
- [42](#): "Deciding what to think about" is an example of such a task!
- [43](#): I'm forced to use colors to create an immediately visible symmetry without overflowing your visual memory. Not only do colors leap more easily to the eye, but they also establish a much stronger binding - one that's much less likely to be a coincidence than a match between Xs and Os. For the bilateral symmetry in XO strings to be visible beyond doubt, the string would have to go on for such a length that it would overflow your visual memory, and your perception of the reflection would perform be conscious rather than intuitive.
- [44](#): In the human brain, the temporal lobe handles object recognition.
- [45](#): Or similar mechanical systems; I mention motorcycles because I have recently been rereading Zen and the Art of Motorcycle Maintenance.
- [46](#): But dangerous - explicit representation of higher levels needs to be handled very carefully, or the °crystalline limits of the representation become the limits of the AI.
- [47](#): Bleah! How do you talk to the AI if you can't identify the stream of consciousness?
- [48](#): Because a seed AI should be able to replay observed events at different speeds, and because a seed AI's stream of consciousness is likely to run either much faster or much slower than the external environment, the seed AI's concept of "simultaneity" may also apply to, for example, observing three supernovas in three weeks. These stellar events may appear "simultaneous" when viewed on a galactic timescale, but no watching human would actually describe them as "simultaneous", since a week is an intrinsically long time to us. A seed AI would be able to actually stretch or slow time to appreciate the galactic scale, and might describe the events as "simultaneous" not metaphorically but literally - actually *perceiving* them as simultaneous on an automatically adjusted subjective timescale.
- [49](#): Why does an uneventful process appear to take longer? (A): Consider the time-as-pathway metaphor; if events are spaced far apart, so that they are passed infrequently, then the time-is-movement metaphor would lead one to think that the observer was moving "slowly". (B): Boredom is unpleasant (because wasting time is an evolutionary disadvantage), and unpleasant processes appear to take longer. Why? Either because it's hardwired in, or because (C): We spend our time wishing that boring events were over, which makes the process appear to take longer. Why? Either because it's hardwired in, or because (D): We pay a lot of attention to how much time is passing. The long intervals move upwards in salience, occupying our immediate minds; and afterwards, we remember all the long intervals as part of the event, which may either stretch out the subjective length of the actual memory, or simply result in the memory being labeled with "took a long time".
- [50](#): The key difference (51) is probably that processes packed full of memorable events will be remembered as being longer, while processes full of immediate, subjective events will occupy all the immediate attention when they occur, while afterwards seeming to have flashed by.
- [51](#): "Key difference" = key variable.
- [52](#): One must be careful to ensure that the maximum resolution of this timeline does not become the new system clock; ideally, the resolution should be finer than the time it takes to process a concept or notice an event, while still enough faster than the system clock that the modality doesn't take up too much overhead.
- [53](#): Actually, there are more requirements: One, for all A and B, one and only one of these relations holds: $A < B$, $A = B$, or $B < A$. Two, if $A = B$, then $A < C$ implies $B < C$, and $C < A$ implies $C < B$. Three, if $A = B$, and $B = C$, $A = C$. Four, if $A = B$, then $B = A$. But that's just legalese to make sure you're using "<" and "=" properly.

- [54](#): Neurons are very good at adapting to patterns; once the pattern is adapted to, it is *expected*, and changes in the pattern can be noticed. This is a *general* property of neurons, applying to *everything* that goes through them. Given a sufficiently long exposure to any sufficiently strong low-level pattern, and no distractions, we will eventually notice the pattern - even if it's a type of pattern we have never seen before.

It may be that this is a genuine instance of a physical property of the underlying neurons that would be very hard to duplicate as an external heuristic, without creating an additional layer of neuronlike interpreted code. However, I think that procedural pattern-detectors, plus the ability to learn heuristics about which pattern-detectors to apply and when, should be able to match the effectiveness of biological neurons at forming expectations and detecting patterns.

Our neural ability to adapt to unexpected new patterns may be simulable by trying to detect identity or covariance in a few thousand entirely random quantities, every now and then.

- [55](#): Trajectory as a visual-temporal experience, objects moving through space, is obviously a ubiquitous topic from a human's perspective. For an AI living in the world of source code, such trajectories are a more esoteric subject.
- [56](#): 200Hz neurons, 100m/s axons, and a 0.1m diameter
- [57](#): I don't remember where I first heard this, but my guess is Raymond Smullyan.